Alexandria Engineering Journal 135 (2026) 273–317

Contents lists available at ScienceDirect

Alexandria Engineering Journal

journal homepage: www.elsevier.com/locate/aej

Original article

Invasive weed optimization based metaheuristic approach for solving constrained risk budgeted portfolio selection problem

Zubair Ashraf [a], Mohammad Shahid [b,*], Lamaan Sami [b,*], Faraz Hasan [c], Mohd Shamim [b]

[a] Department of Computer Science and Artificial Intelligence, College of Computing and Information Technology, University of Bisha, Kingdom of Saudi Arabia
[b] Department of Commerce, Aligarh Muslim University, Aligarh, Uttar Pradesh, India
[c] Department of Artificial Intelligence & Data Science, GITAM School of Technology, GITAM (Deemed to be University) Hyderabad Campus, India

ARTICLE INFO

Keywords:
Portfolio optimization
Risk budgeting portfolio
Sharpe ratio
Invasive weed Optimization
IWO

ABSTRACT

Portfolio management focuses on investing in the financial sector to achieve the highest return while tolerating the lowest risk. The optimal financial allocation has long been considered one of the essential aspects of risk-adjusted financial sector investment. Therefore, many optimization techniques have been established to maximize the return on risk. This paper presents a novel framework named the risk-budgeted portfolio selection (RBPS) model, which allocates the total risk of a portfolio across different securities by incorporating risk budgeting (RB) levels to ensure the portfolio's risk is diversified while maximizing the Sharpe ratio. To address the proposed RBPS model, an invasive weed optimization (IWO) algorithm-based solution method is suggested, and risk budgeting constraints are accommodated using resilient and flexible repairing procedures. Experiments have been performed using two newly created datasets from the Sensex of the Bombay Stock Exchange and the National Stock Exchange from India. The percentage improvement of the maximum Sharpe ratio obtained by IWO is up to 1.95 % at RB% = 12.5 among its peer's algorithms. Moreover, the experiments have been extended to global benchmark datasets to evaluate the proposed approach. Finally, statistical analysis is conducted to test the significance of improvement in the RBPS model.

1. Introduction

Investment is a crucial financial process that involves allocating funds to diverse securities/assets to achieve the highest return at the lowest risk [1]. Investment objectives include the safety of funds, regular return, diversification, and certainty in return. These objectives are directly or indirectly related to maximizing return and minimizing risk. Markowitz [2], the pioneer of modern portfolio theory, developed a mean-variance (M-V) model for optimizing portfolios based on the systematic and scientific trade-off between return and risk. The Markowitz model has gained popularity and has become the basis for developing innovative methods for portfolio optimization [3]. Due to the fluctuating values of securities and the uncertainty of the market, performing an extensive risk management analysis is necessary. The researchers started to understand that tail risk should be more critical than insignificant. Sharpe [4] mentioned that the best way to optimize a portfolio based on its mean variance is not always the same as just adding together the best portfolios for each asset. Several portfolio models, such as Variance-skewness, Value-at-Risk, Minimax, Sharpe ratio, and so on [5–10], were further developed to quantify the risk.

In the literature [1,11–14], the principal constraints of the portfolio are cardinality, boundary, transactional, transaction lot, etc. However, more attention should be given to the risk-budgeted constraints for developing a portfolio model. Risk budgeting [15,16], a crucial strategy in portfolio, is the process of distributing the total risk of a portfolio across various securities. Risk budgeting aims to optimize potential returns while keeping an acceptable risk level. The essential concerns [9, 17,18] of risk budgeting constraints in a portfolio selection are:

- Measurement of risk: The first phase quantifies the risk level associated with each security. It is commonly accomplished using standard deviation, Value at Risk, or other risk assessments.
- Attribution of risk: After quantifying the risk, it is assigned to various origins, such as asset classes, sectors, or individual securities. Risk attribution aids in categorising the main factors that contribute to risk within the portfolio.
- Allocation of risk: The overall risk budget, which represents the allowable degree of risk for the portfolio, is distributed across various

* Corresponding authors.
E-mail addresses: mdshahid.cm@amu.ac.in (M. Shahid), dr.lamaansami@gmail.com (L. Sami).

https://doi.org/10.1016/j.aej.2025.12.067
Received 6 January 2024; Received in revised form 28 November 2025; Accepted 29 December 2025
Available online 7 January 2026
1110-0168/© 2025 The Author(s). Published by Elsevier B.V. on behalf of Faculty of Engineering, Alexandria University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

assets based on risk attribution. Risk attribution assigns a segment of the portfolio's risk to each security.

In light of the above mentioned concern, this work presents a risk-budgeted portfolio selection model abbreviated as the RBPS model. The RBPS divides the total capital into asset classes concerning weight. The entire process of allocating capital into different asset classes was done, considering that the objective is to maximize the Sharpe ratio. A higher Sharpe ratio indicates a better risk-adjusted performance. Moreover, in an RBPS model, as the quantity of securities expands, the number of potential combinations for constructing a portfolio rises exponentially, resulting in computationally difficult-to-assess all feasible portfolios. Also, the connections among risk, return, and portfolio weights may exhibit nonlinear patterns and constraints, which must be considered. When dealing with such constrained portfolio optimization problems, the complexity of the issue rises, making it more challenging to discover the best solutions using conventional optimization approaches. Therefore, artificial intelligence (AI) driven optimization algorithms, such as swarm intelligence and evolutionary algorithms, offer a powerful solution to the challenges of portfolio optimization when dealing with many securities [1,12,13].

Over the past two decades, a significant number of AI-driven optimization techniques have been developed. Among them, the Invasive Weed Optimization (IWO) algorithm, a nature-inspired metaheuristic optimization technique introduced by Mehrabian & Lucas [19], has gained substantial popularity. The IWO, which emulates the behaviour of invasive weed species in the process of colonizing and spreading their colonies, was initially designed for the numerical engineering optimization problem. Its success and widespread adoption in the research community have led to its application in diverse fields, including engineering, medical science, and image processing [20–22]. Moreover, the IWO algorithm is prominent in optimizing financial portfolios and has demonstrated superior performance to other traditional models. Many researchers showed the usefulness of the IWO based on several crucial factors such as algorithmic efficiency, risk budget, decision-making, and market information [23–28]. Combining these factors with IWO ultimately leads to an improved portfolio and a more comprehensive analysis of investments. Importantly, Pouya et al. [24] and Shahid et al. [27,28] utilize the IWO with the P/E ratio for decision-making and the Sharpe ratio with risk-budget, which is more relevant in dynamic markets, producing an efficient portfolio in a shorter period. Such characteristics facilitate IWO to perform better than the traditional models, but the quality of the input data and the market instability make the optimization results less reliable.

This work proposes an improved version of the IWO algorithm to address the risk budgeting portfolio selection (RBPS) model in the financial investment framework while distributing total risk over the various securities to maximize the Sharpe ratio. IWO can effectively distribute risk budgets among assets, optimizing the trade-off between risk and return. As IWO requires no parameter specification, it is an excellent choice for portfolio selection involving many assets. Moreover, in the RBPS framework, which includes the nonlinear and non-convex functions, the reproduction and dissemination techniques allow IWO to escape local optima and identify global optima, which leads to a more stable portfolio. The following is a list of the key contributions that this work has made:

- An RBPS model is proposed to maximize the Sharpe ratio subject to the risk budgeting levels, fully invested, and boundary constraints.
- In addition, we presented a novel IWO-based solution approach to solve the RBPS problem, which utilizes a repairing procedure for constraint handling.
- We conducted a thorough comparative study by putting our proposed IWO algorithm against various well-known optimization methodologies, namely the Genetic Algorithm (GA), Simulated Annealing (SA), Differential Evolution (DE), and Particle Swarm Optimization (PSO).
- Our comprehensive experiments include two newly created datasets from the Sensex of the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE) of the Indian market. We further broadened the analysis by comparing the outcomes with globally recognized data sets such as the DAX, FTSE, Hang Seng, Nikki, and S&P, instilling confidence in the robustness of our findings.
- The comprehensive results are illustrated by comparing the portfolio Shape ratios, Risk, and Return with 20 independent runs of all algorithms and getting the performance indexes max, min, mean, median, and standard deviation for all datasets.
- Statistical analysis was carried out to ensure the validity of our results by comparing the portfolio Sharpe ratio samples obtained with 20 independent runs of all algorithms using SPSS 20.

The remaining paper is as follows. In Section 2, the relevant literature was covered. Section 3 presents the proposed RBPS model together with the problem formulation. The solution methodology and constraint management process are provided in Section 4. The experimental simulation using actual data sets is carried out in Section 5. Finally, Section 6 provides the conclusion.

2. Related work

The present section summarizes various studies conducted to solve portfolio selection problems using evolutionary, swarm intelligence, and machine learning approaches. Moreover, the well-known works in the past on portfolio selection models using risk budgeting are also discussed. Table 1 compares the current study thoroughly with some of the most related works in literature. It summarizes the portfolio selection models presented by several researchers over the decades with the multiple constraint considerations and their solution methods for the different datasets. A comprehensive observation concerning the findings and limitations of all works has been discussed in Table 1.

2.1. Evolutionary based approach

Chang et al. [29] used genetic algorithm (GA) to solve the cardinality-constrained portfolio optimization (CCPO) by applying the actual data collected from the financial market comparing the performance with Simulated Annealing (SA) and Tabu Search (TS). Crama & Schyns [30] presented the SA to solve the complex problems of portfolio optimization by mixing the integer quadratic programming of the M-V model. Baixauli-Soler et al. [6] solved portfolio optimization with value-at-risk taking as a risk measure; the presented model was solved using a multi-objective GA. Anagnostopoulos & Mamanis [31] solved M-V portfolio selection with risk budget constraints, cardinality constraints, floor constraints, and ceiling constraints employing five multi-objective evolutionary algorithms. Yu et al. [32] employed a GA-based non-linear integer programming approach for multi-criteria project portfolio optimization. Ardia et al. [33] suggested differential evolutionary (DE) to demonstrate Amman financial applications' utility in solving non-convex portfolio optimization problems. Cheong et al. [34] used investor information for a clustering-based portfolio optimization scheme using a GA. Meghwani & Thakur [35] solved portfolio selection problems with variance, value-at-risk, and conditional value-at-risk as risk measures. All the models were optimized using NSGAII, GWASFGA, and MOEA/D algorithms. Babazadeh & Esfahanipour [7] solved the value-at-risk portfolio selection problem and reported that the nondominated sorting genetic algorithm-II (NSGA-II) managed the constrained multi-period portfolio optimization problems, resulting in a better tread-off solution to the benchmark model. Pal et al. [36] established a thorough method for choosing and optimizing potential assets. NSGA-II based clustering method is used to identify high quality securities to generate variable-length portfolios. Further, the

Table 1
Comparing the present study with some of the well-known works.

Ref.	Model	Constraints	Methods	Platform	Datasets	Observations	Limitations
Leow et al. [5]	VaR	BC, TL	GA & ML	Unspecified	S&P 500	Models achieved superior results of the tested portfolios	Lacks practical aspects such as CC and TC constraints
Baixauli-Soler et al. [6]	VaR	RB, TC, TL, SS	MOGA	Java	Europe	Method was adequate to get optimum solution	Study lacks comparisons
Anagnostopoulos & Mamanis [31]	M-V	BC, CC, FC, FI	NSGAII, SPEA 2, PESA, NPGA2, e-MOEA	Visual C++	OR Library	All MOEAs outperform the SOEA in all problem instances	Model did not consider TC and transaction TL constraints
Meghwani & Thakur [35]	M-V, VaR, CVaR	BC, CC, SS, FC	NSGAII, GWASFGA, MOEA/D	Unspecified	USA (FF)	NSGAII, GWASFGA algorithms outperformed MOEA/D	More constraints, such as round lot, CC, and chance constraints, should be considered
Babazadeh & Esfahanipour [7]	VaR	BC, CC, FC, TC	NSGAII	R	S&P 100	NSGA-II can efficiently handle constraint multi-period portfolio	Other real trading constraints like TL and RB are not used in this model
Pai [69]	M-V	BC, FC, TC, TL, SS, RB	MODE	Unspecified	BSE 200, NIKKEI 225	Provided optimum portfolios	Lack of comparative
Pai [70]	M-V	BC, FC, RB, SS	MODE	Unspecified	Unspecified	Consistent and outperformed single objective	TC and TL constraints are not considered
Ji & Lejeune [62]	M-V, Minimax	BC, SS, RB, FI	MIP	C++, AMPL	S&P 500, S&P 400, S&P 600	Presented approach is found to be effective and efficient	Needed practical constraints such as TC and market impact costs
Shahid et al. [9]	Sharpe Ratio	BC, RB, FC, SS, FI	SFS	MATLAB	BSE30, NSE50, OR	Study confirmed the better performance of the SFS	The study lacks CC and TC constraints
Shahid et al. [72]	Sharpe Ratio	BC, FC, SS, FI, RB	SFS	MATLAB	BSE 30	SFS approach has superior performance over GA	Empirical testing should have been conducted on a larger dataset.
Konstantinou et al. [73]	M-V	BC, CC, FI, RB, SS	GA & SIO	MATLAB	S&P 500	Approach is better than previous approaches regarding performance and consistency	Study lacks practical constraints like TC, TL
Tollo et al. [10]	M-V	BC, TL, FI, RB, SS	GA	Unspecified	NIKKEI 225	Solve small instances and perform reasonably well for large instances	The size of the instances is the main obstacle to using solvers for large instances
Gandikota et al. [74]	M-V	BC, RB, FI, SS	DE	MATLAB	NSE 50	Solution approach helps in selecting the portfolio returns	Need to verify the benchmark datasets, so results cannot be generalized
Proposed	Sharpe Ratio	BC, FC, FI, SS RB	IWO with repairing procedure	MATLAB	BSE, NSE, DAX, FTSE, Hang Sang, Nikki, S&P	Solution algorithm outperforms GA, SA, DE and PSO.	Does not consider CC, TL and TC constraints

RB: Risk budget, BC: Budget constraints, SS: Short sell, CC: Cardinality constraints, FC: floor and ceiling constraints, FI: fully invested, CCo: Cardinality cost, TC: transaction cost and TL: Transaction lot

authors address the cardinality constraint portfolio model using GA to find the weights of security. Under the assumption of downside risk and diversification Li et al. [37] suggested a random portfolio optimization model. A multi-period three-objective portfolio model proposed by Chen et al. [38] constrains return, securities, liquidity, and turnover, then used a MOEA with orthogonal learning to solve it.

2.2. Swarm intelligence (SI) based approach

Cura [39] proposed the PSO technique in mean and return analysis portfolio optimization for the popular stock markets of the world, like the USA, UK, Germany, Hong Kong, and Japan. Corazza et al. [40] proposed the PSO with non-smooth penalty reformulation methods to handle complex, non-convex, and non-differentiable optimization problems in portfolio selection. Zhu et al. [41] used stocks from the Shanghai Stock Exchange; the returns and standard deviations were calculated, and the PSO algorithm was used to optimize the portfolio. Golmakani & Fazal [42] used a constrained PSO with an improved model for CCPOP. Deng et al. [43] collected weekly data for five years from 1992 to 1997 for some of the world's emerging markets to verify the regulation of the CCPOP. Wang et al. [44] suggested a modified ABC algorithm for the CCPOP and created a proper tradeoff between portfolio risk and return. Chen et al. [45] experimented with the ABC technique on four different markets and found that the model may be superior in complex situations compared to conventional portfolio optimization. Lwin et al. [46] integrated scatter search and hill climbing to solve the constrained portfolio selection, and the performance of the proposed approach was compared with NSGAII, SPEA 2, and PESAII. Hajnoori et al. [23] suggested a model that maximizes income per share to optimize the CCPOP and forecast asset pricing using IWO. Pouya et al. [24] used the IWO technique to optimize the portfolio by transforming the multi-objectives of the portfolio into a single objective. This algorithm was applied to data collected on the first 50 companies based on the Iranian Stock Exchange market capitalization. Tuba & Bacanin [47] presented a Firefly Algorithm, a promising swarm intelligence algorithm, to solve constrained portfolio problems. Suthiwong and Sodanil [48] proposed an improved quick ABC algorithm that has been used to resolve the issues of CCPOP. Furthermore, Tollo [49] demonstrated the application of metaheuristics in portfolio optimization by stating that more than traditional M-V formulations and standard optimization methods may be required to address the challenges of NP-hard and CCPOP. Mansourinia & Momeni [50] employed two versions of the ABC algorithm, namely ABCI and ABCII, to solve the portfolio optimization problem; empirical testing on the stocks of the Tehran stock exchange showed that the ABCII algorithm was more efficient than ABCI in solving the portfolio selection. Shahid et al. [27] suggested the IWO algorithm to allocate the optimum weight of foreign securities in the risk budgeting portfolio optimization. Corazza et al. [51] utilized hybrid PSO and adaptive metaheuristic strategies that combine different crossover techniques to enhance the process. Sahala et al. [52] suggested that the artificial bee colony (ABC) algorithm minimizes the portfolio variances, representing risk in investment.

2.3. Machine learning based approach

Traditional M-V model [3] has difficulty fitting into the non-fixed market circumstances and long-term investment horizon. The approaches using machine learning (ML) and deep learning (DL) are more hopeful, as multiple varieties have been demonstrated to have applicability to different portfolio models [53–55]. Leow et al. [5] employed GA and machine learning algorithms for solving portfolio optimization and found that the suggested models achieved superior performance in standard portfolio performance measures. In [56], Liang et al. present a Support Vector Regression to improve price prediction in the classical portfolio problem. An efficient neural network using the Elman networks was designed by Lin et al. that outperforms M-V models in dynamic portfolio selection. Zimmermann et al. [57] used a feed-forward neural network on top of principles in the Black-Litterman (BL) model [58] to achieve superior results. Bayesian reasoning takes expert opinions into account modern portfolio theory via the BL model. This framework has been recently refined by Ko and Lee [59] and developed its user-friendliness. Perform-Based Regularization (PBR), suggested by Ban et al. [60], restricts portfolio asset selection by risk and return variability. Wang et al. [61] employ a neural network with a Long-Short Term Memory (LSTM) cell to develop an effective portfolio and predict asset prices. Ma et al. [62] create the ideal portfolio composition using Random Forest by fusing return prediction with ML and DL on the Chinese market. Recent progress includes Guarino et al. [63] designed dynamic trading strategies using neuro-fuzzy agents that employ neural networks and fuzzy logic. Park et al. [64] proposed risk-sensitive multi-agent networks to enhance the grade of decisions in the existence of uncertainty. Liang et al. [56] showed that asset combination optimization can be achieved via Policy Gradient approaches. The hybrid CNN-BiLSTM models [65] are used for stock selection with many-objective optimization using DL and NSGA-III. The EvoFolio [66] considers users' stock preferences in the portfolio selection system. L. Yan et al. [67] present a new sine cosine flexible Weibull extension distribution demonstrating the relevance within volatility of the financial market. The authors used the traditional time series models such as the autoregressive, and random walk for forecasting the stock volatility. Z. Ali et al. [68] proposed a cross-market graph modelling to model relationships between stocks within the same market and across different markets using specialized graph layers.

2.4. Risk budgeting portfolio selection based approach

Pai & Michel [69] solved portfolio selection with CC, floor & ceiling (FC), and RB constraints using DE and evolutionary strategy. Empirical testing on Indian and Japanese stock market datasets showed that risk budgeting strategy effectively solves portfolio selection. The authors in [8] solved portfolio selection with FC, RB, and net market exposure constraints using DE and found a robust solution. Pai [70] again employed multi-objective DE and found that the proposed approach outperformed its single-objective version. Further, Pai [71] solved the constrained portfolio selection using multi-objective DE and reported that the proposed approach provides optimum portfolios while satisfying all constraints. Ji & Lejeune [17] presented mixed-integer programming in solving M-V, mean-semi deviation, and minimax portfolio selection using risk budgeting constraints on the USA stock market. Shahid et al. [72] introduced a constrained portfolio selection using the stochastic fractal search approach with risk budgeting constraints. The same authors extended their work in [9] to construct an efficient portfolio by optimizing the Sharpe ratio within the specified risk budgeting limitations. Ashraf et al. [73] solved mean-variance portfolio selection with risk budgeting portfolio problem using a hybrid algorithm integrating SFS and artificial bee inspired optimization. Tollo et al. [10] reported constrained portfolio selection with transaction cost and risk budgeting that can solve small instances and perform reasonably well for significant cases. Gandikota et al. [74] presented an adaptive approach for risk budgeting portfolio optimization using DE strategies to address the constrained problem. Pesenti et al. [75] develop an actor-critic approach, leveraging the elicitability of dynamic risk measures, to solve for risk budgeting strategies using deep learning.

Portfolio optimization is a crucial aspect of financial management, and various algorithms, such as evolutionary approaches and swarm intelligence, have also been used to solve optimization problems. These approaches have been proven to provide optimum portfolios while satisfying several, namely, cardinality, floor and ceiling, fully invested, and short sell constraints. Hence, from the above discussion and Table 1, it's clear that only a few studies have successfully tackled risk-budgeted portfolio selection (RBPS); however, none use the IWO algorithm, a new nature-inspired algorithm that has proven its strength in solving optimization problems. Finally, a novel RBPS model was successfully presented and, using IWO, solved in this study. To implement IWO for risk budgeting in financial portfolios, practitioners can leverage the algorithm's mechanisms to allocate risk while adhering to portfolio constraints and objectives. Assign each asset a maximum risk contribution (e.g., RB% of total portfolio risk), mirroring Sharpe Ratio + Penalty for exceeding risk budget, allowing IWO to explore deviations up to predefined limits. It prioritizes solutions that maximize returns while staying within sector/asset risk limits. Our proposed algorithm's validity was robustly tested on the OR library datasets, which are widely recognized benchmark datasets in the algorithm testing domain. The proposed IWO approach offers a robust framework for addressing the challenges of large-scale non-convex portfolio optimization problems and invaluable tools to navigate modern financial decision-making.

3. Proposed risk-budgeting portfolio selection (RBPS) model

Markowitz [2] and Sharpe [4] established the mathematical framework for portfolio selection, with the objective under certain circumstances that intelligent allocation of each stock's capital investment results in an optimal portfolio with the least risk while ensuring the maximum anticipated rate of return. The portfolio selection emphasizes the importance of diversification to reduce risk [15,18,76] and establish robust risk management. It is imperative to consider and mitigate the unsystematic risks. Unsystematic (non-systematic) risk deals with risk unique to a specific asset, firm, industry, or sector. Fig. 1 is a schematic diagram of risk management through portfolio allocation. The operation of standard deviation (SD) of portfolio return to the relationship with the securities allocation of portfolio ratio is shown in Fig. 1. The region below the dashed red line represents the inherent risk posed by the portfolio's exposure to systemic risk. However, the region above the dashed red line represents the unsystematic risk. Although diversification can assist in reducing unsystematic risks, investors must also contemplate techniques to handle systematic risks, including asset allocation and modifying risk in their investing frame. This study proposed a novel risk-budgeting portfolio selection (RBPS) model to reduce

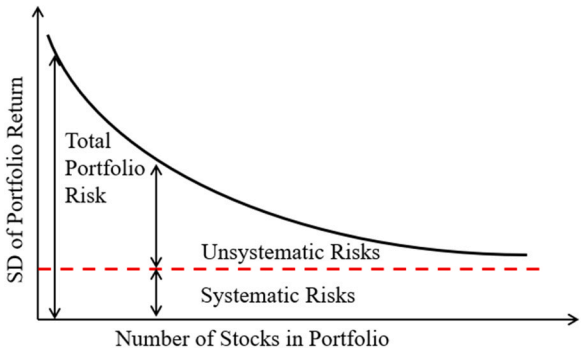


Fig. 1. A schematic diagram of risk management through the portfolio allocation.

the unsystematic. The risk-budgeting approach [9,27,69,70] combines portfolio selection to improve diversification and handle different market constraints to achieve long-term investing objectives.

The proposed RBPS model is based on the following assumptions.

Investors who consider an investment decision do it based on the expected distribution of asset returns over a specific holding period, which maximizes the returns.
Based on the standard deviation of the expected return of the assets, the investor computes the portfolio's risk.
The decisions that investors make are entirely predicated on the level of risk and potential return presented by the assets.
Investors anticipate the highest return at a predetermined threshold for acceptable levels of risk. Accordingly, investors want a fixed rate of return in exchange for accepting a predetermined degree of risk.

The mathematical RBPS model is formulated as follows. Table 2 presents the symbols/notations that frequently appear in mathematical formulations.

A portfolio (P) with K number of assets\securities is expressed as $P = \{ S_1, S_2, \ldots S_K\}$ with weight set as $W=\{ w_1, w_2, \ldots w_K\}$. Suppose the expected return of the portfolio is represented by $r = \{ r_1, r_2, \ldots r_K\}$ with respect to securities $\{ S_1, S_2, \ldots S_K\}$. The total return ($R_P$) of the P may be mathematically evaluated using the Eq. (1) as:

$$R_P = \sum_{i=1}^{K} r_i * w_i \tag{1}$$

Here, r_i and w_i are the i-th expected return and weight value of S_i.

Similarly, the total portfolio risk ($Risk_P$) and the risk contribution ($Risk_{S_i}$) of P may be mathematically evaluated using Eq. (2) and Eq. (3), respectively, as follows.

$$Risk_P = \sqrt{\sum_{i=1}^{K}\sum_{j=1}^{K} w_i * w_j * cv_{ij}} \tag{2}$$

$$Risk_{S_i} = \sqrt{\sum_{j=1}^{K} w_j * cv_{ij}} \tag{3}$$

Here, the statical covariance among the two returns r_i and r_j is represented by cv_{ij}.

In risk budgeting the total portfolio risk ($Risk_P$) is distributed over all securities (S_i). Therefore, the marginal contribution [8,9,15,17] to portfolio risk, aberrated as $MCTR_i$, calculate the contribution of the each S_i in the security risk ($Risk_{S_i}$) to total portfolio risk ($Risk_P$) and may be mathematically evaluated using Eq. (4) as follows.

$$MCTR_i = \frac{Risk_{S_i}}{Risk_P} \tag{4}$$

The absolute contribution to total risk ($ACTR_i$) for the *i*-th security is calculated as:

$$ACTR_i = w_i * MCTR_i \quad i \in K \tag{5}$$

The Sharpe ratio (SP) of the portfolio is a standard measure of excess return per unit of risk [6,12,52,63,64]. Maximizing the Sharpe ratio is the primary objective of investors when allocating funds to securities (S_i). The SP calculation may be performed using the formula below.

$$SR_P = \frac{R_P - R_f}{Risk_P} \tag{6}$$

Where R_f is risk free return. The proposed RBPS deals with finding the optimal portfolio to be invested in by the investor in equity based securities. So, risk free return is considered zero.

The RBPS model is concerned with a long-only portfolio that maximizes its Sharpe ratio SR_P resulting in maximum expected portfolio return R_P and minimum risk $Risk_P$. Risk budgeting (RB) is a proficient investing approach designed to enhance portfolio (P) performance while safeguarding against market volatility. By decomposing total risk $Risk_P$ into its constituent elements, RB enables the formulation of exact percentage risk thresholds for each asset in the portfolio [9,69,70], i.e., $ACTR_i \leq RB_P\% \times Risk_P$. The RB% in RBPS model effectively stating that no single asset is allowed to contribute more than RB% of the total portfolio risk, mathematically, enforces diversification. A smaller RB% forces a higher degree of diversification. To ensure no single asset breaches its risk budgeting level, the proposed RBPS optimization approach spreads the risk across multiple assets. Hence, the RBPS prevents the portfolio from becoming overly reliant on the performance of a few high-risk assets, thereby explicitly limiting concentration risk.

Thus, the proposed RBPS model which decides on the original portfolio allocation has the objective of maximizing its SR_P, i.e., maximizing portfolio return and minimizing risk subject to the constraints of risk-budgeting, fully invested, short-selling, and boundary constraints is formulated as follows.

$$\text{Maximize} SR_P = \sum_{i=1}^{K} r_i * w_i \Big/ \sqrt{\sum_{i=1}^{K}\sum_{j=1}^{K} w_i * w_j * cv_{ij}} \tag{7}$$

$$Subject\ to \begin{cases} ACTR_i \leq RB_P\ \% \times Risk_P & (8) \\ \sum_{i=1}^{K} w_i = 1 & (9) \\ w_i \geq 0 & (10) \\ a_i \leq w_i \leq b_i, i \in K & (11) \end{cases}$$

Here, Eq. (8) presents the risk-budgeting constraint that allocates the portfolio risk over securities having a percentage risk-budgeting level. The $ACTR_i \leq RB\% \times Risk_P$ ensures that the Absolute Contribution to Total Risk (ACTR) from each asset *i* does not exceed a fraction of the $Risk_P$ defined by the risk budgeting level *RB*%. The constraint in Eq. (8) enforces diversification by limiting the concentration risk — that is, it prevents any single asset from dominating the portfolio's total volatility. A high ACTR indicates that the asset is a major source of portfolio risk, leading to concentration. In financial terms, the constraint means each asset's influence on portfolio volatility or downside risk is controlled, helping maintain balance and reducing the likelihood of outsized losses driven by a few assets. Eq. (9) represents a fully invested constraint, meaning that the total budget of the portfolio is allocated to each security. Eq. (10) presents a short-selling constraint, maintaining the non-negativity of the weights. Furthermore, Eq. (11) presents the floor and ceiling constraints or boundary constraints. This constraint sets upper and lower bounds represented as a_i and b_i, respectively, on security weights to ensure that no security is overly represented, avoiding undue risk or underrepresented, which may reduce their relevance within the portfolio.

The proposed RBPS yields a constrained optimization problem that

Table 2
List of symbols/notations used in problem formulation.

Symbols	Details
K	Number of securities or assets in portfolio (P).
S_i	The *i*-th asset in portfolio (P).
r_i	The expected return of *i*-th asset.
w_i	The weights to be invested in *i*-th asset.
a_i	Lower bound of weights to be invested in *i*-th asset.
b_i	Upper bound of weights to be invested in *i*-th asset.
cv_{ij}	The covariance between the returns of assets *i* and *j*.
R_P	Total return of the portfolio (P).
$Risk_P$	Total risk of the portfolio (P).
$Risk_{S_i}$	Risk contribution of the asset S_i.
$MCTR_i$	Marginal contribution to risk by S_i.
$ACTR_i$	Absolute contribution to risk by S_i.
RB_P	Percentage of Risk budgeting of the portfolio Risk.
SR_P	Sharpe ratio of the portfolio (P)

belongs to the class of non-convex nonlinear programming problems due to Eq. (8). Therefore, the meta-heuristic approach-based solution procedure is deemed an intelligent technique for solving the problem. Moreover, the constraints are inflexible and can't be violated, so the solving approach must address them during the execution. To overcome this situation, the constraint repair technique for non-feasible solutions which allows the component weights to be adjusted, limiting them to fit within the feasible region and improving the optimal portfolio.

4. Proposed solution approach

In this section, the solution approach for the proposed RBPS model has been described to determine the best feasible solution and invested weights that maximize the Sharpe ratio objective. One of the critical

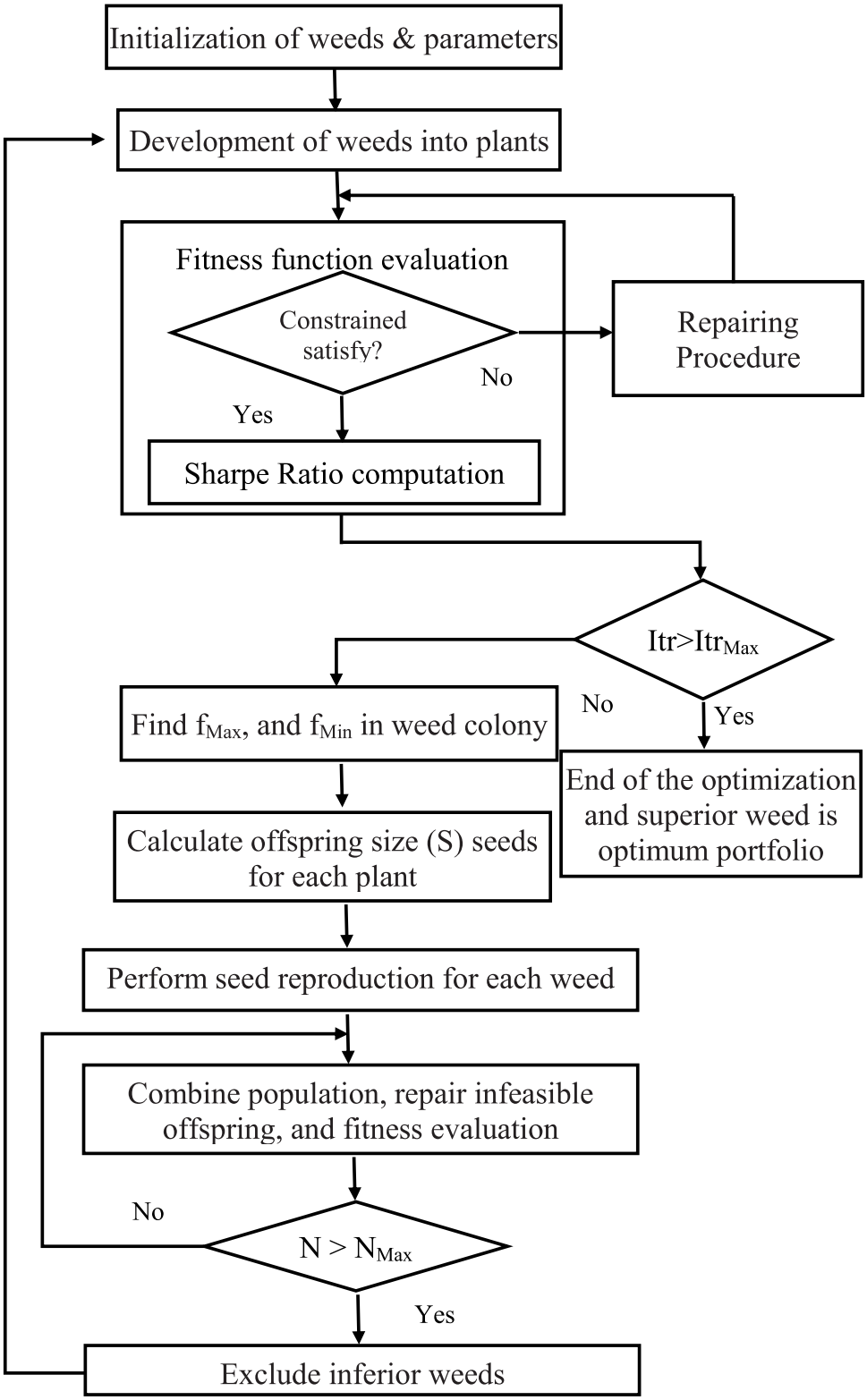


Fig. 2. Flowchart of proposed IWO based solution algorithm.

advancements in the optimization domain is the Invasive Weed Optimization (IWO), introduced by Mehrabian & Lucas [19]. IWO is a numerical stochastic optimization technique motivated by the common weed colony phenomena. Weeds are unwanted plants characterized by aggressive development and adaptability to their natural surroundings. A weed is often provided for assertive species and poses a significant danger to desired plants. Due to their very resilient and adaptable character, weeds have evolved into unwanted plants in farming.

IWO is distinguished from other optimization algorithms by its reproduction method, spatial dispersal, and competitive exclusion. Following are some of the fundamental characteristics [20,22,24] of the technique that imitates the weeds' colonization behaviour:

- Limited quantities of seeds are dispersed throughout the region that is being searched.
- Each seed will eventually develop into a blooming plant that, if healthy enough, will produce further seeds.
- The created seeds are scattered randomly around the search region so that they may develop into new trees.
- The procedure will be terminated after the maximum number of plants has grown to their capacity.

Only the most robust plants can survive and continue the species, whereas less vigorous plants perish and do not produce offspring. The procedure repeats until the predetermined limit of successful repetitions is reached. Therefore, the plant with the highest fitness level is the one that gets the closest to the optimal solution. The present study proposes an improved version of the conventional IWO to maintain the fully invested, risk-budgeted, and boundary-constrained issues with continuous domain adaptation and find the best optimal solutions for the RBPS model. The pseudocode of the suggested solving approach is presented in Algorithm 1, and the flowchart is shown in Fig. 2. The detailed description of solving the RBPS model using the IWO-based solution procedures is as follows.

4.1. Initialization

The initialization process of the IWO algorithm provides the random initial population, representing the solution space by the number of seeds (N). Each seed in the population characterizes the decision variable, the invested weights (w) of the RBPS. The vector of weights involved in the portfolio consist of $K-$ dim representing the number of assets, as presented in Algorithm 1 (Step 6–11). Eq. (12) is used to uniformly distribute random population of candidate weight (w_j) that will be permitted to fall within the permissible lower (a_j) and higher (b_j) boundaries.

$$w_{ij} = a_j + rand(0,1) * \quad (b_j - a_j) i = 1,2,3,\ldots,N \text{and} j = 1,2,3,\ldots,K \tag{12}$$

4.2. Evaluate fitness function

The evaluation of the Sharpe ratio (SR_P) value of the portfolio in the proposed RBPS model is the fitness function, see Eq. (7). The $f = SR_P$ values corresponding to the weight vectors (w_{ij}) are investigated to maximize the Sharpe ratio, as shown from Step 13–16 in Algorithm 1. Before the fitness function evaluation, the feasibility of the weight vectors (candidate solutions) are tested one by one against the constraints considered in the RBPS problem (as in Eq. (7) - (11)), and infeasible individuals are repaired by the constraints repair scheme presented in Section 4.6.

4.3. Seed reproduction phase

The seed reproduction procedure of IWO corresponds to reinforcement of profitable strategies during specific market regime shifts. The adaptive reproduction of seeds exploits shifts due to macroeconomic shocks, policy changes, and structural market breaks. In this process, each colony member participates in generating several seeds via a predetermined region regarding its position to remain robust across changing market states. Mathematically, it pursues a linear relationship between the minimum number of allowed seeds (S_{Min}) and the maximum number of permitted seeds (S_{Max}) according to the fitness function $f = SR_P$ (Sharpe ratio of the RBPS). The number of seeds (S) can be computed with a linear function of S_{Max} and S_{Min} by each weed (Step 20–29 of Algorithm 1) that participated in reproduction over iteration using Eq. (13) as follows.

$$S_i = \lfloor S_{Max} - (S_{Max} - S_{Min}) \frac{f_i - f_{Min}}{f_{Max} - f_{Min}} \rfloor \tag{13}$$

Where, $i = 1,2,\ldots,N$, and f_i is the fitness function for $i-$ th weed in iteration, f_{Max} is the minimum fitness value, and f_{Min} is the maximum fitness value in the weed colony. All the weeds generate S_i number of new seeds to create the offspring. The seeds with higher fitness values (higher Sharpe ratios) are more likely to reproduce and spread more seeds.

4.4. Spatial dispersal

The dispersal of seeds in IWO serves as an analog for exploring new portfolio configurations in response to changing market regimes (e.g., bull, bear, high volatility). The seed dispersal phase introduces randomness and adaptation into the solution approach to foster diversity and assesses IWO's ability to explore new regions. After seed reproduction, the seeds of new offspring are randomly distributed across the search space to create new weeds, using normally distributed random numbers with zero mean and varying variance. It explores multiple regions of the return–risk landscape to avoid being trapped in regime-specific optima. The non-linear alteration to estimate the varying standard deviation (σ = SD) in an iteration is written using Eq. (14) as follows.

$$\sigma = \sigma_{final} + (\sigma_{initial} - \sigma_{final}) \frac{(Itr_{\max} - itr)^n}{(Itr_{\max})^n} \tag{14}$$

The standard deviation of σ decreases from $\sigma_{initial}$ to σ_{final} and n is the non-linear modulation index. The nonlinear reduction of the standard deviation for seed dispersal is a natural fit for volatility clustering. The algorithm starts with broad exploration and high volatility during initialization and fine-tunes its solution with low volatility as it converges.

4.5. Competitive exclusion

The competitive exclusion phase of IWO excludes the weeds beyond the maximum weeds required in the colony. The weeds with a lower fitness function value, that is, the Sharpe ratio of the portfolio, are discarded; only higher fitness value weeds survive to live into higher iteration. In financial terms, stronger assets receive greater weight, while weaker assets lose influence, following the economic principle of survival of the fittest in competitive markets. The offspring reallocated capital toward stronger-performing portfolios that offer better Sharpe ratios at risk-budgeting levels. As mentioned in step 30 to step 35 of Algorithm 1, infeasible individuals cannot reproduce the seeds for offspring generation. However, the better offspring produced by infeasible weeds will have a chance to survive in the next iteration. Thus, the infeasible offspring are excluded from the population in each iteration for faster convergence.

The algorithm repeats the reproduction, dispersal, and selection process until a termination condition is met, such as a maximum number of iterations. The presented IWO-based solution approach is improved and addresses those constrained RBPS optimization problems that are

closed and bounded in the continuous feasible space. Further, to handle the risk-budgeting constraint in Eq. (8), the fully invested constraint in Eq. (9), restricts the short sell in Eq. (10), and boundary conditions in Eq. (11), we propose a modification of the IWO-based solution algorithm by calling as repairing procedure.

Algorithm 1. Invasive weed optimization

Input: Dataset for RBPS, Parameters of IWO, and Risk-Budgeting constant.
Output: Optimal weight vector w and the objective value SR_P.

1. w_{ij}: Weight of $i-$th population and $j-$th asset; $i = 1, \dots, N$ and $j = 1, \dots, K$
2. S_{Max}, S_{Min}: Maximum and minimum no. of seeds
3. $\sigma_{initial}$, σ_{final}: Maximum and minimum value of standard deviation (SD)
4. Itr_{Max}: Maximum number of iterations
5. **Begin**
6. **Initialization ()**
7. **for** $i = 1$ to N **do**
8. **for** $j = 1$ to K **do**
9. Initial population of weeds w_{ij} *// Using Eq. (12)*
10. **end for**
11. **end for**
12. **Repairing_Procedure ()** *// Using Algorithm 2.*
13. **Fitness_function ()**
14. **for** $i = 1$ to N **do** *//Each individual seed*
15. Calculate fitness value $f_i = SR_P$ *// Sharpe ratio of portfolio Eq. (7)*
16. **end for**
17. Obtain the best (f_{Max}) and worst (f_{Min}) solution in the colony
18. **for** $Itr = 1,2, \dots, Itr_{Max}$ *// Number of iterations*
19. Update Standard deviation σ_{itr} *// Spatial dispersal using Eq. (14)*
20. **Reproduction_phase()**
21. Get the f_{Max} and f_{Min} in the weed colony
22. Initialize offspring size by computing the S *// Using Eq. (13)*
23. **for** $i = 1$ to N **do**
24. **for** $s = 1$ to S **do**
25. Generate offspring $seed_s = w_i * rand_i * \sigma_{itr}$
26. **Repairing_Procedure ()** *// Using Algorithm 2.*
27. Calculate fitness value $f_{i,s} = SR_P$ for each $seed_s$
28. **end for**
29. Add offspring to the population
30. **end for**
31. **Competitive_exclusion()**
32. Merge all individual weeds population
33. **Repairing_Procedure ()** *// Using Algorithm 2.*
34. **Fitness_function()**
35. Sort weeds and seeds together as per fitness values
36. **if** $N > N_{Max}$ **then**
37. Best fitness is included in the population in next iteration
38. **end if**
39. **end for**
40. Print (Best weight vector for investment in K assets.)
41. **End**.

4.6. Repairing procedure: constrained handling

This section delves into the essential process of implementing constraints in the proposed RBPS problem. The proposed solution approach is pivotal in ensuring that the limitations outlined by Eqs. (8) - (11) of RBPS formulation are fulfilled.

- Risk-budgeting constraint (Eq. (8)): This constraint relates to the risk associated with each asset and ensures that the overall portfolio risk is within a specified budget or limit.
- Fully invested constraint (Eq. (9)): This constraint is a fundamental requirement in portfolio optimization. It stipulates that the sum of all portfolio weights should be equal to 1, ensuring that the entire investment capital is fully allocated across the assets. This constraint is crucial as it prevents any unallocated or idle capital, thereby maximizing the potential returns of the portfolio.
- Boundary constraints (Eq. (10−11)) define the lower and upper bounds for the portfolio weights assigned to each asset. In other words, they ensure the portfolio weights are within a specified range.

The repair process [9,27,29,77] described in Algorithm 2 is a mechanism to ensure that the solutions generated by the IWO algorithm satisfy these constraints. In the repair process to address the risk-budgeting constraint (Steps 30–50), the portfolio weights are adjusted proportionally based on the risk contributions of each asset and the overall portfolio risk. This means that assets with higher-risk contributions will reduce their weights, while assets with lower-risk contributions will increase their weights. This adjustment continues until the overall portfolio risk is within the specified budget or limit. Repairing the bounds of the weights (Steps 7–14) involves changing the portfolio weights if they violate the boundary constraints: If a weight is below the lower bound, it is set to the lower bound value, and if a weight is above the upper bound, it is set to the upper bound value. Finally, Algorithm 2 ensures the fully invested constraint (Steps 15–29) by ensuring that the sum of all portfolio weights is equal to 1: if the sum of weights is less than 1, the weights are proportionally increased, to sum up to 1, and if the sum of weights is greater than 1, the weights are proportionally decreased, to sum up to 1. After applying the repair process in Algorithm 2, the IWO algorithm-based solution procedure continues its optimization process using the repaired or adjusted portfolio weights (Wr) that satisfy all the constraints.

The overall goal is to improve the search capabilities of the conventional IWO algorithm by incorporating these constraint-handling mechanisms. It allows the algorithm to explore the feasible solution space more effectively and find optimal or near-optimal solutions that satisfy all the constraints of the portfolio optimization problem. The numerical investigation mentioned at the end of the paragraph likely involves testing and evaluating the performance of the modified IWO algorithm in solving constrained portfolio optimization problems, possibly by comparing it with other optimization algorithms or benchmark instances.

Algorithm 2. Repairing Procedure

K: Number of securities in portfolio selection problem
W: Weights of portfolio before repairing
w_i: weight of $i-$th security before repairing
w_i^r: weight of $i-$th security after repairing
W^r: Weights of portfolio after repairing
Begin
Boundary_Constraints( )
    while $i \leq K$ **do**
        if ($w_i \in [0, 1]$) **then**
            do-nothing
        else
            Adjust weight such that $w_i \in [0, 1]$
        end if
    end while
Fully_Invested_Constraints( )
    Calculate $SUM = \sum w_i$
    If ($SUM < 1$) **then**
        Find $DEFICIT = 1 - SUM$ *//compute deficit*
        Find $W_{INC} = DEFICIT\ /K$ *// find increment for the weights*
        for $i = 1$ to K **do**
            $w_i^r = w_i + W_{INC}$
        end for
    else if ($SUM > 1$) **then**
        Compute $EXCESS\ =\ SUM - 1$ *//computer excess*
        Compute $W_{DEC} = EXCESS\ /K$ *//compute decrement for the weights*
        for $i = 1$ to K **do**
            $w_i^r = w_i - W_{DEC}$
        end for
    end if
Risk_Budgeting_Constrains( )
    Set $FEASIBLE\ =\ \{\emptyset\}, INDEX\ =\ 0$
    Calculate $ACTR_j$ of securities S_j for x_{ij} individual *//Using Eq. (5)*
    for $i = 1$ to N **do** *// For all population N*
        for $j = 1$ to K **do** *// For all securities S_j*
            if ($ACTR_j \leq RB\%\ of\ Risk_P$) **then**
                do-nothing
            else
                assigning $INDEX\ =\ 1$
            end if
        end for
        if ($INDEX\ ==\ 0$) **then**
            $FEASIBLE\ = FEASIBLE + x_{ij}$ *// Add to feasible solution set*
        else
            Discard solution x_{ij}
        end if
    end for
    if ($FEASIBLE\ ==\ \emptyset$) **then**
        Generate new solution x_{ij} *// Using Eq. (12)*
        Go to Step-7
    end if
Return w_i^r and W^r
End

5. Experimental study and comparative analysis

In this section, the performance of the proposed IWO-based solution technique is evaluated for the RBPS problem. The efficacy of the proposed solution approach is tested by comparing it with four well-known metaheuristic approaches, i.e., GA, SA, DE, and PSO. Each algorithm uses a different approach to determine optimal solutions. The details of implementations procedure for GA, SA, DE, and PSO to address the RBPS model are presented in Appendix.

5.1. Simulation setup

The experiment simulations are carried out on a machine with an Intel(R) Core (TM) i7 CPU 3.20 GHz CPU and 16 GB of RAM using MATLAB software R2016b. The GA, SA, DE, and PSO implementations in MATLAB can be found from the code.[1] However, some essential changes have been made to address the RBPS problem.

5.2. Datasets descriptions

For the experimental study of the proposed RBPS model using IWO based solution algorithm, we have considered the following datasets regarding the worldwide stocks concerned in various capital markets [9, 29,77].

i. DAX: The Deutscher Aktien Index (DAX), also known as the German Stock Index, monitors the progress of prominent German corporations listed on the Frankfurt Stock Exchange. It is among the most significant European stock market indices.
ii. FTSE: The Financial Times Stock Exchange (FTSE) index is a share index comprising the most significant market capitalization companies listed on the London Stock Exchange. It is extensively employed to monitor the performance of United Kingdom-listed blue-chip companies.
iii. Hang Seng: The performance of the major corporations in the Hong Kong stock market is primarily monitored by the Hang Seng Index. It is the preeminent indicator of the performance of the Hong Kong stock market.
iv. S&P: The Standard & Poor (S&P) Index is a stock market index that measures the shares of publicly traded large-cap companies in the United States of America. It is one of the most widely observed equity indices worldwide and a prominent indicator of the performance of the US stock market.
v. Nikkei: The Nikkei is a Tokyo Stock Exchange-traded price-weighted index of Japan's leading blue-chip corporations. Nikkei is a substantial metric for evaluating the performance of major Japanese corporations.
vi. BSE: The Bombay Stock Exchange (BSE) is Asia's oldest and premier stock market. Originally known as the Native Share & Stock Brokers Association, in 1875. The Sensex 30 was introduced in 1986 as the foundation for determining the thirty most prominent stocks across different sectors of India.
vii. NSE: The National Stock Exchange (NSE) introduced the NIFTY 50 Index and initiated trading and settlements of stocks in dematerialized form 1995–1996. The NIFTY index comprises of fifty most actively traded companies on the NSE stock exchange.

The DAX, FTSE, Hang Seng, S&P, and Nikkei datasets were taken from OR-Library [9,29,77] and were formed to receive the weekly price of stocks from March 1992 to September 1997. For BSE30 and NSE50,[2] the monthly returns were calculated and extracted from 1st April 2010–31 st March 2020. The details of all seven financial data sets used in the experimental study are summarized in Table 3. In Table 3, the mathematical symbol K represents the number of securities or assets associated with the datasets. For example, K= 30 in the BSE SENSEX is a free-float market-weighted stock market index of 30 well-established and financially sound companies listed on the Bombay Stock Exchange. Similarly, K= 50 in the NIFTY is a benchmark of the Indian stock market index representing the weighted average of 50 largest Indian companies listed on the National Stock Exchange.

5.3. Parameter sensitivity analysis

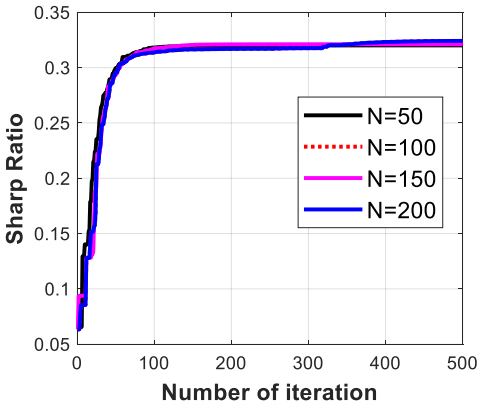
Parameter selection plays a crucial role in the algorithm's performance, as it can significantly impact the algorithm's exploration and exploitation capabilities, convergence behavior, and overall effectiveness. To obtain the most efficient utilization of the proposed IWO-based solution algorithm, we need to find a suitable set of values for the parameters that maximize the performance of IWO. The parameters are population size (N), the number of iterations (Itr), the minimum and maximum number of seeds (S_{Min} and S_{Max}), and standard deviation ($\sigma_{initial}$ and σ_{final}), which are used to calculate σ. The proposed RBPS model in Eq. (7) - (11) is addressed to find the maximum Sharpe ratio subject to the defined constraints and tested for the BSE 30. The best parameter values were found using the trial-and-error method for all the various combinations of parameters of the IWO algorithm. In other words, the parameter value N determines the number of initial weeds (solutions) in the population; it may be obtained from Fig. 3(a), which shows the convergence plots for the IWO algorithm for N = 50, 100, 150, and 200 for the RBPS model using BSE30 data at RB% = 12.5. From Fig. 3(a), it is evident that the immense N value slightly improves the objective Sharpe ratio value but increases the computational run time of the model. Therefore, the most suited value of N is 100. Similarly, maximum iterations (Itr) specify the number of iterations or generations the algorithm should run, which can be obtained from Fig. 3(c). In Fig. 3 (c), the optimal Sharpe ratio achieved by IWO at Itr = 50, 100, 150, 200, 250, 300, 400, and 500 are shown. Hence, selecting an Itr of more than 500 is unreasonable because it will exponentially increase the computational run time. To further demonstrate the robustness, we extend the parameter sensitivity analysis to the DAX data (widely studied in literature for portfolio optimization) and repeated as described for BSE. Fig. 3(b) and 3 (d) showcasing the sensitivity analysis on population size, and number of iterations parameters using DAX and confirms that the parameter settings remain effective across different datasets.

From Fig. 3 and the literature [9,22,29,42,77], the best value of S_{Max} is 10, S_{Min} is 1, and n is 3, enabling IWO to achieve better convergence. Moreover, N is 100, maximum iteration is 500, $\sigma_{initial}$ is 0.002, and σ_{final} is 0.00001. Fig. 3(e) presents the tuned parameter data showing a convergence evaluation of GA, SA, DE, and PSO with the IWO technique on the best Sharpe ratio. From Fig. 3, it is evident that the IWO solving method exceeds all other strategies regarding convergence capabilities. Table 4 lists the parametric characteristics of the stated and other actualized algorithms for a comprehensive study.

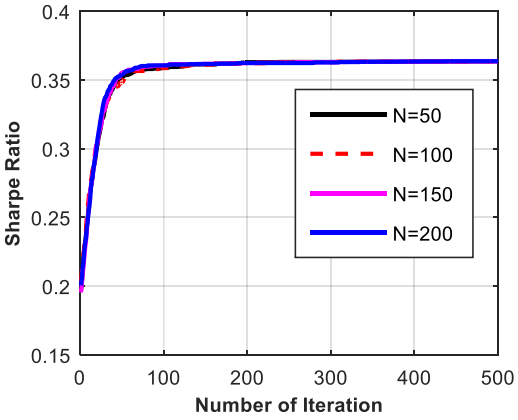
Table 3
Details of financial data sets used in the experimental study.

S. No.	Name of Dataset	Aberration	Country	*K*
1	German Stock Index	DAX	Germany	85
2	Financial Times Stock Exchange	FTSE	United Kingdom	89
3	Hong Kong stock market	Hang Seng	Hong Kong	31
4	Tokyo Stock Exchange	Nikkei	Japan	255
5	US stock market	S&P	USA	99
6	Bombay Stock Exchange	BSE	India	30
7	National Stock Exchange	NSE	India	50

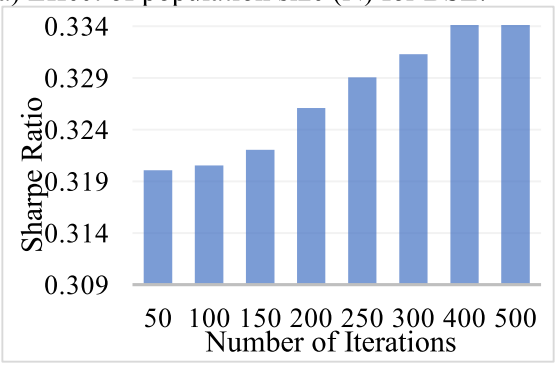
[1] https://yarpiz.com/
[2] http://people.brunel.ac.uk/mastjjb/jeb/orlib/portinfo.html.



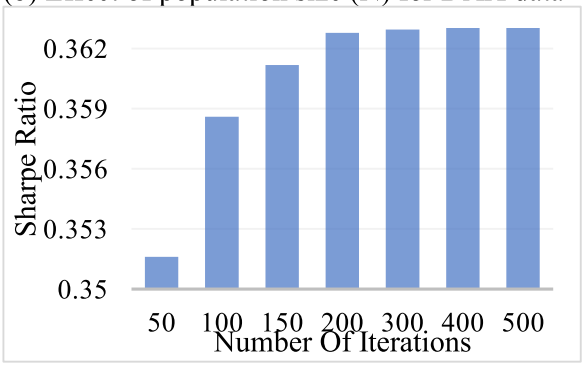
(a) Effect of population size (N) for BSE.



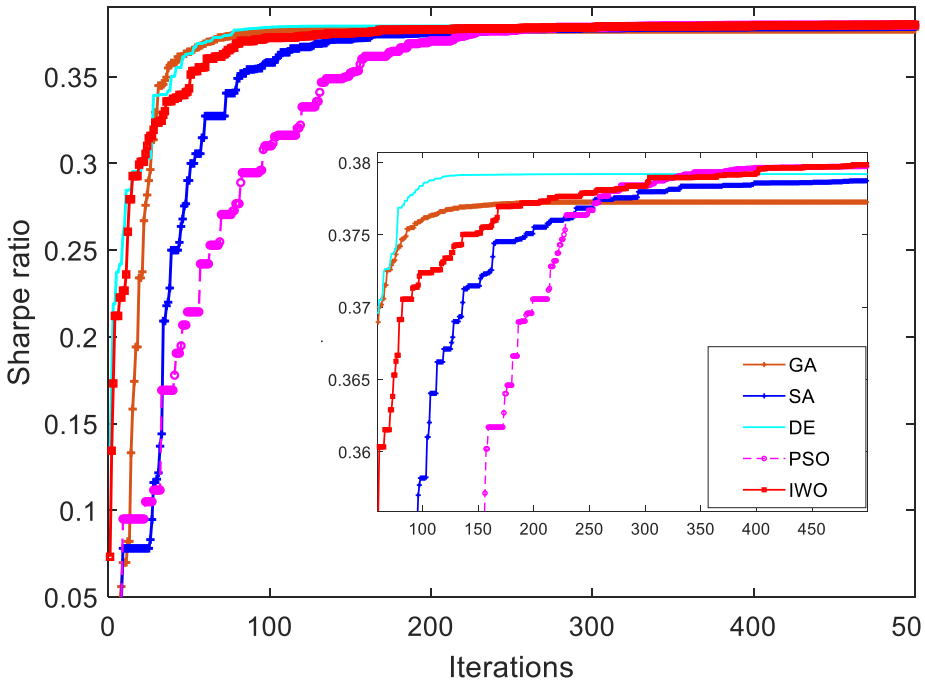
(b) Effect of population size (N) for DAX data



(c) Effect of number of iterations for BSE.



(d) Effect of number of iterations for DAX data



(e) Convergence plots of GA, SA, DE, PSO and IWO algorithms.

Fig. 3. Parameter sensitivity analysis.

5.4. Experimental outcomes

The extensive simulations of the GA, SA, DE, and PSO, as well as the proposed IWO-based algorithms are performed to evaluate the proposed RBPS problem as formulated in Eq. (7). The optimal asset weights achieving the maximum Sharpe ratio for BSE, and NSE along with the benchmark data of DAX, FTSE, Hang Seng, S&P, and Nikkei are obtained. The percentage of risk-budgeting (RB%) of the total portfolio risk ($Risk_P$), to evaluate the actual contribution to the risk (ACTR) of each security or asset, is arbitrarily considered to satisfy the risk budgeting

Table 4
Parameter specifications of GA, SA, DE, PSO and proposed IWO algorithms.

Algorithms	Parameter specification	Parameter value
IWO	Number of initial solutions	N = 100
	Number of iterations	Itr = 500
	Non-linear modulation index	n = 3
	Initial value of SD	$\sigma_{final} = 0.00001$
	Final value of SD	$\sigma_{initial} = 0.002$
	Max. seeds	$S_{Max} = 10$ $S_{Min} = 1$
	Min. seeds	
GA	Number of initial solutions	N = 100
	Number of iterations	Itr = 500
	Encoding type	Real-coded
	Simulated binary crossover	Probability = 0.7
	Polynomial mutation	Probability = 0.3
	Selection	Elitism
SA	Number of initial solutions	N = 20
	Number of iterations	Itr = 500
	Encoding type	Real-coded
	Number of neighbors	Nbh = 5
	Mutation	Probability = 0.45
	Selection	Elitism
PSO	Number of initial solutions	N = 100
	Number of iterations	Itr = 500
	Acceleration coefficients	$c_1 = c_2 = 1.5$
	Weight	uniform distribution $(0, 1)$
DE	Number of initial solutions	N = 100
	Number of iterations	Itr = 500
	Scaling factor	F = 0.5,
	Recombination probability	Probability = 0.8

constraint as presented in Eq. (8). The values of RB% levels are 12.5, 17.5, 22.5, 27.5, 35, 50, 75, and 100. Choosing the minimum risk of RB % value less than 12.5 is illogical because the $ACRT_i$ is evaluated from the marginal contribution to the risk ($MCTR_i$), which is the ratio of individual security risks ($Risk_i$) to the total portfolio risk ($Risk_P$). Therefore, considering a lower value of risk budgeting of each security starts violating the basics of portfolio selection, which is, achieving the highest return at a certain level of risk. And hence obtaining the lowest return value with the lowest risk in capital market investment doesn't make sense. Further, to overcome the problem of randomly generated initial solutions to the optimization algorithms, we have conducted 20 independent runs of all algorithms and obtained the Max, Min, Mean, Median, and SD of the performance measures. The detailed results and comparisons of each data set corresponding to all the mentioned algorithms are discussed in the following subsections.

5.4.1. Results of BSE

In BSE data sets, the monthly returns of the top 30 companies listed in the Bombay Stock Exchange have been considered for the financial year from April 2010 to March 2020. The dimension of the decision variable or weight vector in the RBPS model is 30, i.e., k = 30. The objectives of Sharpe ratio, return, and risk of the portfolio have been rigorously evaluated by solving the GA, SA, DE, PSO, and IWO algorithms for the different values of RB% are 12.5, 17.5, 22.5, 27.5, 35, 50, 75, and 100. These levels represent the percentage of the total risk budget allocated to each security in the RBPS model. The results of the max, min, mean, median, and SD of the Sharpe ratio, risk, and return of the proposed RBPS model for comparing GA, SA, DE, PSO, and IWO algorithms using the BSE dataset are presented in Table B1 of the Appendix.

The maximum Sharpe ratio obtained during the 20 runs using GA, SA, DE, PSO and IWO algorithms are compared in Fig. 4 at RB% = 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100 of RBPS models. From Fig. 4, we observe that when the absolute contribution risk of an individual asset at an RB level is 12.5 % of the total portfolio risk, the proposed IWO approach efficiently outperforms the other algorithms. As the values of the RB levels increase from 17.5 to 35.0, the IWO, PSO and DE start converging to the best Sharpe ratios of the RBPS models. Yet, at RB

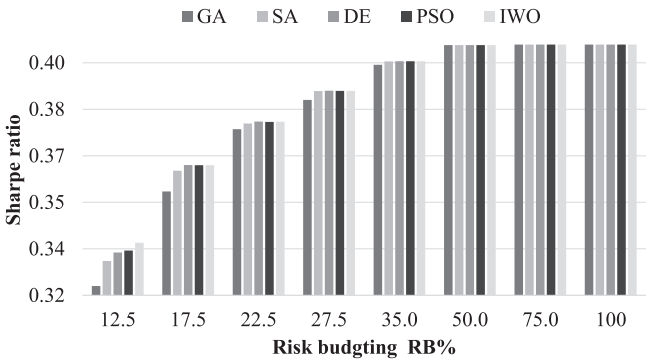


Fig. 4. Comparison of the best Sharpe ratio of BSE data by GA, SA, DE, PSO and IWO algorithms at RB% = 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100 of RBPS model.

of 50, 75 and 100, the RBPS model presents no difference in the optimal Sharpe ratios, hence, validates the importance of risk budgeting constraints. Moreover, it is evident from Fig. 4 that RB significantly impacts the Sharpe ratio and increases as the RB increases, thereby underlining the crucial role of risk budget levels in financial modelling. The effectiveness of the proposed IWO approach may be elaborated by comparing the % improvement in the maximum Sharpe ratio obtained by it over each of the other algorithms. After computing the mean % improvement across all RB levels, IWO had the most excellent enhancement over GA at 1.05 %. IWO has a 0.31 % increase when compared to SA and compared to DE is 0.12 %. IWO exhibits an improvement of 0.10 % compared to PSO, consistent with the observed trend. The results show, overall, that in terms of performance, IWO outperforms GA, SA, DE and PSO.

To provide a more comprehensive analysis of the results, we examined the algorithm behaviour from the perspectives of search space characteristics, exploration and exploitation, and statistical significance. We present Fig. 5, Fig. 6, and Fig. 7, which are box plots representing the comparisons of the Sharpe ratio, risk, and return, respectively, achieved by different algorithms (GA, SA, DE, PSO, and proposed IWO) at various RB% levels. The essential observations are drawn from the results of BSE as in Table B1, Fig. 5, Fig. 6, and Fig. 7, as discussed below.

- At a lower percentage of RB levels, e.g., RB% = 12.5 and 17.5, the search space of the RBPS model is highly constrained, with fewer feasible weights of securities that satisfy the risk budgeting constraints. In such scenarios, the ability of optimization algorithms to explore the search space effectively and identify promising regions becomes crucial. From Fig. 5(i) and Fig. 5(ii), we observe the max, min, median, and mean values of the Sharpe ratios are the highest for the proposed IWO in comparison to the GA, SA, DE, and PSO. A similar pattern can also be seen for the IWO from Fig. 6(i)-(ii) and Fig. 7(i)-(ii) for the comparison of return and risk of the RBPS at the RB% are 12.5 and 17.5, respectively. Hence, the proposed IWO-based solution procedure demonstrates superior exploration capabilities compared to others at these RB% levels, potentially due to its adaptive dispersion and reproduction processes. These processes refer to the algorithm's ability to dynamically adjust the dispersion of solutions and the reproduction rate of promising solutions, which allows it to explore the search space effectively and converge to optimal solutions. The lower SDs observed for IWO at RB% = 12.5 and 17.5 further suggest that its search strategies result in more consistent and reliable performance across multiple runs.
- As the risk budgeting level increases, that is, RB% = 22.5, RB% = 27.5, and RB% = 35.0, the search space becomes comparatively less restricted, and the objective function (Sharpe ratio) increases concerning increment in RB levels. In these scenarios, the algorithms' ability to exploit the expanded search space and converge toward near-optimal solutions becomes more critical. From Fig. 5 (iii), Fig. 5

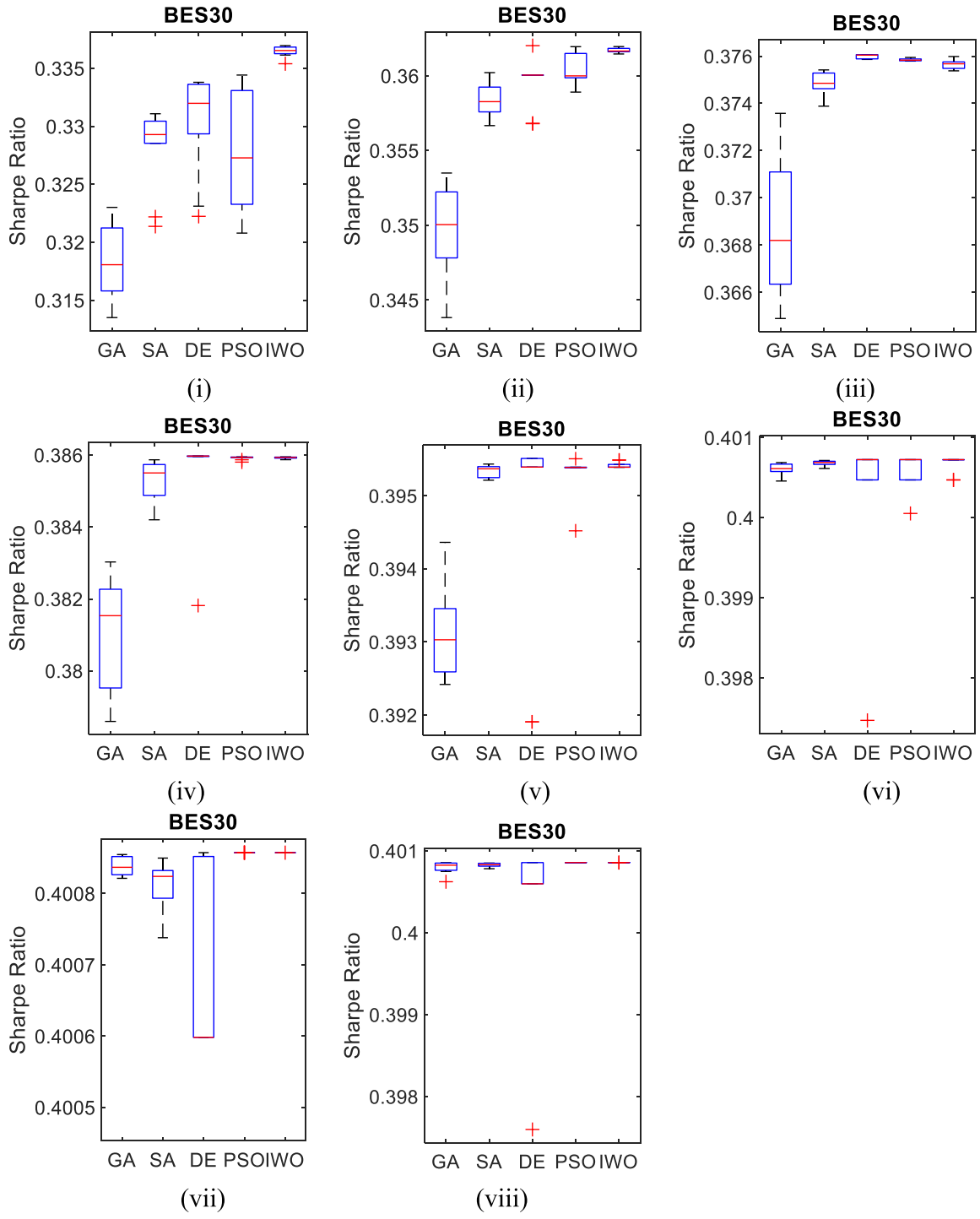


Fig. 5. Comparative results of BSE for the Sharpe ratios of the RBPS models obtained by GA, SA, DE, PSO and IWO for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35.0, (vi) RB% = 50.0, (vii) RB% = 75.0, and (viii) RB% = 100.

(iv), and Fig. 5(v), it is evident that the performance metrics values of max, median, and mean are highest for the IWO in comparison with others. Further, the box plots of the return and risk shown in Fig. 6 (iii)-(v) and Fig. 7(iii)-(v), the best optimal weights obtained from the IWO are better compared with the GA, SA, DE, and PSO. While IWO still exhibits an advantage regarding maximum and median Sharpe ratio values, the performance differences among the algorithms become smaller, indicating that they converge towards similar portfolios.

- At higher risk budgeting levels, e.g., RB% = 50.0, RB% = 75.0, and RB% = 100.0, the search space of the RBPS models is highly unrestricted, and all the algorithms appear to have converged to the optimal portfolio. It is evidenced by the virtually identical performance values of maximum, average, and medium Sharpe ratios across all algorithms; see Fig. 5(vi), Fig. 5(vii), and Fig. 5(viii). A similar pattern in the return and risk of the portfolios are expected in such unrestricted scenarios from Fig. 6(vi)-(viii) and Fig. 7(vi)-(viii) at higher RB% levels (50.0, 75.0, and 100.0). Therefore, all the

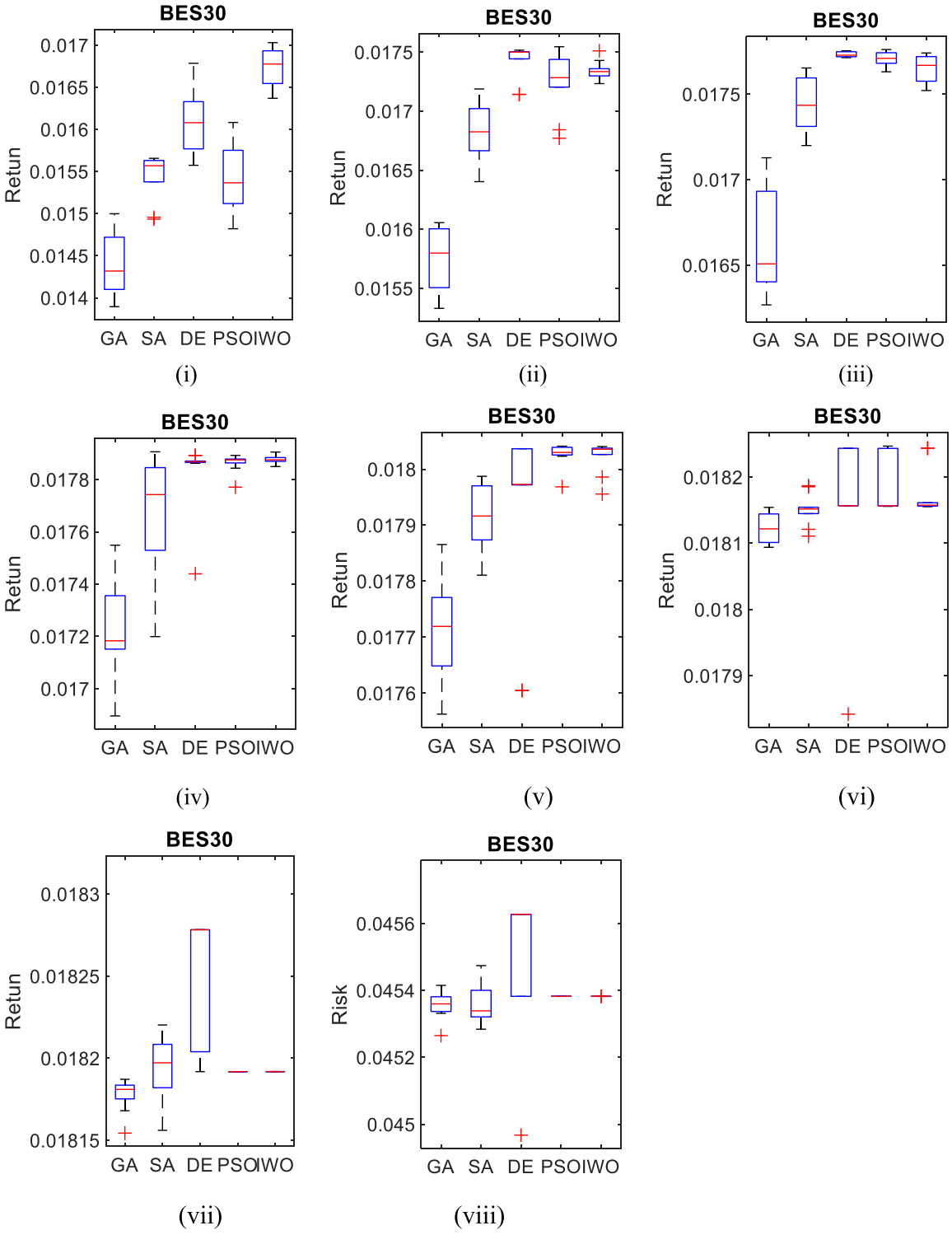


Fig. 6. Comparative results of BSE for the Returns of the RBPS models obtained by GA, SA, DE, PSO and IWO for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35.0, (vi) RB% = 50.0, (vii) RB% = 75.0, and (viii) RB% = 100.

algorithms are likely to find the global optimum weights of securities under risk budgeting, regardless of their specific search mechanisms.

Finally, all algorithms demonstrate an increasing trend in Sharpe ratios as the RB% level increases. This indicates that taking on higher risks may lead to better risk-adjusted returns, a standard expectation in investment strategies. Hence, the RBPS model obtains a high portfolio return as more and more risk levels are associated. However, after attaining the highest portfolio return, the further increment of the risk budgeting is miserable. Therefore, the results follow the general phenomenon of selecting the portfolio and the trade-off between risk and return. \

5.4.2. Results of NSE

The monthly returns of the top 50 enterprises listed on the National Stock Exchange (NSE) of India are considered in the NSE data sets over

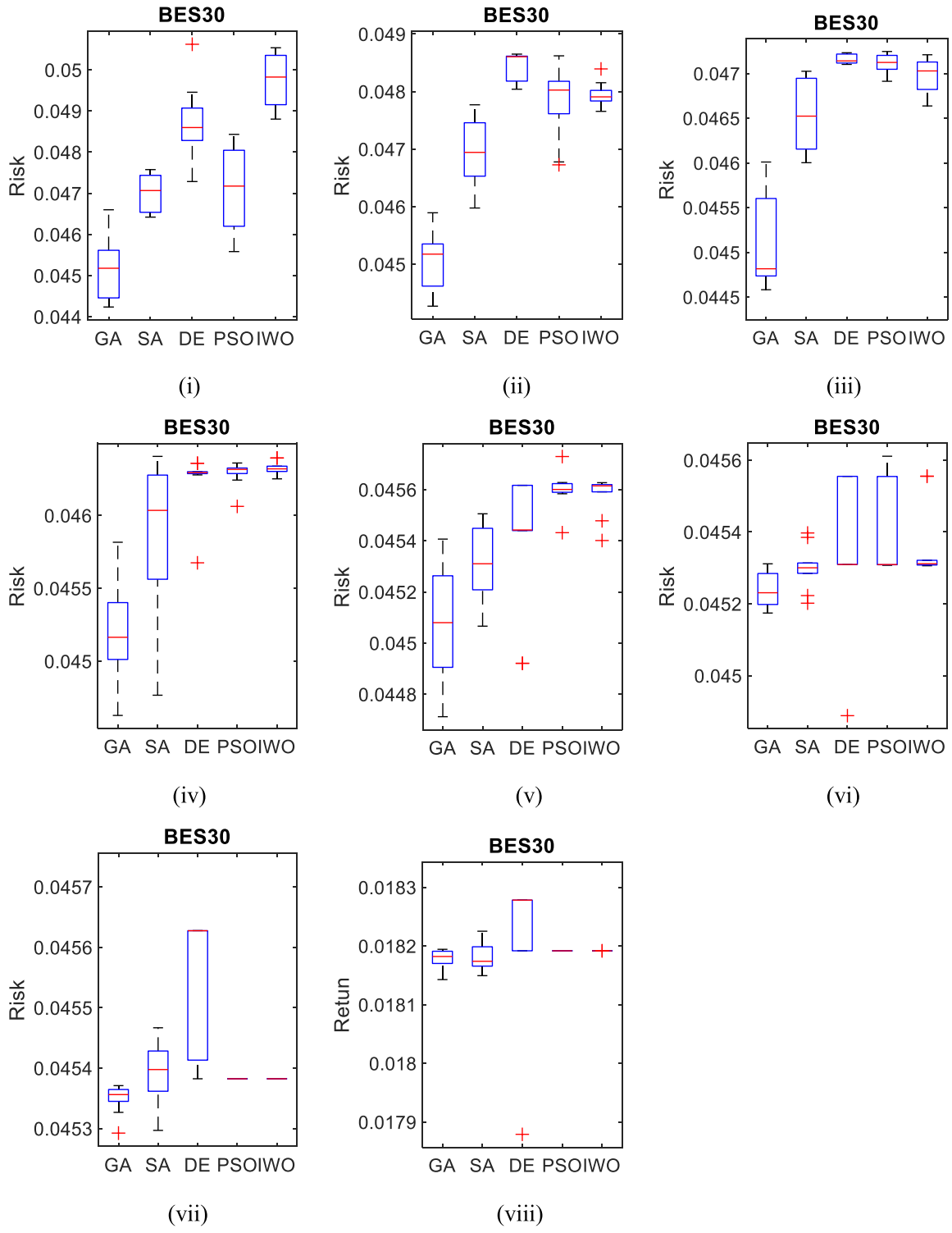


Fig. 7. Comparative results of BSE for the Risk of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

the calendar year from April 2010 to March 2020. The proposed RBPS model in Eq. (7) now has decision weights with a dimension of 50, k = 50. The portfolio's Sharpe ratio, return, and risk goals were thoroughly analyzed using GA, SA, DE, PSO, and IWO algorithms for RB% values of 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100.0. These findings in Table B2 of the Appendix provide valuable insights into the max, min, mean, median, and SD of the RBPS model's Sharpe ratio, risk, and return for all algorithms using the NSE dataset.

Fig. 8 compares the highest Sharpe ratio values from 20 different runs for GA, SA, DE, PSO, and IWO procedures with RB% = 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100 of RBPS models. At RB% = 12.5 in the RBPS model of the NSE data, Fig. 8 shows that the proposed IWO procedure efficiently surpasses the other algorithms by the highest Sharpe ratio, a high portfolio returns even at a lower risk level. However, as the values of the RB% are 17.5, 22.5, and 27.5, the IWO, PSO, SA and DE establish convergence. In addition, for RB% values of 35, 50, 75, and

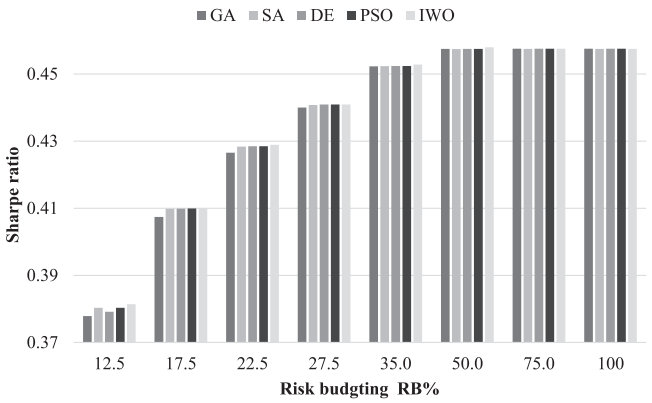


Fig. 8. Comparison of the best Sharpe ratio of NSE data by GA, SA, DE, PSO and IWO algorithms at RB% = 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100 of RBPS model.

100, the RBPS model exhibits no change in optimal Sharpe ratios. As a result, Fig. 8, like BSE Fig. 4, demonstrates the significance of risk budgeting constraints, which enable increasing flexibility in the absolute contribution risk of securities to achieve a good portfolio return regardless of whether the investment entails more significant risk. Moreover, it is evident from Fig. 8 that increasing the RB levels will directly improve the Sharpe ratio in the portfolio selection, highlighting the significance of risk budgeting levels. The importance of the IWO was elaborated by comparing the improvement in the obtained Sharpe ratio over each of the other algorithms. From Fig. 9, we can compute the average gains in the overall RB% level. The IWO provides the most elevated improvement over the GA, with an improvement of 0.24 %. When compared to SA, IWO achieves an improvement of 0.17 %. Compared with DE and PSO, IWO's improvement is 0.04 % and 0.01 %, respectively. Hence, the findings suggest that IWO has a clear performance advantage over GA, SA, DE, and PSO.

Similar to BSE results, we present box plots of the statistical metrics (maximum, minimum, median, mean, and standard deviation) of the Sharpe ratio, risk, and return achieved by different algorithms at various RB% levels in Figs. 9, 10, and 11. The following discussions present the key findings from the NSE dataset, which are detailed in Table B2 of the Appendix, Fig. 9, Fig. 10, and Fig. 11.

- The RBPS model is highly restricted at lower RB% levels, such as 12.5 and 17.5, with fewer securities that meet the risk budgeting limitations. Figs. 9(i) and 9(ii) demonstrate that the IWO has excellent Sharpe ratios compared to GA, SA, DE, and PSO. Also, Figs. 11(i)-(ii) and 12(i)-(ii) show a similar trend for the IWO, comparing the RBPS model's return and risk at RB% = 12.5 and 17.5, respectively. The IWO-based solutions have better performance metrics at these RB% levels, following a similar trend to the results obtained on BSE data.
- As RB% levels rise (RB% = 22.5, 27.5, and 35.0), search space becomes less confined, and the objective function (Sharpe ratio) increases. The flexibility of constraints makes algorithms' capacity to utilize the more prominent search area and converge to near-optimal solutions. Fig. 9(iii), 9(iv), and 9(v) show that the IWO has the best max, median, and mean performance measures. The return and risk presented in Fig. 10(iii), 10(iv), and 10(v) and 11(iii), 11(iv), and 11 (v) for these RB% reveal that the IWO is best compared to the GA, SA, DE, and PSO.
- RBPS models have a much-unbounded search space at higher risk budgeting levels, such as RB% = 50.0, 75.0, and 100.0, and all algorithms have converged to the optimum portfolios. Fig. 9(vi), 9 (vii), and 9(viii) show that all algorithms have almost equal maximum, average, and medium Sharpe ratios. A similar trend in portfolio return and risk is projected in unconstrained scenarios from Fig. 10(vi)-(viii) and Fig. 11(vi)-(viii) at higher RB% levels. Hence, all algorithms likely discover the global optimal security weights under budgeting, independent of their search strategies.

Hence, our findings confirm the universal principle of portfolio selection and risk-return trade-off for NSE data. All algorithms demonstrate an increasing Sharpe Ratio with increasing RB %, indicating that higher risk can lead to higher risk-adjusted returns.

5.4.3. Results of DAX

The proposed IWO-based solving algorithm for the RBPS model was tested on the DAX, a renowned European stock market index with 80 assets and a decision vector dimension of k = 80. The portfolio's Sharpe ratio, return, and risk goals were thoroughly analyzed using GA, SA, DE, PSO, and IWO algorithms for RB% values of 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100.0. The findings presented in Table B3 of the Appendix provide valuable insights into the max, min, mean, median, and SD of the RBPS model's Sharpe ratio, risk, and return for all algorithms using the DAX dataset. The essential observations drawn from the results of DAX, as in Table B3, Fig. 12, Fig. 13, and Fig. 14, are discussed below.

The RBPS model search space is significantly tightened at 12.5 % and 17.5 % RB levels, with few weight combinations of securities that fulfil risk budgeting limitations. Figs. 12(i) and 12(ii) show that the proposed IWO algorithm outperforms the GA, SA, DE and PSO algorithms in Sharpe ratios, maximum, median, and mean values. In Figs. 13(i)-(ii) and 14(i)-(ii), IWO surpasses the other algorithms in RBPS return and risk at these RB% levels. Moreover, IWO's reduced SDs at RB% = 12.5 and 17.5 show that its search algorithms provide more consistent and dependable running performance.

As RB% increases to 22.5, 27.5, and 35.0, search space becomes less confined, and the fitness function (Sharpe ratio) increases with RB% levels. These situations allow optimization algorithms to utilize the more prominent search area and converge to near-optimal solutions. Fig. 12(iii), 12(iv), and 12(v) show that the IWO method has the most excellent maximum, median, and mean performance measures compared to the benchmark algorithms. The box plots of return and risk in Fig. 13(iii)-(v) and 14(iii)-(v) for these RB% levels indicate that IWO finds the most optimum solutions compared to GA, SA, DE, and PSO. However, the IWO still has the edge in maximum and median Sharpe ratio values, but the algorithms' performance disparities shrink, showing they converge towards identical portfolios.

At RB% levels of 50.0, 75.0, and 100.0, the RBPS models' search area becomes unconstrained, and all algorithms have converged to the optimum portfolio. Fig. 12(vi), 12(vii), and 12(viii) indicate that all algorithms have almost similar maximum, average, and median Sharpe ratios. In risk-free situations, the portfolios' return and risk are presented in Fig. 13(vi)-(viii) and 14(vi)-(viii) for the higher RB% values. All algorithms are expected to discover the global optimal security weights under RB limitations, independent of their search techniques. Therefore, all algorithms show an increasing Sharpe ratio as RB% increases, supporting the investment strategy idea that greater risk levels may provide superior risk-adjusted returns.

5.4.4. Results of FTSE

The proposed IWO algorithm for the RBPS model was tested on the FTSE Index, a prominent stock market index with 89 securities and a decision vector dimension of k = 89. The portfolio's Sharpe ratio, return, and risk objectives were extensively examined utilizing GA, SA, DE, PSO, and IWO methods with RB% values ranging from 12.5 to 100.0. The results in Table B4 of the Appendix provide significant insights into the maximum, minimum, mean, median, and standard deviation of the Sharpe ratio, risk, and return of the RBPS model for all algorithms applied to the FTSE dataset. The key findings derived from the results reported in Table B4, as shown in Fig. 15, Fig. 16, and Fig. 17, are as follows.

At 12.5 and 17.5 of the RB% level, the RBPS model's search space is very confined, with few weight combinations of securities that meet the

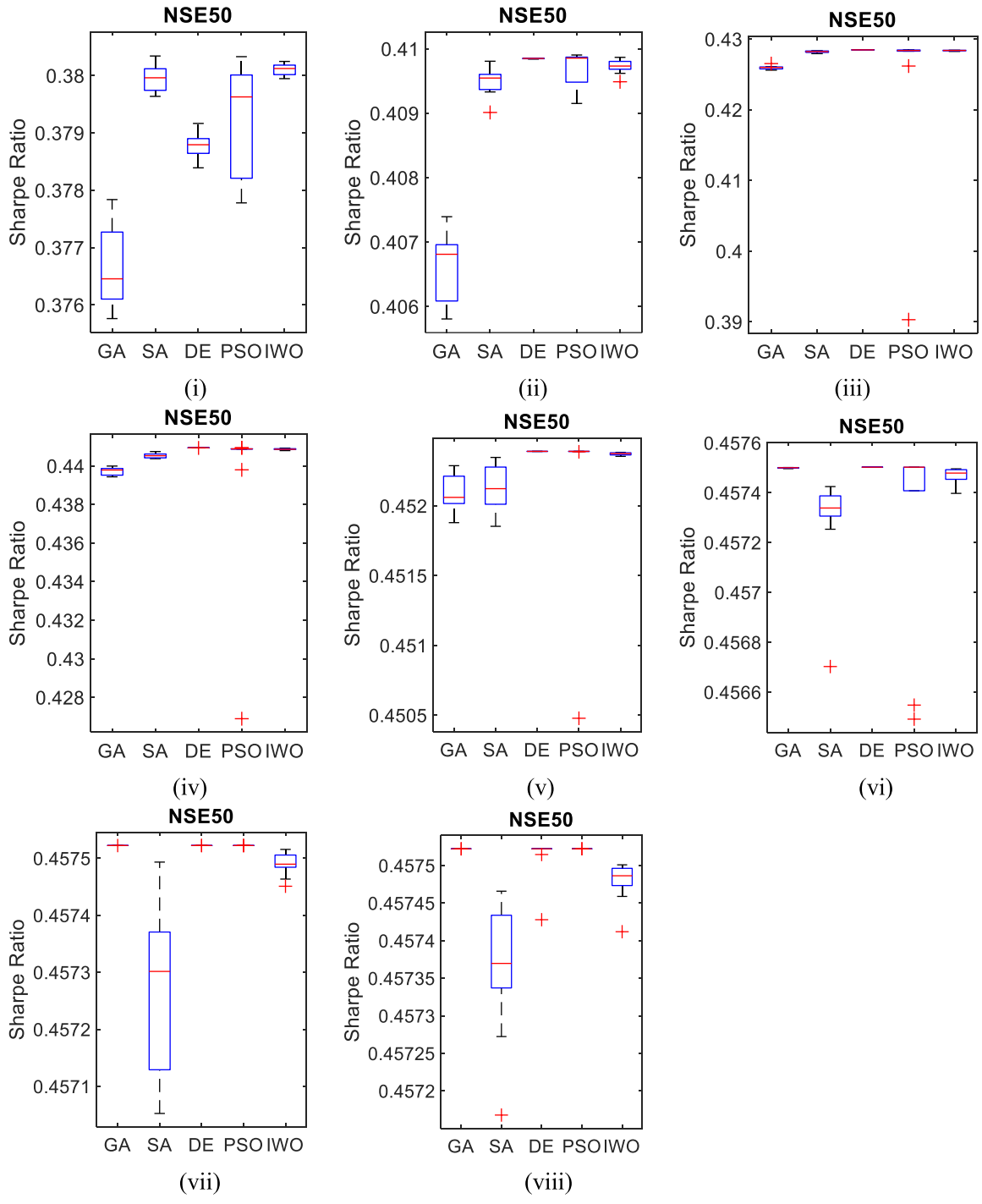


Fig. 9. Comparative results of NSE for the Sharpe ratio of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

RB criteria. Optimization algorithms' exploration skills to travel the search space and find interesting areas are vital in such cases. Figs. 15(i) and 15(ii) show that the proposed IWO algorithm outperforms the benchmark algorithms in Sharpe ratios, maximum, and mean. Figs. 16 (i)-(ii) and 17(i)-(ii), it is evident that IWO outperforms the other algorithms in RBPS return and risk at the RB% levels. IWO's adaptive dispersion and reproduction processes allow dynamic adjustment of solution dispersion and the reproduction rate of promising solutions, enabling compelling search space exploration and convergence to optimal solutions. IWO's lower SDs at RB% = 12.5 and 17.5 show that its search algorithms provide more consistent performance across runs.

As RB% goes to 22.5, 27.5, and 35.0, search space becomes less confined, and the objective function (Sharpe ratio) increases with RB% level. Such relaxation in optimization algorithms allows us to utilize the more prominent search area and converge to near-optimal solutions. Fig. 15(iii), 15(iv), and 15(v) show that the IWO method has the most excellent maximum and mean performance metrics compared to the benchmark algorithms. The box plots of return and risk in Fig. 16(iii)-(v) and 17(iii)-(v) indicate that IWO finds the most optimum solutions compared to GA, SA, DE, and PSO. IWO has the edge in maximum and median Sharpe ratio values, but the algorithms converge towards identical portfolios.

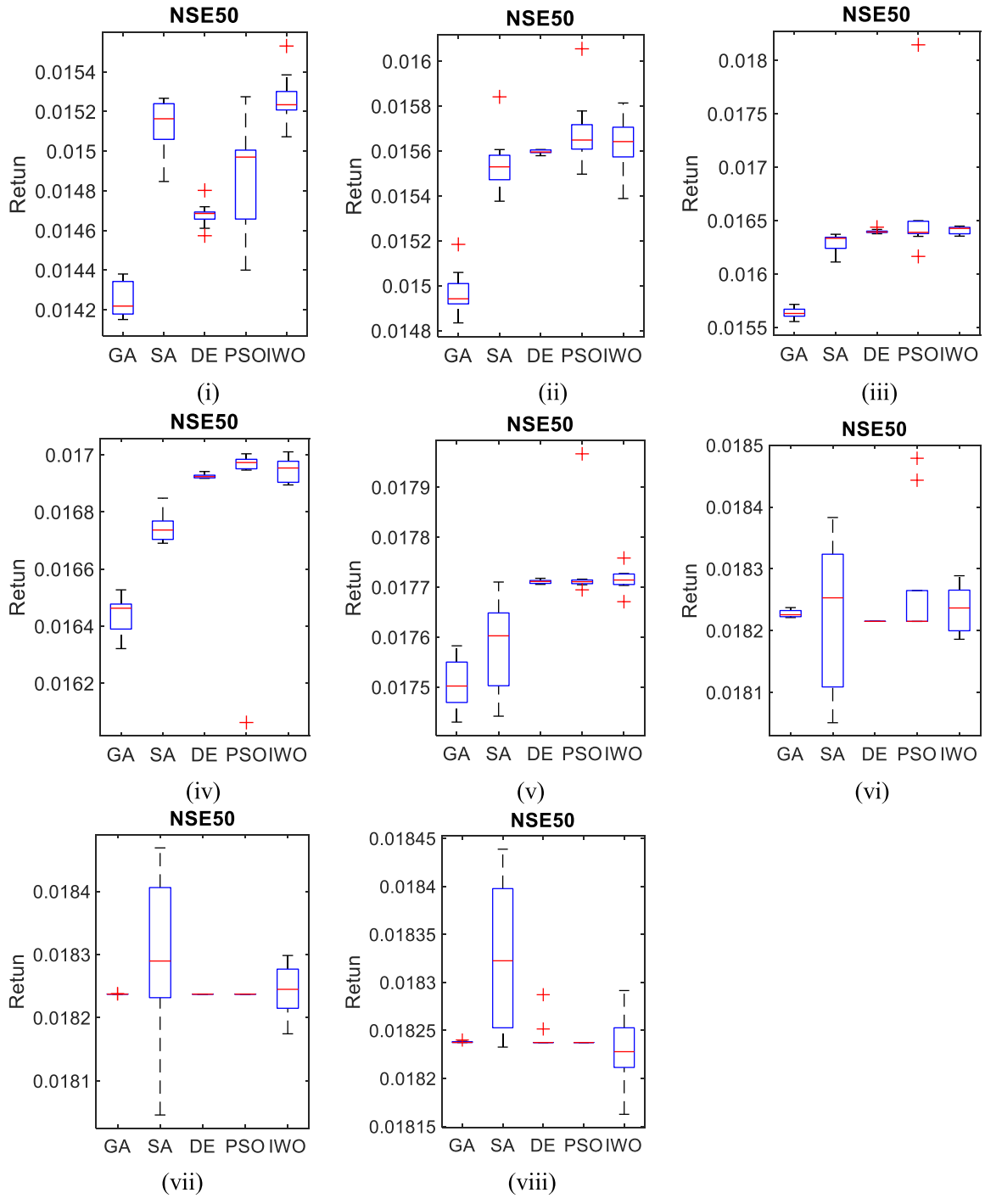


Fig. 10. Comparative results of NSE for the Return of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

At RB% levels of 50.0, 75.0, and 100.0, the RBPS models' search area becomes unrestricted, and all algorithms have obtained the optimum portfolio. Fig. 15(vi), 15(vii), and 15(viii) indicate that all algorithms have almost equal maximum, average, and median Sharpe ratios. The portfolios' return and risk are shown in Fig. 16(vi)-(viii) and 17(vi)-(viii) for the higher RB% values. As expected in investing strategies, all algorithms show an upward trend in Sharpe ratios as RB% rises, suggesting that greater risk levels may provide superior risk-adjusted returns.

5.4.5. *Results of HangSeng*

HangSeng data of the Hong Kong Index monitors significant companies with 31 assets; the HangSeng dataset was used to evaluate the proposed IWO method and benchmark algorithms for the RBPS model having k = 31 decision vector dimensions. The GA, SA, DE, PSO, and IWO algorithms calculated the portfolio's Sharpe ratio, return, and risk for RB% of 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100.0. Table B5 of the Appendix provides the max, min, mean, median, and SD of the RBPS model's Sharpe ratio, risk, and return for the HangSeng dataset, providing a comprehensive overview of the algorithm's performance.

The RBPS model's search area is restricted by the number of security

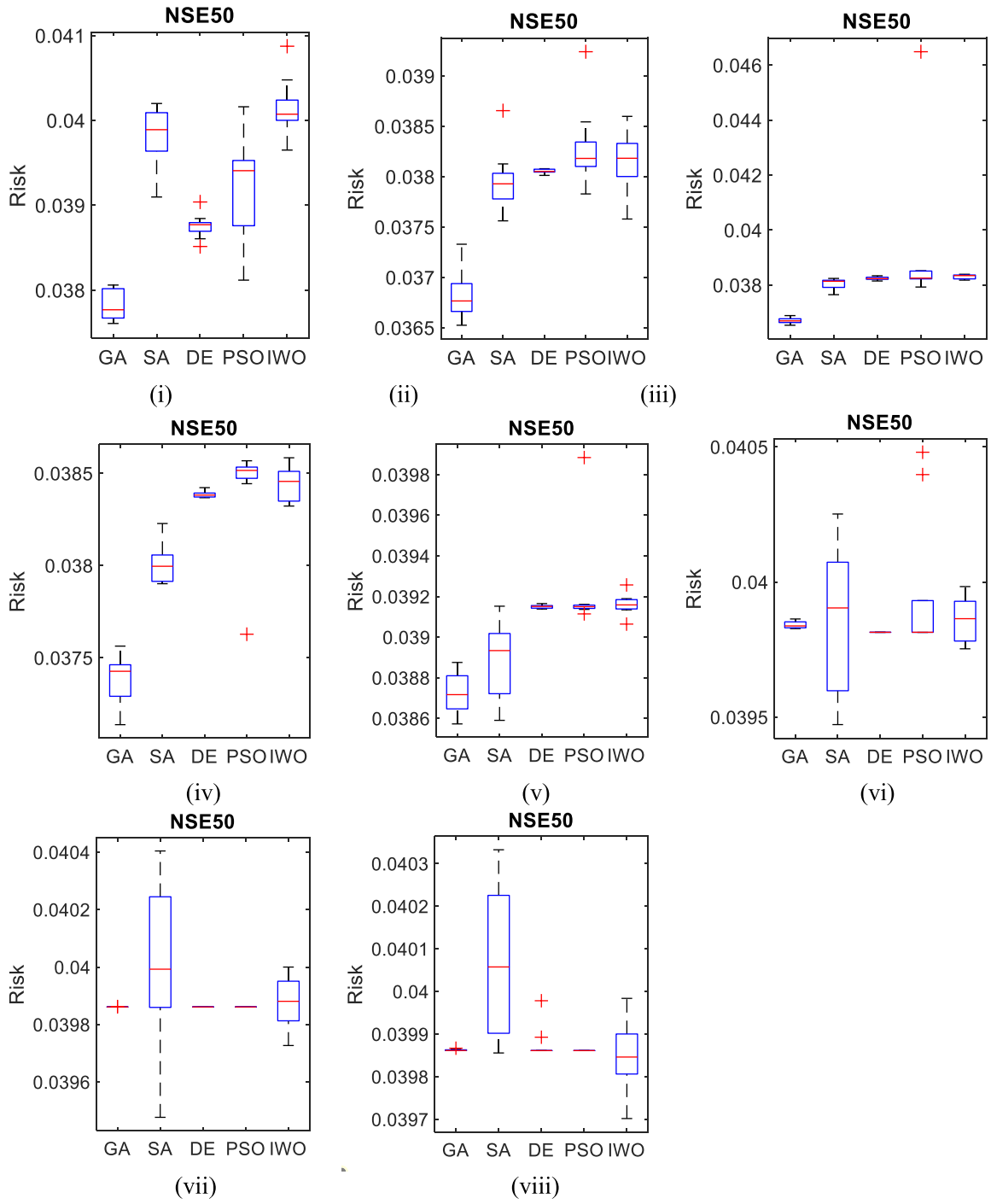


Fig. 11. Comparative results of NSE for the Risk of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

weight combinations that meet RB% constraints at 12.5 and 17.5. In such circumstances, optimization algorithms' search space exploration abilities are vital in finding the best optimal solutions. Figs. 18(i) and 18 (ii) indicate that the proposed IWO algorithm outperforms benchmark algorithms in comparing the maximum and mean of the Sharpe ratios. Figs. 19(i)-(ii) and 20(i)-(ii) show that the IWO algorithm outperforms the other algorithms in RBPS return and risk at these RB% levels. IWO's adaptive dispersion and reproduction procedures dynamically alter solution dispersion and promising solution reproduction rates, allowing search space exploration and convergence to optimum solutions.

As the RB% level rises to 22.5, 27.5, and 35.0, search space opens up, and the Sharpe ratio rises. These conditions need optimization methods to use the broader search area and converge to near-optimal solutions. IWO has the highest Sharpe ratio with maximum and mean performance measures compared to benchmark algorithms, as shown in Fig. 18(iii), 18(iv), and 18(v). Further, Fig. 19(iii)-(v) and 20(iii)-(v) show that IWO provides the best risk and return of the RBPS model for these RB% values compared to GA, SA, DE, and PSO.

At RB% levels of 50.0, 75.0, and 100.0, the search area for RBPS models becomes unbounded, and all algorithms appear to have found the optimal portfolio. Fig. 18(vi), 18(vii), and 18(viii) show that all algorithms have equal maximum, average, and median Sharpe ratios. The

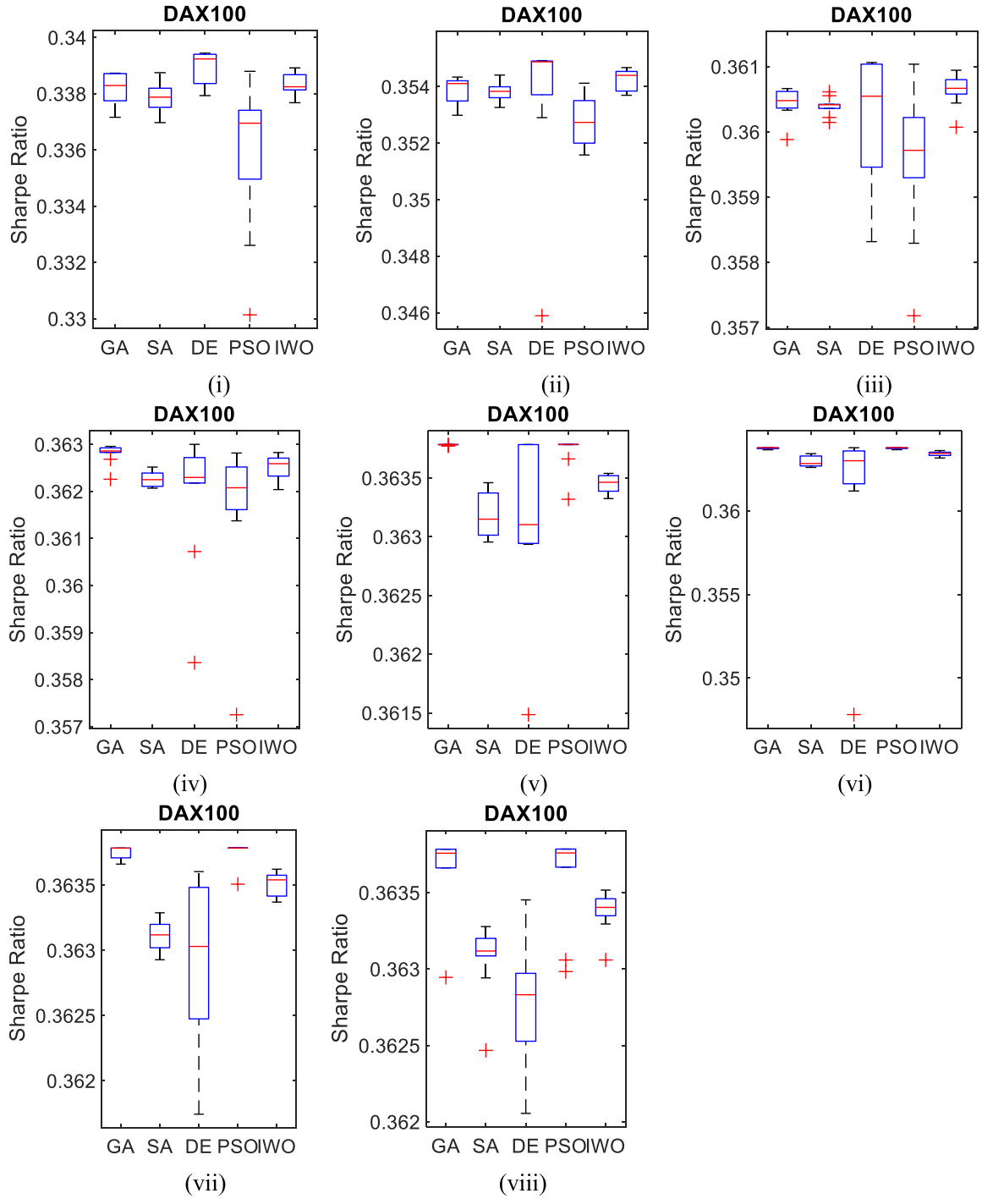


Fig. 12. Comparative results of DAX for the Sharpe ratio of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

portfolios' return and risk are shown in Fig. 19(vi)-(viii) and 20(vi)-(viii). Regardless of the search method, all algorithms find the global optimum security weights within the allowed risk budget. As predicted in investment techniques, all algorithms exhibit an increasing trend in Sharpe Ratios as RB% increases. Higher risk levels may provide better risk-adjusted returns.

5.4.6. Results of Nikkei

The Nikkei data of the Tokyo Stock Exchange, a benchmark dataset of Japanese companies, was utilized in our evaluation of the RBPS model, with a decision vector dimension of k = 225, reflecting the 225 assets. The Nikkei index was used to compare the performance of the proposed IWO algorithm with benchmark algorithms such as GA, SA, DE, and PSO. The RBPS model is a widely used framework in financial modelling aiming to allocate a portfolio's risk budget optimally. For RB % levels of 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100.0, which represent different levels of risk budgeting, the Sharpe ratio, return, and risk of the portfolio were determined using the GA, SA, DE, PSO, and IWO algorithms. The results shown in Table B6 of the Appendix help understand the max, min, mean, median, and SD of the RBPS model's Sharpe ratio, risk, and return for all methods using the Nikkei dataset.

Due to a limited number of potential weight combinations of

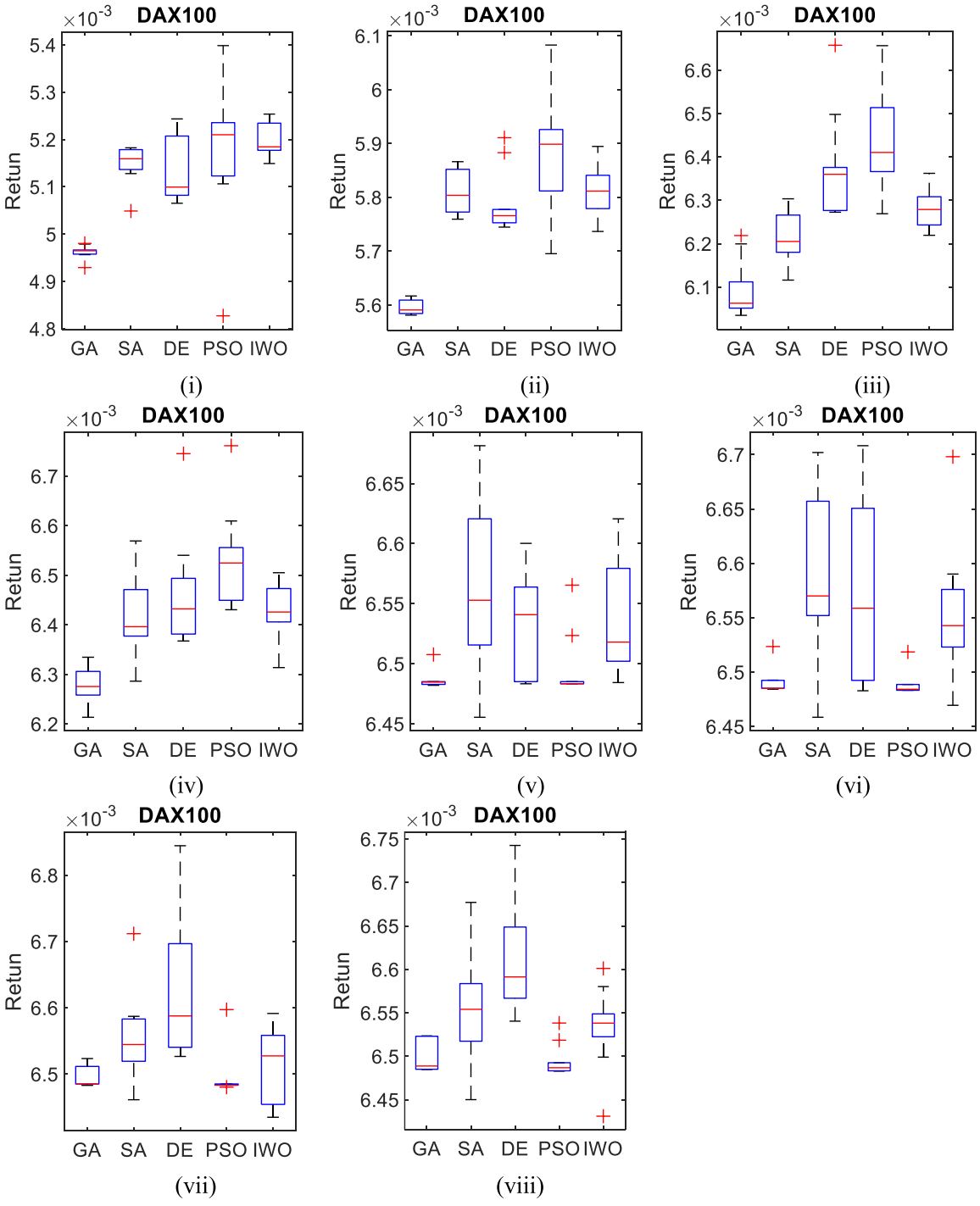


Fig. 13. Comparative results of DAX for the Return of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

securities that fulfil RB% at 12.5 % and 17.5 %, the RBPS model's search space is unrestricted. Optimization algorithms' exploration skills to travel the search space and find interesting areas are crucial in such cases. Figs. 21(i) and 21(ii) show that the proposed IWO algorithm outperforms the benchmark algorithms in Sharpe Ratios, maximum, and mean. From Figs. 22(i)-(ii) and 23(i)-(ii), it is evident that IWO surpasses the other algorithms in RBPS return and risk at the RB% levels. IWO's adaptive dispersion and reproduction processes allow dynamic adjustment of solution dispersion and the reproduction rate of promising solutions, enabling compelling search space exploration and convergence to optimal solutions. IWO's SDs at RB% = 12.5 and 17.5 show that its search algorithms provide more consistent and dependable performance across runs.

As the RB% level increases to 22.5, 27.5, and 35.0, search space becomes less confined, and the Sharpe ratio increases with RB% levels. Fig. 21(iii), 21(iv), and 21(v) show that the IWO method has the most significant maximum and mean performance measures for portfolio Sharpe ratio in comparison to the benchmark algorithms. The box plots of return and risk in Fig. 22(iii)-(v) and 23(iii)-(v) for these RB% levels indicate that IWO finds the most optimum solutions compared to GA, SA, DE, and PSO.

At RB% levels of 50.0, 75.0, and 100.0, the RBPS models' search area

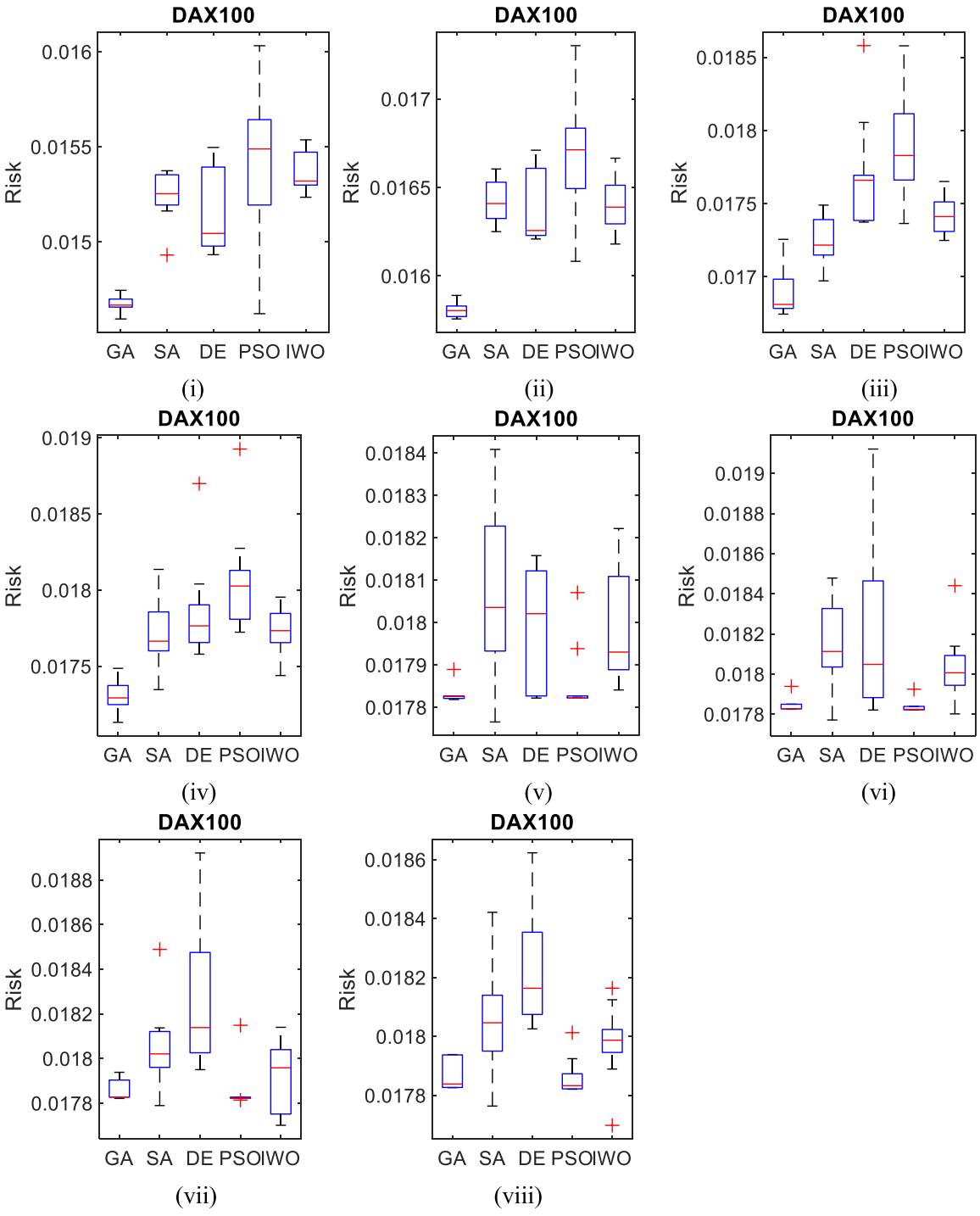


Fig. 14. Comparative results of DAX for the Risk of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB % = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

becomes open, and all algorithms have converged to the optimum portfolio. Fig. 21(vi), 21(vii), and 21(viii) indicate that all algorithms have almost similar maximum, average, and median Sharpe ratios. Moreover, the portfolios' return and risk are compared in Fig. 22(vi)-(viii) and 23(vi)-(viii) for the higher RB% values.

5.4.7. *Results of S&P data*

The Standard & Poor's (S&P) Index is the large cap of US corporations and has 98 assets, giving it a decision vector dimension of k = 98. To evaluate the proposed IWO algorithm and benchmark algorithms, GA, SA, DE, and PSO, for the RBPS model, S&P data was used. For RB% levels of 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100.0, which represent different levels of risk budgeting, the Sharpe ratio, return, and risk of the portfolio were determined using the GA, SA, DE, PSO, and IWO algorithms. The results shown in Table B7 of the Appendix are beneficial for understanding the max, min, mean, median, and SD of the RBPS model's Sharpe ratio, Rrisk, and return for all methods that use the S&P dataset.

RBPS model search space is significantly constricted at 12.5 and 17.5 RB% levels owing to a limited number of weight combinations of

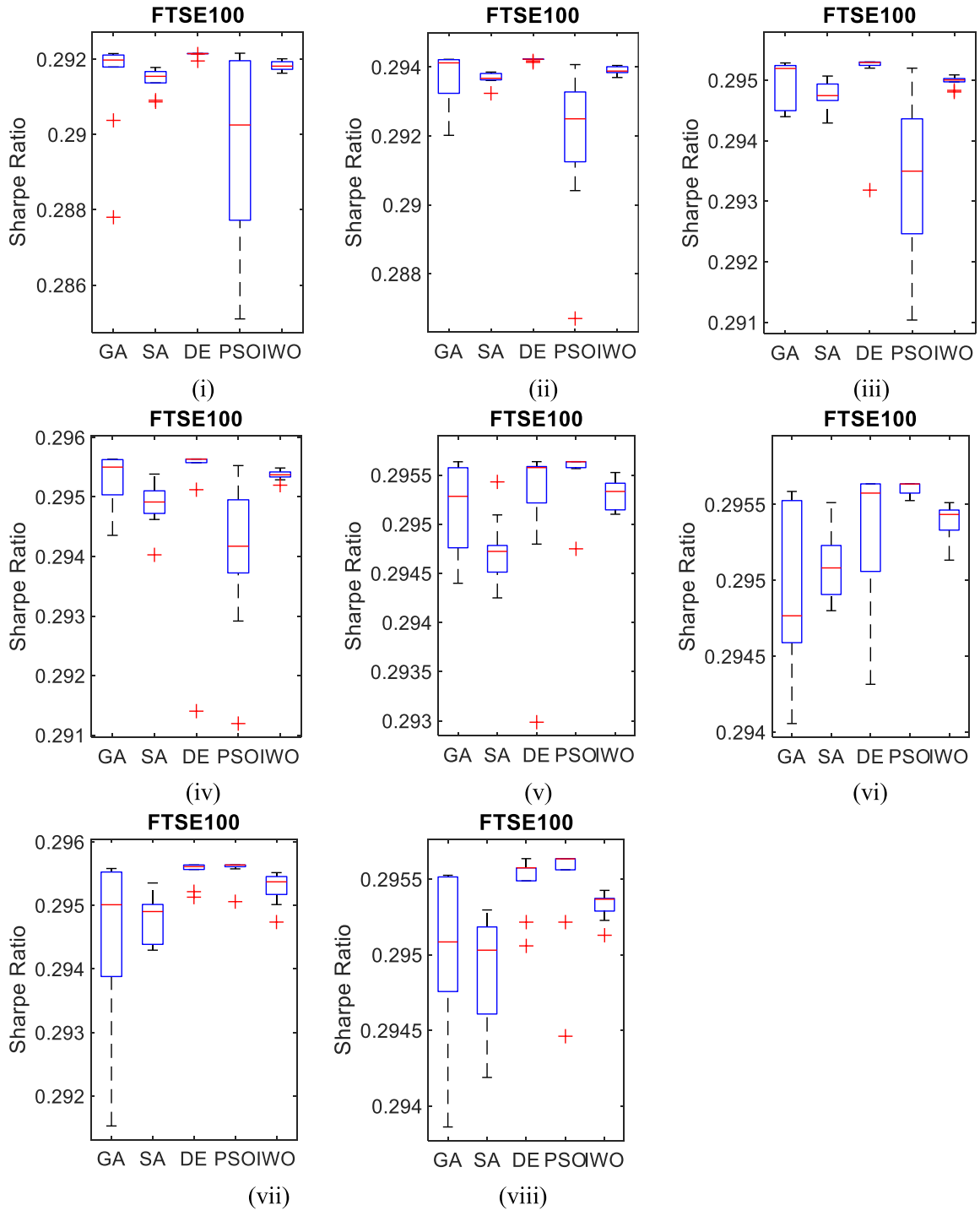


Fig. 15. Comparative results of FTSE for the Sharpe ratio of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

securities that fulfil risk budgeting limitations. Optimization algorithms' ability to explore the search space and find feasible solutions is crucial in such cases. As shown in Figs. 24(i) and 24(ii), the proposed IWO algorithm outperforms the benchmark algorithms by achieving Sharpe ratios best maximum and mean. IWO surpasses the other algorithms in return and risk of the RBPS at the RB% levels shown in Figs. 25(i)-(ii) and 26(i)-(ii). Due to its adaptive dispersion and reproduction processes, IWO can effectively explore the search space and converge to optimal solutions by dynamically adjusting solution dispersion and promising solution reproduction rates. Smaller SDs for IWO at RB% = 12.5 and 17.5 show that its search algorithms provide more consistent and dependable results over several runs.

The search area gets less confined when RB% increases to 22.5, 27.5, and 35.0, while the Sharpe ratio value increases with RB%. As displayed in Fig. 24(iii), 24(iv), and 24(v), the IWO method has the greatest maximum and mean performance metrics compared to the benchmark algorithms. The return and risk representations of box plots in Fig. 25 (iii) -(v) and 26(iii)-(v) for these RB% values indicate that IWO finds the most optimum solutions compared to GA, SA, DE, and PSO. While IWO still has an edge in maximum and median Sharpe ratio values, the

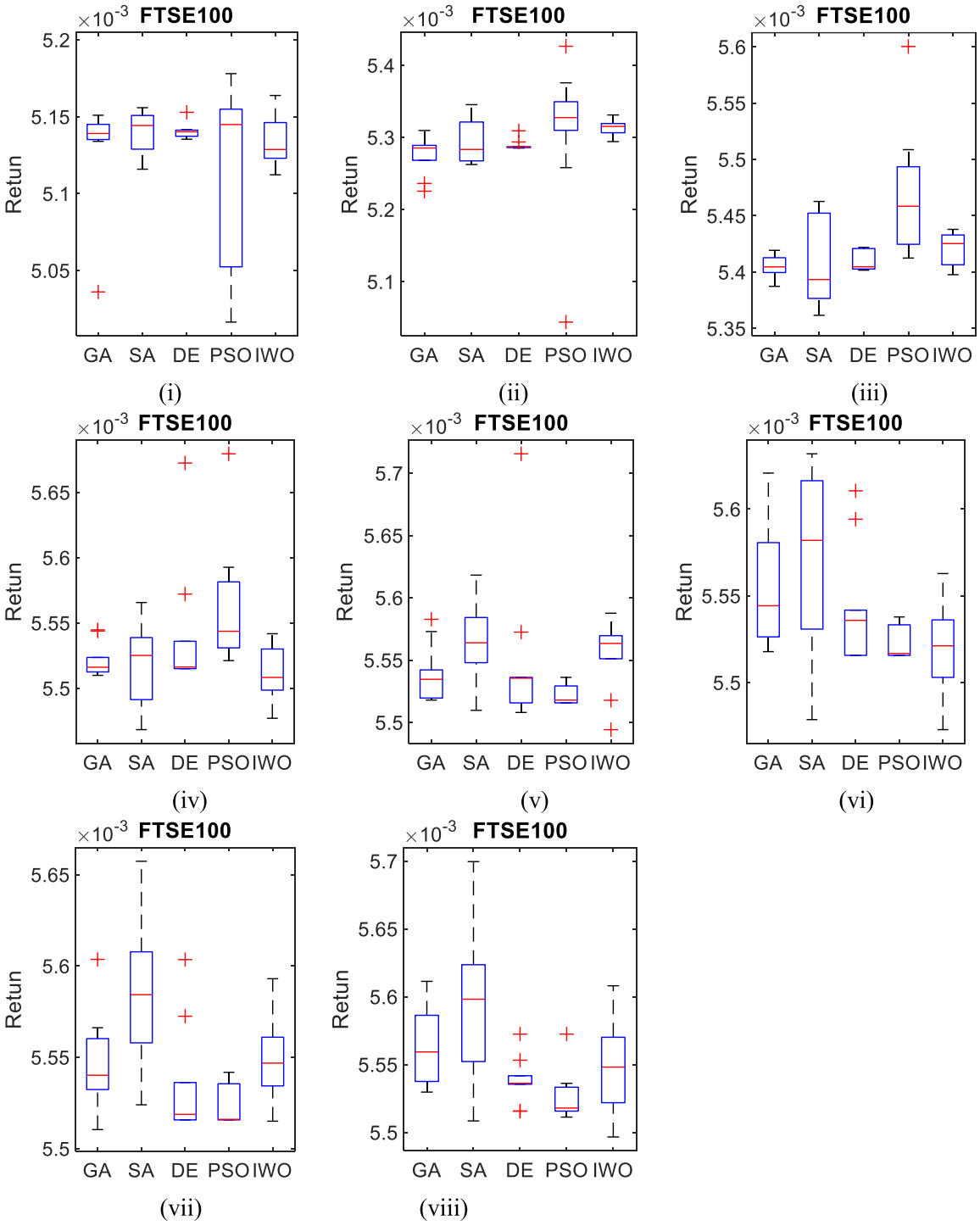


Fig. 16. Comparative results of FTSE for the Return of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

algorithms converge towards identical portfolios.

At 50.0, 75.0, and 100.0 values of RB %, the RBPS models' search area becomes unbounded, and all algorithms have converged to the optimum portfolio. As seen in Fig. 24(vi), 24(vii), and 24(viii), all algorithms have almost similar maximum, average, and median Sharpe ratios. Return and risk of the portfolios demonstrated in Fig. 25(vi)-(viii) and 26(vi)-(viii) for the higher RB% levels also follows the similar pattern. Regardless of their search methodologies, all algorithms will likely discover the global optimal security weights under budgetary limitations.

5.5. Statistical Testing

The statistical analysis tests the hypothesis associated with the findings presented in Section 5.4. The primary objective of statistical testing is to verify the normality of the data samples of objective function values, that is, the Sharpe ratio. Subsequently, it analyzed the algorithms with a confidence level of 95 % to compare the effectiveness of GA, SA, DE, PSO, and IWO. The assumptions considered for normality, homogeneity, and comparability are as follows:

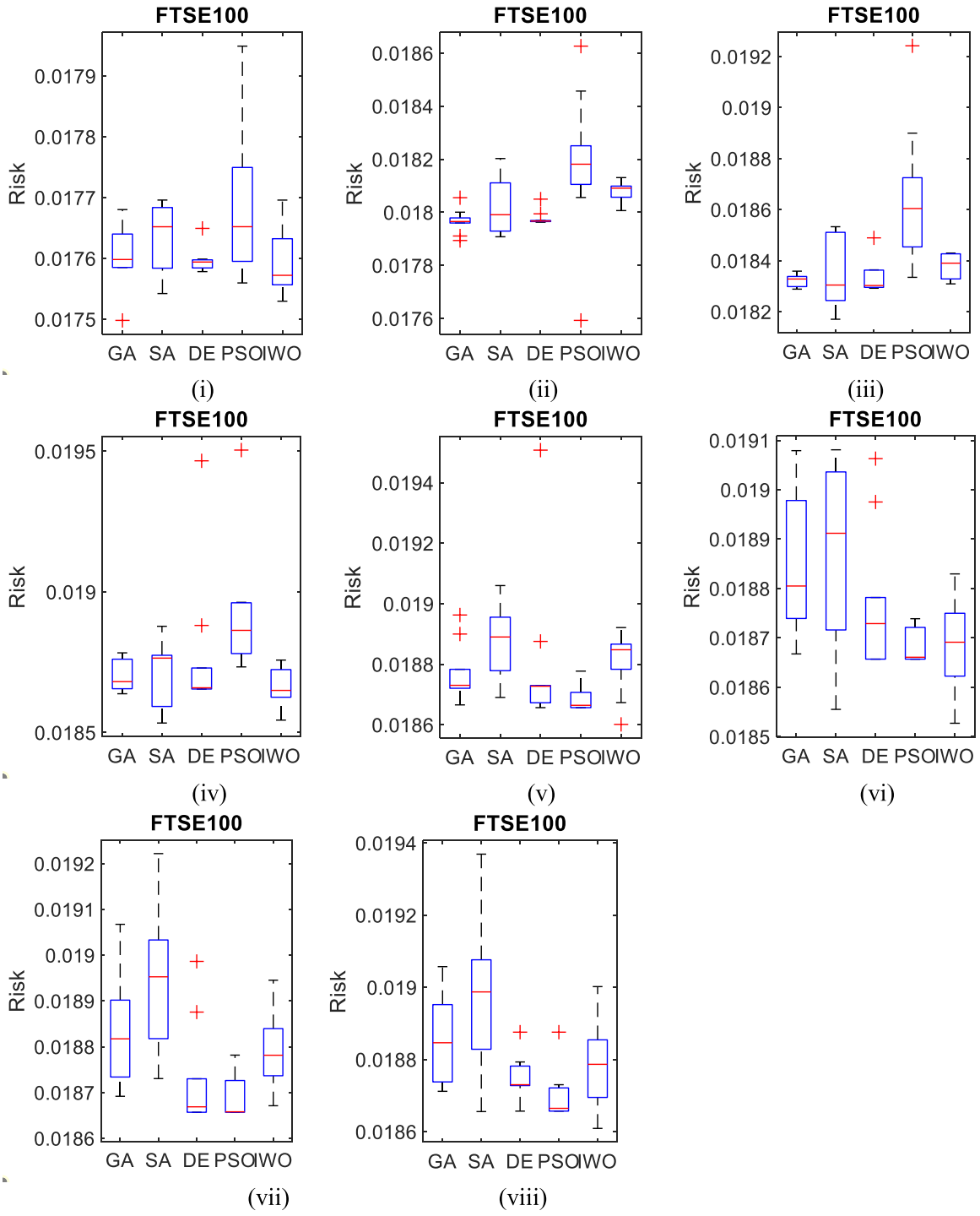


Fig. 17. Comparative results of FTSE for the Risk of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB % = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

Hypothesis 1. The sample data exhibits negligible variation and follows a normal distribution.

Hypothesis 2. The sample data exhibits homogeneity, meaning it follows an identical distribution of points.

Hypothesis 3. There is no statistically significant difference in the values of GA, SA, DE, PSO, and IWO on X, where X represents the vector of the objective function.

The Kolmogorov-Smirnov (K-S) test was performed as the Normality test on the Sharpe ratio data samples obtained by 20 independent runs. The significance value in the K-S test is a measure of the probability that the data is drawn from a normal distribution. If the significance value is more than 0.05, then Hypothesis 1 is invalid, which means the sample does not follow the normal distribution.

Table 6 presents the results of the Normality and Homogeneity test for GA, SA, DE, PSO and IWO algorithms corresponding to each dataset. The significance value in the K-S tests is calculated to determine the performance consistency of the meta-heuristic algorithms over multiple runs for the BSE, NSE, DAX, FTSE, Hang Seng, S&P and Nikkei datasets. Table 5 presents the 20 sample data of fitness function (Sharpe ratio) for

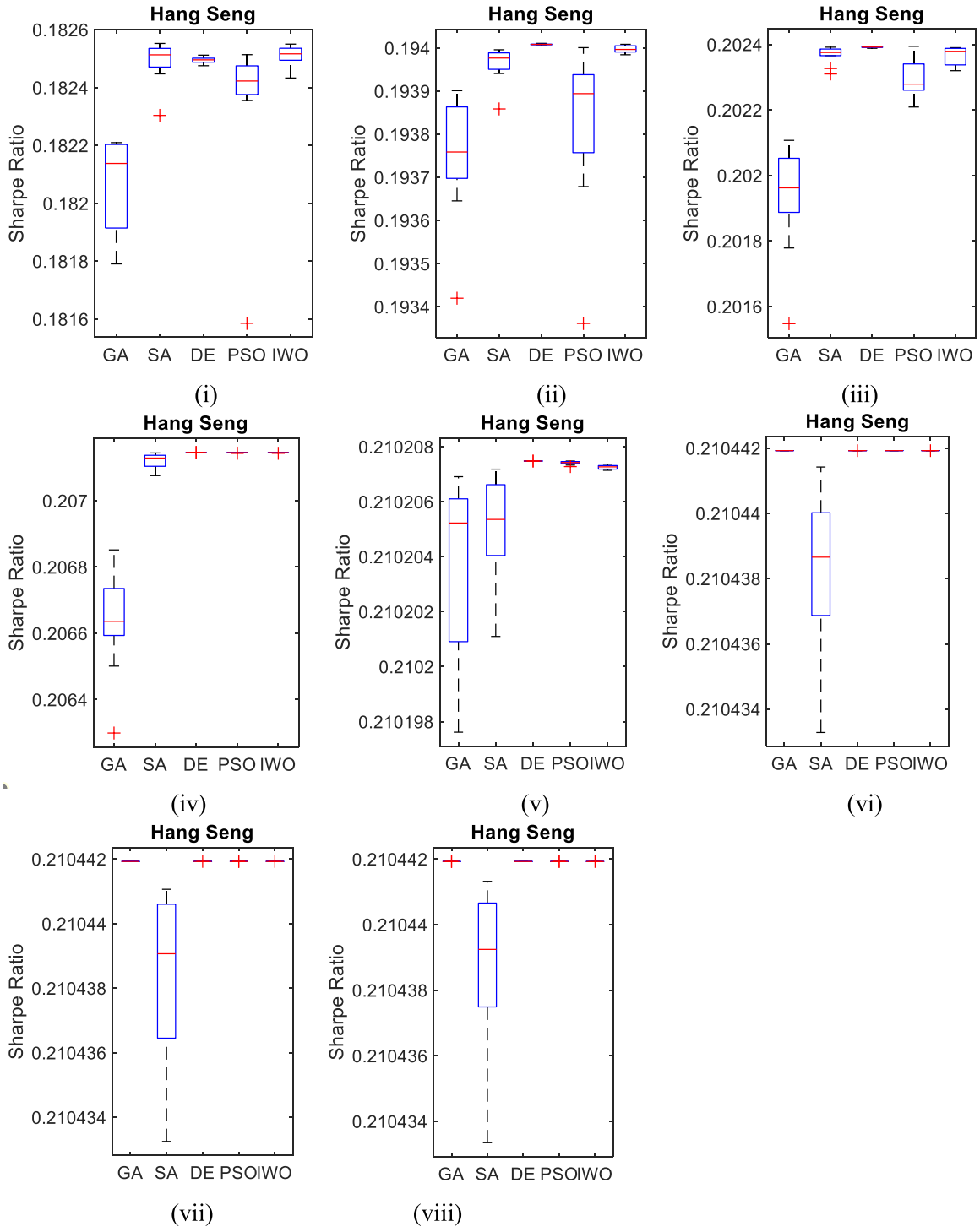


Fig. 18. Comparative results of HangSeng for the Sharpe ratio of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

BSE at RB% of 12.5 using all algorithms. In Table 6, the significance values in the K-S test obtained by IWO are more than 0.05 for all the datasets. Hence, Hypothesis 1 is rejected, suggesting that the IWO algorithm has statistically different performance distributions and consistently performs better than the other. However, the Normality test (Hypothesis 1) is validated regarding GA for FTSE, SA for BSE, DE for DAX and FTSE, and PSO for Hang Seng, showing no statistical difference in their performances.

Further, the Levene test, a statistical test used to assess the homogeneity of variances, was used to ensure the homogeneity of the data samples. To ensure data homogeneity, the significance value of the Levene test, calculated using the mean, should be more than 5 % (0.05). The significance values of the Levene test for the Sharpe ratio samples of BES, NSE, DAX, FTSE, Hang Seng, S&P, and Nikkei using GA, SA, DE, PSO, and IWO algorithms are shown in Table 6. Table 6 reveals that all significance values obtained from the Levene test are less than 5 %. Therefore, Hypothesis 2 is invalidated, indicating that the samples are not homogeneous. Hence, the Kruskal-Walli's test is necessary to assess

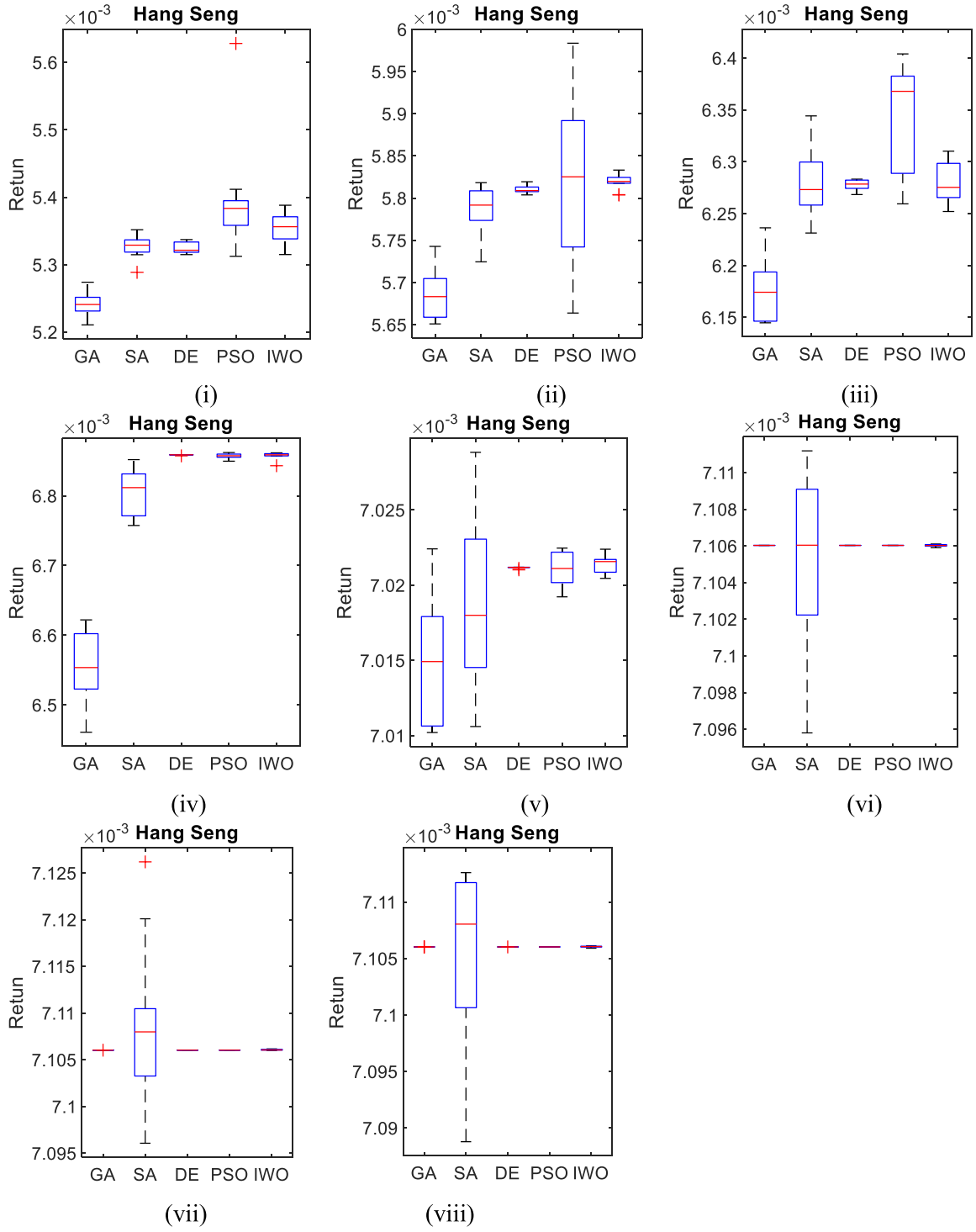


Fig. 19. Comparative results of HangSeng for the Return of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

the significant disparity in the Sharpe ratios across algorithms. The statistical outcomes of the Kruskal-Walli's test are shown in Table 7. In Table 7, all algorithms obtain the Sharpe ratio with a statistically significant value of 0.000, below the 0.05 criterion at a 5 % significance level. Hence, Hypothesis 3 is rejected. From Tables 6 and 7, we argue that there is considerable variation in the Sharpe ratios obtained by solving the RBPS model using GA, SA, DE, PSO, and IWO, i.e., across all datasets, the differences are statistically significant.

In the next step, post hoc analysis (multiple comparisons) was used to determine whether individual groups of methods were present. We used Dunn's post hoc test with the Bonferroni correction to compare IWO with GA, SA, DE, and PSO. The results of the post-hoc analysis are shown in Table 8. Based on the results of the post-hoc analysis in BSE, the significance level of 0.05 of the IWO is statistically significantly lower compared to the GA, SA, DE, and PSO. The integration of these statistical results into the performance comparison justifies the claims of IWO's superior effectiveness.

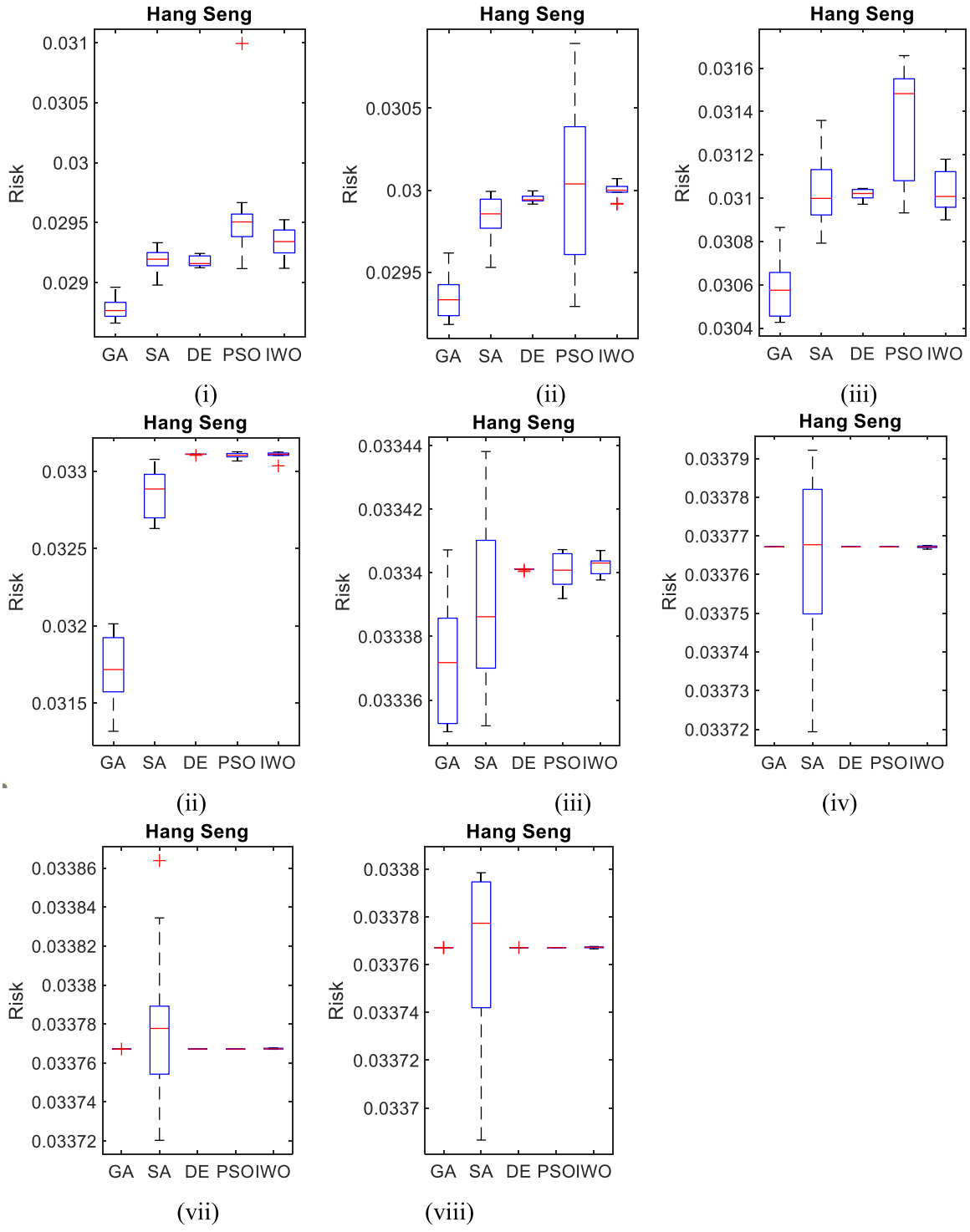


Fig. 20. Comparative results of HangSeng for the Risk of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

5.6. Analysis and reflection on the findings

5.6.1. Alignment with Stochastic dynamic financial market

The proposed IWO solution algorithm is well-suited for financial RBPS optimization problems. Financial markets exhibit stochastic and adaptive behaviours, characterised by volatility clustering, regime shifts, and mean-reversion patterns. The proposed solution mechanism inherits random initialization, adaptive reproduction, spatial dispersal, and competitive exclusion, which perfectly align with the characteristics of financial time series.

- Randomly initializing the seed enables the algorithm to start exploring new portfolio configurations based on market conditions.
- The adaptive reproduction phase of the IWO stems market regime shifts due to macroeconomic shocks, policy changes, and structural market breaks. Bull, bear, and high-volatility states each produce distinct risk and return patterns during this phase.

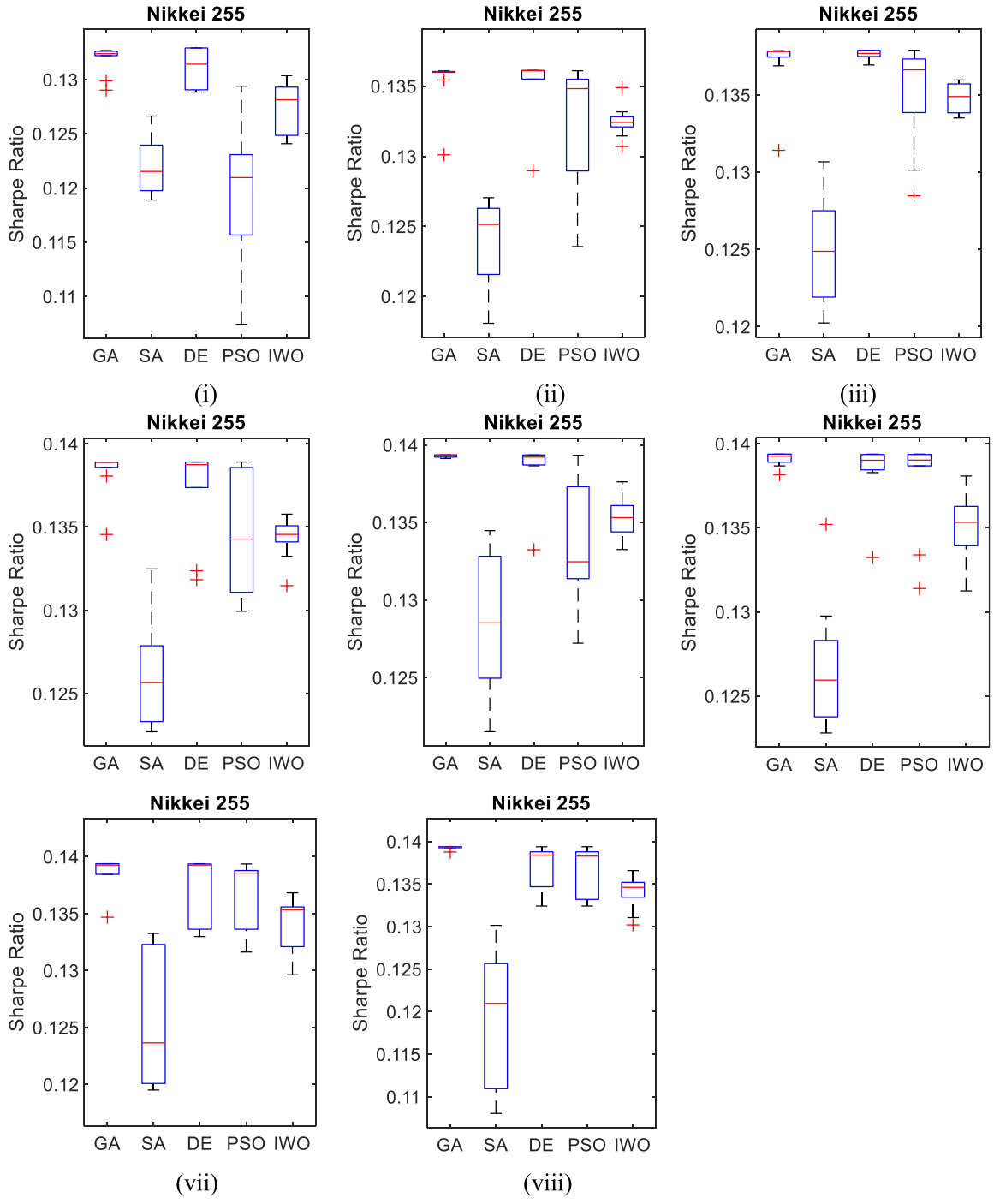


Fig. 21. Comparative results of Nikkei for the Sharpe ratio of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

- Spatial dispersal helps the algorithm to create cyclical patterns around equilibrium levels. It supports exploiting the mean-reversion benefit of adaptive repositioning. Continuous dispersal keeps the search active across shifts and reduces the risk of locking.
- Volatility clustering occurs when large price changes tend to be followed by large ones, and small changes tend to be followed by small ones. The nonlinear reduction of the standard deviation for seed dispersal is a natural fit for volatility clustering. The dispersal starts its search with a broad exploration (high volatility) and fine-tunes its solution (low volatility) over the iterations.
- Competitive exclusion results in strong asset selection for the fitter plants under hard pressure in financial markets, as fitter plants produce more offspring. The process reflects the risk-adjusted performance that drives allocation decisions for the institutional portfolio management environment.

5.6.2. *Balance between exploration and exploitation*

The proposed solution approach in Section 4 utilizes the IWO technique, which exhibits an equilibrium between exploitation (searching for a feasible region) and exploration (pursuing the best optimal) for the RBPS model. Mathematically, the normal distribution of seed-dispersing

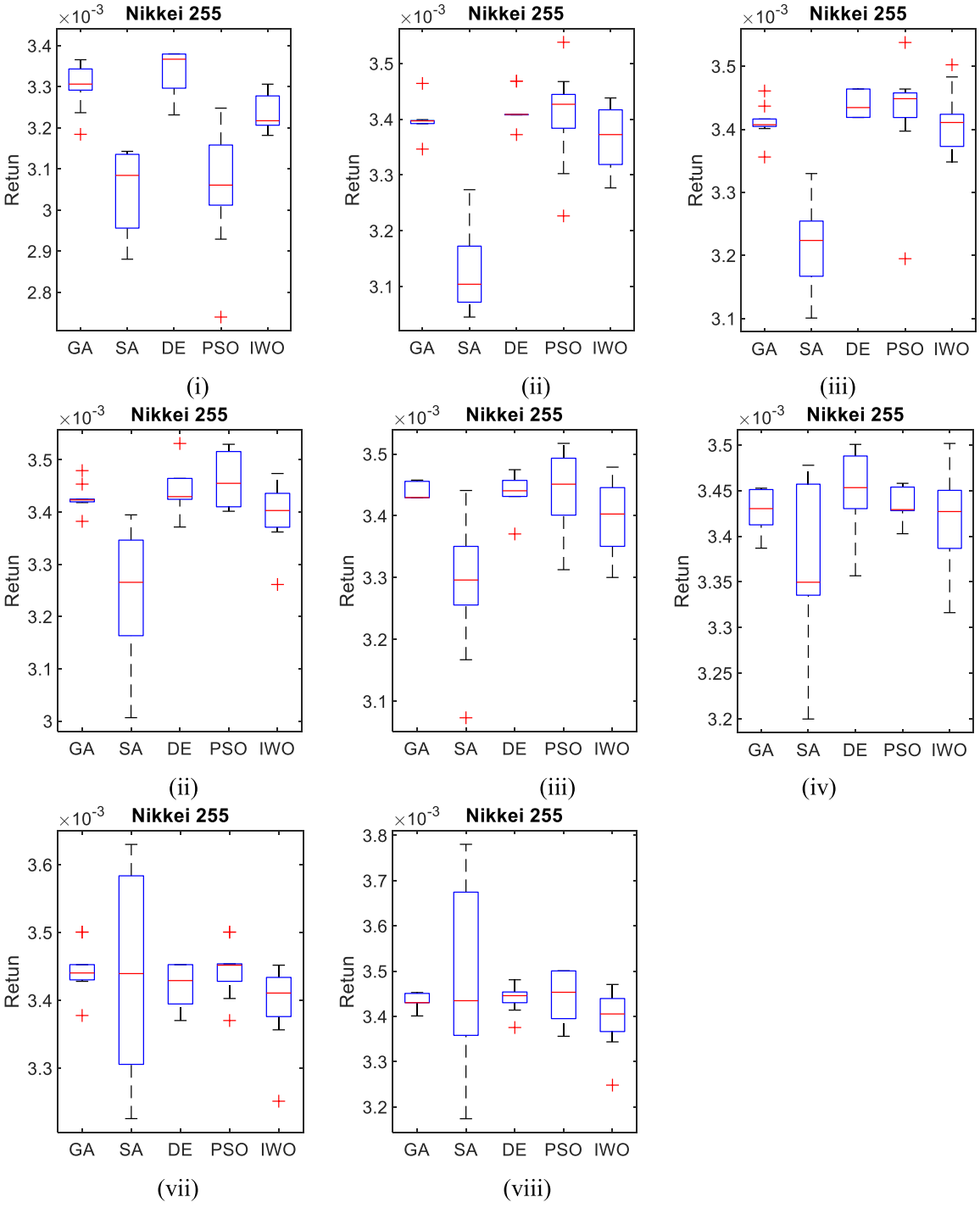


Fig. 22. Comparative results of Nikkei for the Return of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

around parent plants creates a gradient of solutions resembling a multi-scale search strategy, reducing premature convergence. By dynamically varying the standard deviation (σ) of seed dispersal over generations, the IWO navigate the solution space more efficiently and reach optimal or near-optimal solutions more quickly [78]. On the contrary, the GA, SA and DE algorithms, as described in Appendix A.1, A.2 and A.3, tend to prioritize exploration rather than exploitation, which hinders the convergence process.

5.6.3. Few numbers of parameters required for optimizations

Several parameters, including population size, crossover and mutation rates, and selection strategies, are typically optimized during GA. In the same way, a greater number of parameters must be adjusted for DE and PSO, as described in Appendix Sections A.3 and A.4, respectively. Modifying these parameters may require a significant time to attain maximum performance. In contrast, the IWO technique comprises fewer parameters, contributing to its simplified and more efficient implementation [19,79].

5.6.4. Outmatching genetic operations

GA and DE utilize two of the most common genetic operations: mutation and crossover. These two operations require a significant

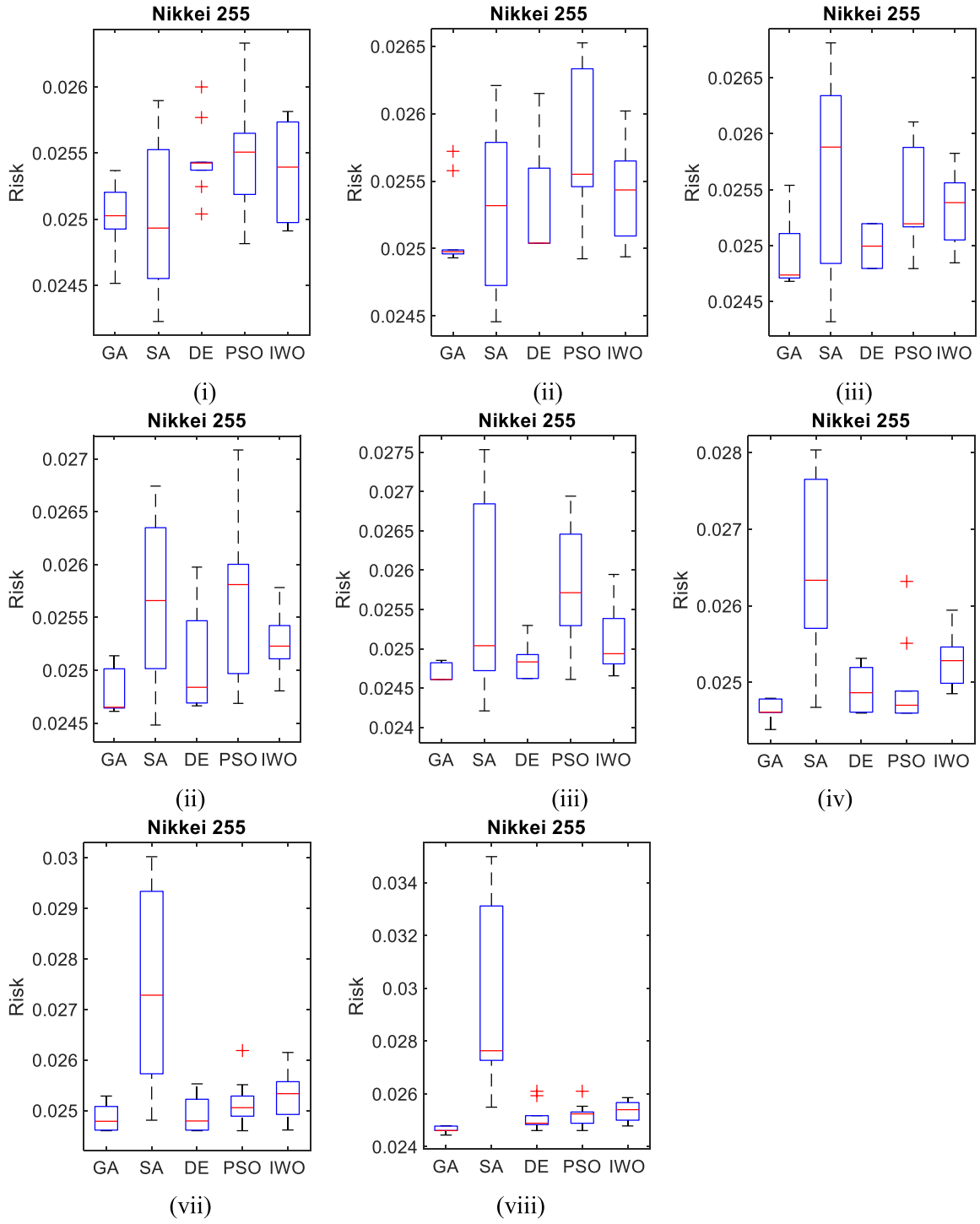


Fig. 23. Comparative results of Nikkei for the Risk of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

computational time over the generations, and the initial population influences the search. In contrast, the IWO employs an independent search for each weed, creating an implicit parallel spatial dispersal strategy, which is more effective in terms of convergence speed and computation time.

The effectiveness of the IWO method is more suited due to its balancing between exploration and exploitation, independent of the initial population, with fewer parameters, and surprising problems with crossover and mutation operations. Such factors make an IWO a faster alternative to GA, PSO, DE, and SA in solving problems.

5.7. *Computational complexity analysis*

The proposed IWO-based solution algorithm to solve the RBPS is accomplished by two procedures: IWO and repairing. The IWO consists of three sub-procedures: generating the initial weeds, seed production, and spatial dispersal. The computational time complexity of weed generation procedures for invested k-dimensional weights (w) is O(N), where N represents the total number of weeds and O(N + S) gives the time complexity of the seed production procedure, where the S represents the number of seeds produces from the weed [19,78]. Moreover,

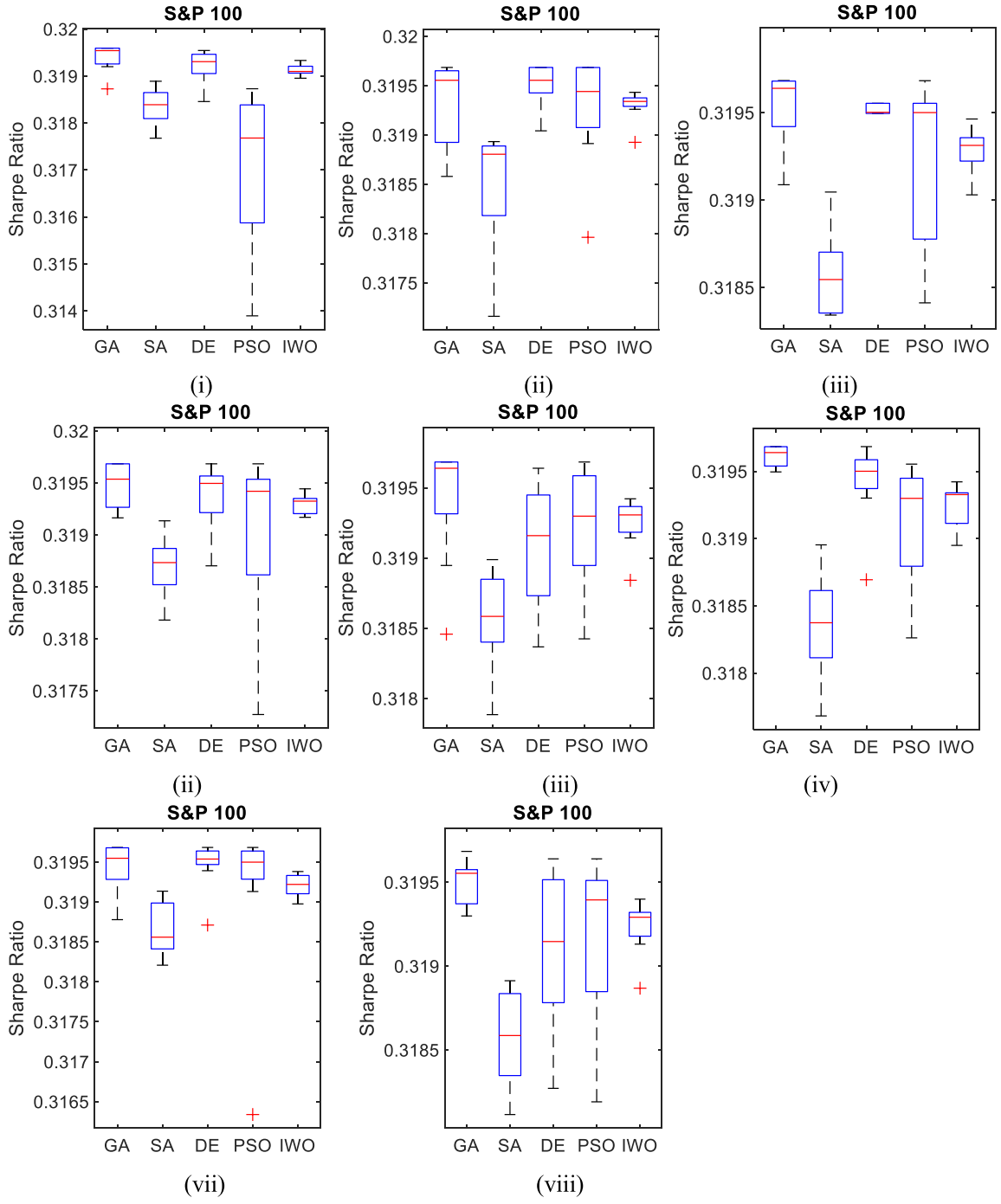


Fig. 24. Comparative results of S&P for the Sharpe ratio of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

the time complexity of spatial dispersal and reconstruing is O(N). The combined time complexity of the IWO procedure is O(N)+O(N + S) +O (N) ≃ O(N) as N > S. The time complexity of the repairing procedure is O(N). Now, the time complexity of iterating the IWO and repairing procedures over the T number of iterations is O(T × [N + N]) ≃ O (T × N). Hence, the worst-case time complexity of the IWO-based approach is O(T × N). Furthermore, for the N number of population and T number of maximum generations (or iteration), the worst-case time complexity of GA, PSO, and DE is O(T × N), and SA is O(T). Finally, the above dissection shows that the computational costs for IWO, GA, PSO and DE are the same. Still, IWO is more efficient due to adaptive dispersal (σ) that requires fewer iterations (T) for convergence.

5.8. Advantages and disadvantages of implementing IWO

The IWO algorithm's primary advantage lies in guaranteeing that all feasible candidates participate in the reproduction process, a feature absent in most meta-heuristic algorithms. This characteristic makes IWO more efficient than algorithms like GA and DE, which do not allow individuals with lower fitness levels to have offspring. The IWO method is also relatively easy to understand and implement, making it a practical choice. For instance, the PSO procedure, which is more complex than the IWO algorithm, involves additional computations to determine the best local and global positions for each particle and the entire population.

Despite its advantages, the IWO algorithm encounters difficulties

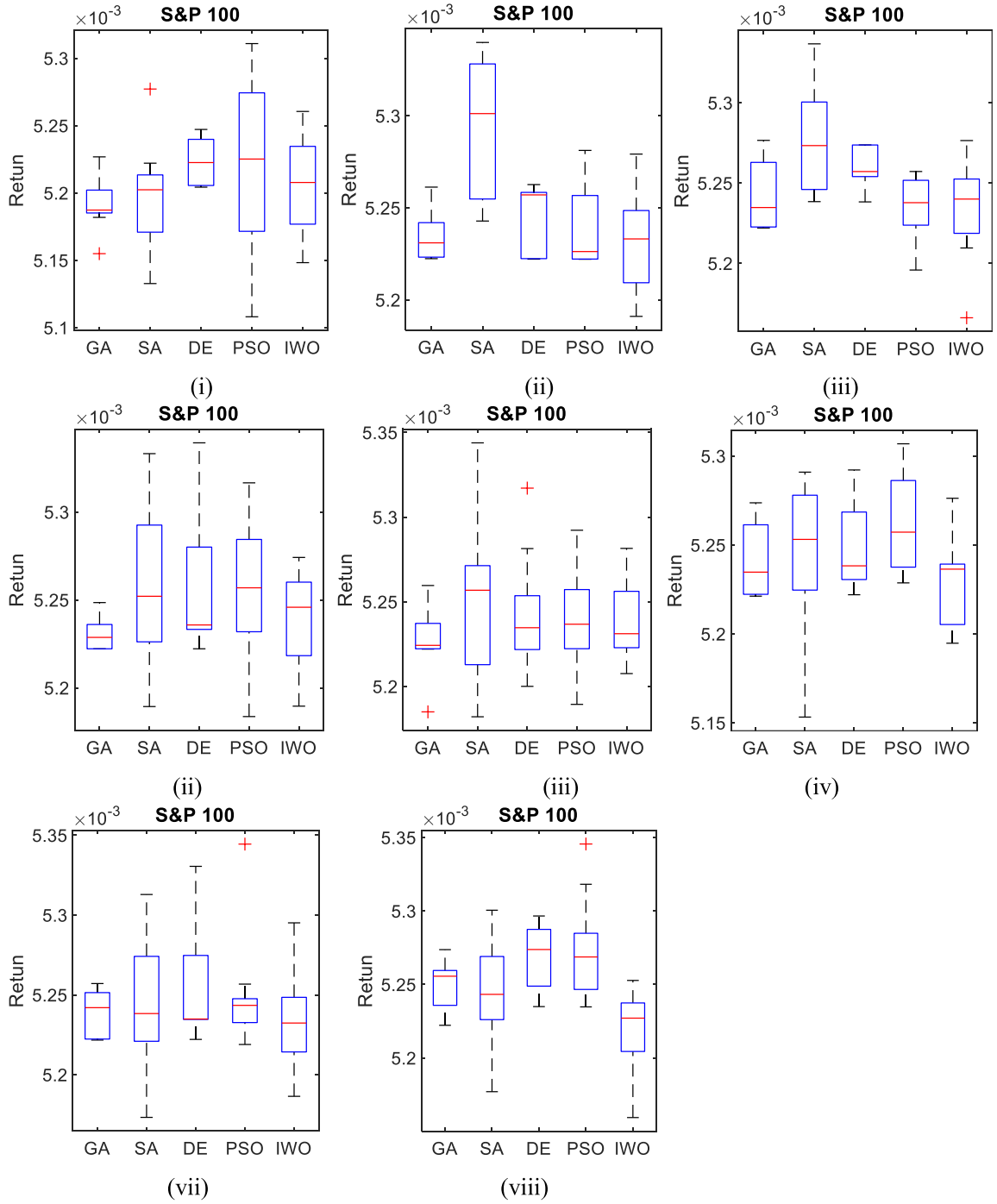


Fig. 25. Comparative results of S&P for the Return of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB% = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

when confronted with problems that feature a vast search space. The IWO requires numerous seeds for each plant to scour the feasible region. This can lead to a significant increase in computation time, especially when the number of variables to be adjusted rises in an extensive multi-dimensional space. Additionally, reducing standard variance, a crucial component of the IWO technique, can lead to premature convergence, where the algorithm must explore the search space, potentially missing the global optimum. This necessitates additional efforts to balance the imprecision and prevent convergence to local optima.

It is worth mentioning that the IWO-based solution approach outperforms GA, SA, DE, and PSO for our proposed RBPS model. Regardless, we cannot claim that it will be better for all problem collections since there is no assurance that an optimization technique thriving on one set of problems would be equally sufficient on another set of problems, as reported by "No Free Lunch theorems" [80].

6. Conclusions

This paper introduces a risk-budgeting portfolio selection (RBPS) optimization problem. The novel RBPS model maximizes the portfolio Sharpe ratio within a risk-budgeted, fully invested, and boundary-constrained framework. The RBPS model is designed to construct a portfolio that achieves the desired level of risk while maximizing returns through strategic risk budget allocation. To address the RBPS model, a

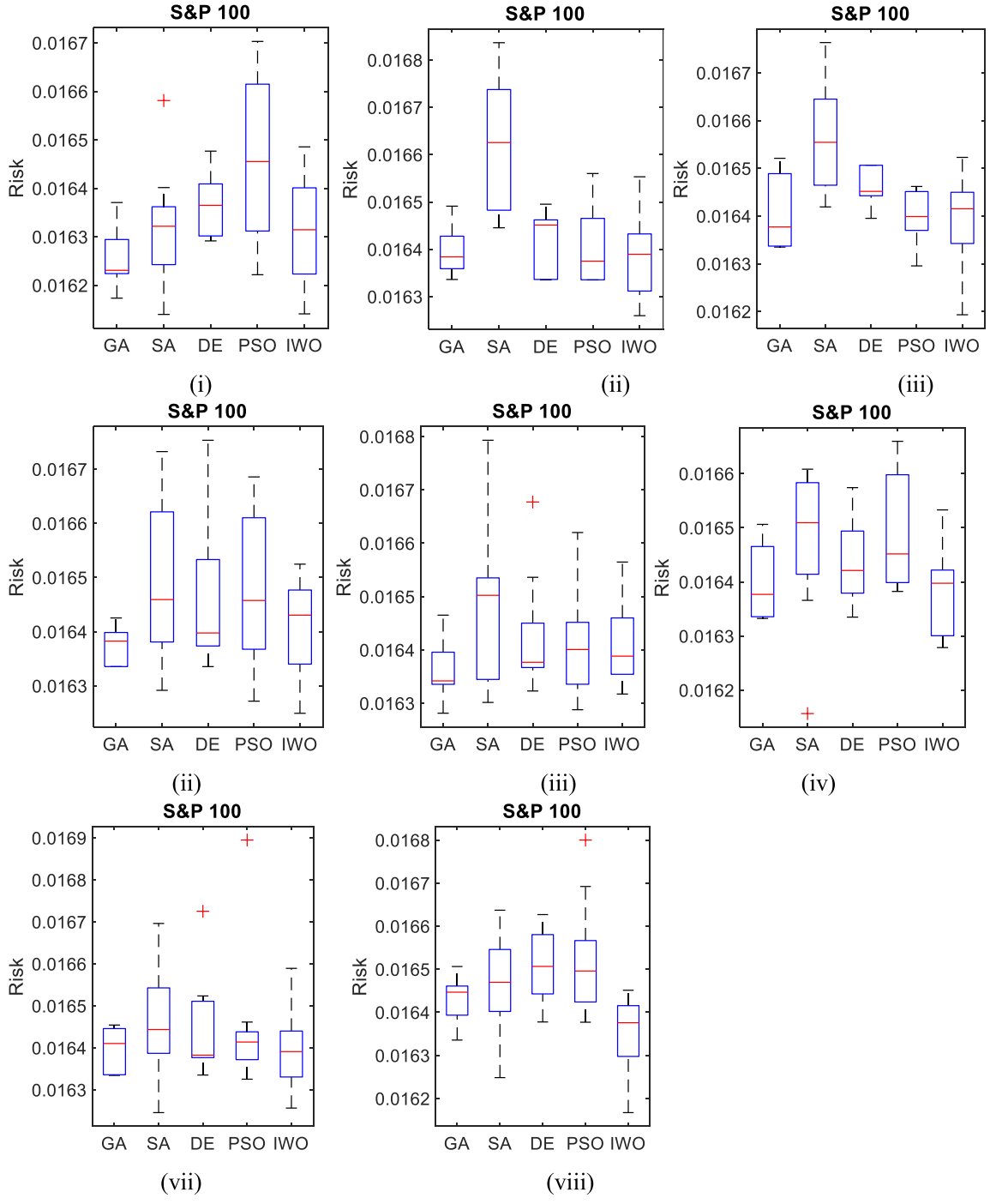


Fig. 26. Comparative results of S&P for the Risk of the RBPS models obtained by GA, SA, DE, PSO and IWO algorithms for: (i) RB% = 12.5, (ii) RB% = 17.5, (iii) RB % = 22.5, (iv) RB% = 27.5, (v) RB%= 35, (vi) RB% = 50, (vii) RB% = 75, and (viii) RB% = 100.

new invasive weed optimization (IWO) based solution algorithm is proposed that utilizes the repairing procedures to handle the constraints efficiently. The experimental study utilizes two newly created real-world datasets from the Indian stock market Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE). Additionally, a comprehensive analysis is conducted using the widely used datasets of global stock exchanges, namely, DAX, FTSE, Hang Seng, Nikki and S&P, to demonstrate the robustness and applicability of the proposed IWO-based solution approach in various international contexts beyond the Indian market. The results suggested that the proposed model provided beneficial outcomes as compared with other algorithms like Genetic Algorithm (GA), Simulated Annealing (SA), Differential Evolutionary (DE), and particle swarm optimization (PSO). The experimental outcomes are presented, and comparisons of the results using GA, SA, DE, and PSO algorithms are conducted over various risk budgeting levels RB% = 12.5, 17.5, 22.5, 27.5, 35, 50, 75, and 100 to determine the best Sharpe ratio of all datasets.

The results demonstrate that the RBPS model achieves a high portfolio Sharpe ratio as more risk budgeting levels are associated. However, once the highest Sharpe ratio is reached, further increasing the risk budgeting becomes impractical. This finding aligns with the general phenomenon of portfolio selection and the trade-off between risk and return, highlighting the practical implications of our research. The 20 independent runs of the GA, SA, DE, PSO, and IWO algorithms in the

Table 5
Simulation samples of Sharpe ratio (objective values) for solving RBPS Model at RB% = 12.5 level GA, SA, DE, PSO, and IWO algorithms using BSE data set.

Run	GA	SA	DE	PSO	IWO
1	0.318976	0.331090	0.332753	0.333091	0.336492
2	0.323013	0.328537	0.331552	0.322474	0.336576
3	0.315825	0.330420	0.332426	0.325216	0.336510
4	0.317158	0.322193	0.322240	0.331687	0.336964
5	0.313506	0.329375	0.323117	0.334112	0.336153
6	0.319849	0.330449	0.329355	0.329346	0.336272
7	0.320254	0.317258	0.319169	0.332189	0.336635
8	0.324151	0.313879	0.320061	0.333609	0.336065
9	0.321123	0.326964	0.334841	0.323124	0.336470
10	0.316746	0.312546	0.331297	0.332610	0.336422
11	0.312662	0.313619	0.334594	0.334043	0.336618
12	0.322781	0.330532	0.330452	0.334427	0.336860
13	0.315806	0.321375	0.333628	0.323280	0.336839
14	0.321236	0.329217	0.333625	0.323988	0.336832
15	0.314119	0.328675	0.333802	0.320793	0.335403
16	0.315658	0.319126	0.320526	0.331956	0.336658
17	0.319653	0.315818	0.322049	0.332092	0.335553
18	0.320098	0.324420	0.334830	0.323433	0.335636
19	0.315349	0.323217	0.329041	0.331021	0.336268
20	0.319765	0.316319	0.326094	0.333881	0.336806

Table 6
Results of Normality Test (Significance values of K-S test) and Homogeneity (Significance values of Levene Test) for Sharpe ratio samples.

Approach/ Dataset	Sig. values K-S Test					Sig. values of Levene Test Based on Mean
	GA	SA	DE	PSO	IWO	
BSE	0.200	0.002	0.200	0.200	0.200	0.000
NSE	0.200	0.200	0.200	0.130	0.200	0.000
DAX	0.200	0.200	0.011	0.132	0.200	0.000
FTSE	0.000	0.192	0.000	0.200	0.200	0.000
Hang Seng	0.057	0.161	0.200	0.000	0.200	0.014
S&P	0.117	0.200	0.200	0.061	0.200	0.000
Nikkei	0.200	0.116	0.200	0.055	0.200	0.000

experimental study demonstrate the consistency of the performance analysis in terms of finding the best mean, max, min, medium, and SD. Running multiple independent runs of the proposed IWO-based algorithms on all RB% problem instances shows the consistency of the results and the ability to converge to the best optimal portfolio for all datasets. The significance of the proposed approach is elaborated by comparing the percentage improvement in the maximum Sharpe ratio obtained by it over each of the other algorithms. The mean of percentage improvement values of the maximum Sharpe ratio of BSE data obtained by IWO over GA, SA, DE, and PSO are 1.05 %, 0.31 %, and 0.12 % at RB% = 12.5, 17.5, 22.5, 27.5, 35.0, 50.0, 75.0, and 100, respectively. Similarly, for NSE data, the mean percentage improvement values of the maximum Sharpe ratio of NSE data obtained by IWO over GA, SA, DE, and PSO are 0.24 %, 0.17 %, 0.04 % and 0.01 %, respectively.

Further, the statistical testing procedures, namely, the Kolmogorov-Smirnov, Kruskal-Wallis and Dunn's post-hoc Tests, have been applied on Sharpe ratio samples of size 20 obtained from the GA, SA, DE, PSO, and IWO algorithms validate our findings, i.e., the superiority of the IWO based solution algorithm over other computing algorithms. While the proposed IWO-based solving technique has shown promising results, it also opens several opportunities for future research and improvement. Some potential areas for further exploration are as follows:

- Other IWO variants and recently developed metaheuristics can be utilized to obtain better optimality and faster results. Thus, investigating the comparative performance capabilities of such available optimization approaches on the current problem may be a good topic for further study.
- Proposed RBPS framework consists of a few numbers of constraints. To address institutional applications, we will improve the model by incorporating constraints on minimum turnover, liquidity, capitalization, transaction lots, and transaction costs [1,13].
- Incorporating systemic risk measures such as Conditional Value-at-Risk (CoVaR) and Marginal Expected Shortfall (MES) [11,12] will enhance the RBPS model. Furthermore, integrating stress testing through simulated crisis scenarios facilitates the assessment of portfolio resilience to systemic market disruptions.
- The portfolio optimization environment is dynamic with changing stock pricing and market conditions. Future work may focus on developing adaptive selection algorithms that dynamically adjust the risk budgeting allocation in real time.
- IWO combined with machine learning models (PCA, Random Forests, Support Vector Machines) may enhance portfolio selection by forecasting asset returns, volatility, market regimes, feature importance, and allocation trend.
- The IWO-based algorithm is evaluated on a limited scale of financial stock market datasets. Future research can investigate its scalability to a larger stock market with more capital investments, and more performance measures can be considered for evaluation.

Table 8
Dunn's post-hoc Test results for Sharpe ratio samples.

Dataset	Pairwise Comparison	z-score	p-value	p (Bonferroni correction)
BSE	IWO vs GA	-3.89	0.0000	0.0000
	IWO vs SA	-6.19	0.0000	0.0000
	IWO vs DE	-4.02	0.0001	0.0006
	IWO vs PSO	-3.70	0.0002	0.0022
NSE	IWO vs GA	-5.34	0.0000	0.0000
	IWO vs SA	-0.72	0.0740	0.0001
	IWO vs DE	-3.36	0.0008	0.0078
	IWO vs PSO	-2.01	0.0445	0.0044
DAX	IWO vs GA	-4.49	0.0000	0.0000
	IWO vs SA	-1.52	0.0128	0.0010
	IWO vs DE	-1.50	0.0132	0.0001
	IWO vs PSO	-3.01	0.0025	0.0251
FTSE	IWO vs GA	-4.09	0.0000	0.0000
	IWO vs SA	-4.33	0.0000	0.0002
	IWO vs DE	-4.28	0.0001	0.0002
	IWO vs PSO	-2.70	0.0069	0.0069
Hang Sang	IWO vs GA	-4.71	0.0000	0.0000
	IWO vs SA	-2.74	0.0135	0.0135
	IWO vs DE	-1.83	0.0067	0.0067
	IWO vs PSO	-2.85	0.0043	0.0433
S&P	IWO vs GA	-3.93	0.0001	0.0008
	IWO vs SA	-4.68	0.0000	0.0000
	IWO vs DE	-3.01	0.0026	0.0257
	IWO vs PSO	-3.77	0.0002	0.0016
Nikkei	IWO vs GA	-3.90	0.0000	0.0000
	IWO vs SA	-4.19	0.0011	0.0010
	IWO vs DE	-2.02	0.0022	0.0025
	IWO vs PSO	-3.18	0.0018	0.0149

Table 7
Kruskal-Walli's Test for Sharpe ratio samples.

Dataset	BSE	NSE	DAX	FTSE	HangSeng	S&P	Nikkei
K-W	36.778	36.438	22.898	25.224	30.830	32.267	33.255
df	4	4	4	4	4	4	4
Asymp Sig.	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Ethical and informed consent for data used

We are not using any data for this consent is necessary.

CRediT authorship contribution statement

Lamaan Sami: Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization. **Mohammad Shahid:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Conceptualization. **Mohd Shamim:** Writing – review & editing, Writing – original draft, Methodology. **Faraz Hasan:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Zubair Ashraf:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare no conflict of interest.

Acknowledgements

The authors are thankful to the Deanship of Graduate Studies and Scientific Research at the University of Bisha, Bisha, for supporting this work through the Fast-Track Research Support Program. This work is supported by the major research project funded by ICSSR with sanction No. F.No.-02/47/ 2019–20/MJ/RP.

Appendix A. Risk-Budgeted Portfolio Selection using existing well-known metaheuristic

Heuristic/metaheuristic techniques show significant advantages when handling large-scale optimization problems, non-deterministic or stochastic problems, NP-hard problems, and other real-world applications where traditional optimization methods might prove inefficient or impractical. Due to the vast search space and the number of feasible portfolio distributions, the RBPS is classified as an NP-class problem. Therefore, apart from the proposed solving approach, this paper uses four alternative metaheuristic approaches—GA, SA, DE, and PSO—to tackle the RBPS problem.

Genetic Algorithm (GA)

Genetic algorithms (GA) is a metaheuristic approach that applies Darwin's idea of survival of the fittest to a population of potential solutions that develops from one generation to the next. GA determines the optimal solution for optimization problems with constraints, which involves a non-linear and complex objective function. The real-coded GA is applied to solve the RBPS problem, which belongs to the NP-class of problem. The solving approach of the RBPS problem using GA is explained in detail below.

Step 1: An initial population of real-coded chromosomes is randomly created for the RBPS problem.

Step 2: The individual candidate of the population is refined using the repairing procedure based on the constraints defined in Eqs. (8) - (11) to get a population that meets the risk-budgeting criteria.

Step 3: Evaluate the fitness value (Sharpe ratio) of each individual in the valid population using Eq. (7).

Step 4: The minimum fitness value is stored in an array to generate the convergence graph of the proposed RBPS problem.

Step 4: The individuals are sorted in ascending order based on their fitness value, and the highest fitness value is recorded to show the convergence of the proposed RBPS problem.

Step 5: The tournament selection procedure selects individuals as parent solutions for undergoing crossover procedures.

Step 6: The simulated binary crossover process produces offspring.

Step 7: Polynomial mutation introduces favourable individual variants.

Step 8: Following the crossover and mutation process, the offspring (generated from steps 5–7) undergo a repair process to verify their legitimacy.

Step 9: The offspring and population are merged to ensure elitism, and the new population is forwarded to the subsequent generation.

Step 10: The process outlined in steps 3–9 is continued until the termination condition is satisfied, often when the maximum number of iterations.

Simulated Annealing (SA)

Simulated annealing (SA) is a metaheuristic optimization technique introduced by Kirkpatrick et al. in 1983 that simulated the annealing process of metal. It is a probability-based method that mimics the behavior of heating a material to an adequate temperature and, afterward, gradually cooling it. Since a temperature rises during heating, the solid interior molecules become chaotic, which causes an increase in the potential energy of the solid. After attaining sizeable potential energy, the dropping of internal energy makes molecules progressively decrease to a minimum as they slowly get ordered, reaching stability at each degree drop and, eventually, achieving the equilibrium state at ambient temperature.

SA is used to solve the RBPS problem because of its necessity of a few numbers of parameters. The steps involved in solving the RBPA using SA are steps explained below.

Step 1: Initialize the randomly generated set of solutions S for the RBPS problem, provide the initial temperature is set to t_0, and establish the final temperature t_{end}. The values of the iteration index $Itr = 0$ and $t_{Itr} = t_0$.

Step 2: The randomly generated solutions are refined using the repairing procedure based on the constraints defined in equations (8) - (11) to get a population that meets the risk-budgeting criteria.

Step 3: Evaluate the fitness value (Sharpe ratio) of each individual in the valid population using Eq. (7).

Step 4: Randomly create a solution S_{nbh}, demonstrate the neighborhood of S_i, and undergo a repair process to verify its legitimacy. Determine the difference in the fitness function values, $\Delta F = F_{S_{nbh}} - F_{S_i}$.

Step 5: If $\Delta F > 0$, then S_i will be selected and go to Step 6; that is, S_i is reasonably good, then S_{nbh}. Otherwise, create a uniformly distributed number (χ) between 0 and 1; if $\exp(\Delta F/t_{Itr}) > \chi$, then $F_{S_{nbh}}$ is greater than F_{S_i}, that is, solution S_{nbh} is better than solution S_i and will be selected with probability $\exp(\Delta F/t_{Itr})$.

Step 6: Move forward to Step 7 if the number of neighborhood solutions exceeds the limit; else, go to Step 4.

Step 7: Set $Itr = Itr + 1$ and lower t_{Itr} by using the cooling approach as $t_{Itr+1} = t_{Itr} * q$, where $q \in (0.95, 0.99)$. If $t_{Itr} < t_{end}$, then stops the loop; else, go to Step 4.

Differential Evolution (DE)

Storn and Price developed differential evolution (DE) in 1996. It is a straightforward real parametric stochastic, population-based evolution optimization algorithm. In DE, the discrete trial individuals are called parameter vectors or chromosomes because they construct a population. DE performs the same analytical operations as a typical evolutionary computation, such as GA. Regardless, DE employs divergences in the individuals to examine the landscape of goal functions, contrasting conventional GA. Specifically, it considers the Nelder-Mead algorithm and the controlled random search method, which used differential vectors to modify the existing trial individuals.

DE starts with a k-dimensional real-valued parameter vector population generated randomly for the RBPS problem. The chromosomes of the population are refined using the repairing procedure based on the constraints defined in Eqs. (8) - (11) to get a population that meets the risk-budgeting criteria. Further, the fitness value (Sharpe ratio) of each individual in the valid population is evaluated using Eq. (7). In the next step of DE, the chosen individual is the parent individual from the present generation (using tournament selection), the donor individual is the mutated individual produced by the differential mutation operation, and the trial individual is the offspring created by combining the existing donor and the target individual. After completing the donor vector by mutation, a binomial crossover operation is performed to increase the population's potential genetic diversity. The principle point of the algorithm involves adjustment to decide whether the target or the trial individual persists until the next iteration to maintain the population size consistent throughout succeeding generations. The process continues till the termination condition is satisfied, often when the maximum number of iterations is reached.

Particle Swarm Optimization (PSO)

PSO, proposed by Kennedy and Eberhart, is a population-based stochastic search approach to the foraging behavior of swarms to identify optimal pathways via group cooperation. PSO treats each unique solution as a particle, and the optimum value in the problem space is selected to represent the behavior of the particle at this location. Each particle can locate a position and speed that are optimum for that particular place using its memory, which then determines the direction and distance of the subsequent step. The variable $X_i = (x_1, x_2, ..., x_{\dim})$ may be used to symbolize the location of the particle $-i$ in dim $-$ dimensional space, while the variable $V_i = (v_1, v_2, ..., v_{\dim})$ can describe the particle's velocity. The objective function determines the fitness value of each particle. Additionally, it is aware of its present location (X_i), which may be thought of as a particle's own flight experience, and the personal best position (*pbest*) it has discovered. Furthermore, every particle learns the optimum global best position for all particles found in the population (*gbest*, *gbest* is the best value in *pbest*), which can be considered particle partners' flight experience. Particles use their past actions and the collective knowledge of their neighbors to plan their future steps and arrive at the greatest possible global solution.

Suppose, $X_i(t)$ presents position at instant t and updated to $X_i(t+1)$ by adding a velocity $V_i(t)$ as follows.

$$X_i(t+1) = X_i(t) + V_i(t) \tag{15}$$

At the next instance $(t + 1)$, the velocity $V_i(t+1)$ is updated by the current location $X_i(t)$ and velocity $V_i(t)$ of the particle as:

$$V_i(t+1) = \chi[V_i(t) + c_1 * r_1(pbest_i(t) - X_i(t)) + c_2 * r_2 * (gbest_i(t) - X_i(t))] \tag{16}$$

Here, χ is constriction coefficient constant, c_1 and c_2 present the acceleration factors, r_1 and r_2 are random number between 0 and 1. Also, $= 2\omega/|2 - \varphi - \sqrt{\varphi(\varphi-4)}|$, $\omega \in [0, 1]$, $\varphi_1 = c_1 * r_1$, $\varphi_2 = c_2 * r_2$ and $\varphi = \varphi_1 + \varphi_2$.

PSO is also used here to solve the RBPS problem and the steps involved are explained below.

Step 1: Randomly initialize the position (X_i) and velocity (V_i) of each particle in swam for the RBPS problem. The personal best, $pbest_i = X_i$ and global best, $gbest = 0$.

Step 2: The randomly generated positions are refined using the repairing procedure based on the constraints defined in Eqs. (8) - (11) to get a population that meets the criteria.

Step 3: Evaluate the fitness value (Sharpe ratio) of each particle in the swarm using Eq. (7).

Step 4: Find the $pbest_i$ and *gbest* value: If fitness value of a particle X_i is better than the previous one, then $pbest_i$ is set to X_i. Also, at current iteration, *gbest* is the global best solution in the swarm; otherwise, no changes are made in $pbest_i$ and *gbest*.

Step 5: Updated the position (X_i) and velocity (V_i) of particle using Eq. (15) and Eq. (16), respectively and are refined using the repairing procedure.

Step 6: The $pbest_i$ and *gbest* are evaluated from the updated position (X_i) and velocity (V_i) and stored for plotting the convergence diagram.

Step 7: The process outlined in steps 2–6 is continued until the termination condition is satisfied, often when the maximum number of iterations.

Appendix B. Experimental results

Table B1
Computational values of Sharpe ratio, Return, and Risk for RBPS model using proposed IWO approach compared with GA, SA, DE, and PSO for BSE data for different RB % values

RB%		Sharpe Ratio					Return					Risk				
		GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO
12.5	**Max**	0.323013	0.331090	0.333802	0.334427	**0.336964**	0.014998	0.015656	0.016784	0.016082	**0.017029**	0.046599	0.047576	**0.050622**	0.048432	0.050537
	Min	0.313506	0.321375	0.322240	0.320793	**0.335403**	0.013895	0.014934	0.015574	0.014820	**0.016369**	0.044236	0.046421	0.047285	0.045583	**0.048803**
	Median	0.318067	0.329296	0.331989	0.327281	**0.336543**	0.014318	0.015568	0.016079	0.015366	**0.016775**	0.045178	0.047067	0.048598	0.047175	**0.049825**
	Mean	0.318227	0.328186	0.330295	0.327841	**0.336490**	0.014392	0.015436	0.016094	0.015431	**0.016733**	0.045219	0.047033	0.048724	0.047069	**0.049727**
	SD	0.003461	0.003483	0.004268	0.005253	**0.000466**	0.000393	0.000275	0.000383	0.000392	**0.000235**	0.000849	0.000440	0.000886	0.001028	**0.000647**
17.5	**Max**	0.353497	0.360211	**0.362017**	0.361948	0.361955	0.016056	0.017188	**0.017516**	0.017546	0.017510	0.045894	0.047770	**0.048650**	0.048621	0.048393
	Min	0.343828	0.356668	0.356797	0.358908	**0.361460**	0.015330	0.016403	0.017142	0.016771	**0.017233**	0.044271	0.045977	**0.048041**	0.046727	0.047656
	Median	0.350047	0.358270	0.360047	0.359994	**0.361657**	0.015798	0.016825	0.017500	0.017283	**0.017335**	0.045175	0.046943	**0.048605**	0.048025	0.047908
	Mean	0.349516	0.358380	0.359597	0.360476	**0.361698**	0.015751	0.016811	0.017425	0.017247	**0.017346**	0.045064	0.046906	**0.048458**	0.047843	0.047958

(*continued on next page*)

Table B1 (*continued*)

RB%		Sharpe Ratio					Return					Risk				
		GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO
	SD	0.003261	0.001154	0.001591	0.001029	**0.000165**	0.000277	0.000268	0.000151	0.000259	**0.000077**	0.000492	0.000625	0.000258	0.000644	**0.000202**
22.5	Max	0.373578	0.375417	0.376064	0.375945	**0.375984**	0.017128	0.017653	0.017754	**0.017761**	0.017741	0.046010	0.047025	0.047232	**0.047246**	0.047210
	Min	0.364892	0.373882	**0.375862**	0.375791	0.375378	0.016268	0.017200	0.017713	**0.017631**	0.017521	0.044582	0.046003	**0.047102**	0.046913	0.046637
	Median	0.368198	0.374850	**0.376044**	0.375830	0.375676	0.016508	0.017435	0.017728	**0.017710**	0.017669	0.044817	0.046524	**0.047141**	0.047124	0.047029
	Mean	0.368686	0.374856	**0.375984**	0.375845	0.375659	0.016625	0.017449	0.017732	**0.017711**	0.017650	0.045091	0.046548	**0.047162**	0.047122	0.046983
	SD	0.002885	0.000464	0.000092	**0.000049**	0.000179	0.000306	0.000161	**0.000015**	0.000042	0.000075	0.000531	0.000389	**0.000049**	0.000107	0.000189
27.5	Max	0.383029	0.385868	**0.385975**	0.385949	0.385951	0.017549	0.017906	0.017892	0.017893	**0.017905**	0.045816	0.046405	0.046358	0.046360	**0.046395**
	Min	0.378596	0.384201	0.381817	0.385804	**0.385871**	0.016896	0.017199	0.017440	0.017771	**0.017850**	0.044628	0.044767	0.045675	0.046062	**0.046251**
	Median	0.381535	0.385503	**0.385972**	0.385937	0.385934	0.017183	0.017743	0.017868	0.017875	**0.017875**	0.045164	0.046035	0.046294	0.046316	**0.046320**
	Mean	0.381083	0.385294	**0.385554**	0.385919	0.385927	0.017226	0.017683	0.017830	0.017863	**0.017878**	0.045202	0.045894	0.046244	0.046288	**0.046324**
	SD	0.001526	0.000549	0.001313	0.000047	**0.000025**	0.000181	0.000216	0.000137	0.000035	**0.000018**	0.000325	0.000500	0.000202	0.000086	**0.000046**
35.0	Max	0.394363	0.395435	**0.395512**	0.395507	0.395491	0.017866	0.017987	0.018037	**0.018041**	0.018040	0.045407	0.045506	0.045617	**0.045729**	0.045627
	Min	0.392419	0.395213	0.391907	0.394521	**0.395390**	0.017563	0.017811	0.017605	**0.017969**	0.017956	0.044712	0.045066	0.044921	**0.045432**	0.045401
	Median	0.393030	0.395369	0.395393	0.395388	**0.395392**	0.017719	0.017916	0.017973	0.018030	**0.018036**	0.045080	0.045310	0.045442	0.045601	**0.045615**
	Mean	0.393144	0.395335	0.394743	0.395313	**0.395415**	0.017715	0.017914	0.017925	**0.018026**	0.018022	0.045061	0.045314	0.045407	**0.045598**	0.045578
	SD	0.000681	0.000087	0.001496	0.000281	**0.000041**	0.000091	0.000056	0.000171	**0.000021**	0.000028	0.000233	0.000136	0.000269	**0.000072**	0.000076
50.0	Max	0.400690	0.400716	0.400726	0.400726	**0.400726**	0.018154	0.018187	0.018244	0.018247	0.018244	0.045311	0.045397	0.045555	0.045611	0.045556
	Min	0.400459	0.400616	0.397471	0.400054	**0.400473**	0.018094	0.018111	0.017842	0.018156	0.018155	0.045174	0.045202	0.044889	0.045307	0.045306
	Median	0.400614	0.400690	0.400726	0.400726	**0.400726**	0.018122	0.018152	0.018157	0.018157	0.018158	0.045231	0.045300	0.045310	0.045309	0.045312
	Mean	0.400608	0.400681	0.400325	0.400608	**0.400675**	0.018123	0.018151	0.018151	0.018183	0.018175	0.045239	0.045301	0.045341	0.045388	0.045360
	SD	0.000073	**0.000034**	0.001010	0.000221	0.000107	0.000022	0.000024	0.000116	0.000043	0.000036	0.000049	0.000060	0.000197	0.000129	0.000103
75.0	Max	0.400855	0.400850	0.400857	0.400857	**0.400857**	0.018187	0.018220	0.018278	0.018192	**0.018192**	0.045371	0.045467	0.045628	0.045382	**0.045382**
	Min	0.400821	0.400738	0.400598	0.400857	**0.400857**	0.018154	0.018156	0.018192	0.018192	**0.018192**	0.045292	0.045297	0.045382	0.045382	**0.045382**
	Median	0.400837	0.400824	0.400598	0.400857	**0.400857**	0.018181	0.018197	0.018278	0.018192	**0.018192**	0.045356	0.045398	0.045627	0.045382	**0.045382**
	Mean	0.400838	0.400814	0.400691	0.400857	**0.400857**	0.018178	0.018193	0.018250	0.018192	**0.018192**	0.045350	0.045389	0.045547	0.045382	**0.045382**
	SD	0.000013	0.000034	0.000123	0.000000	**0.000000**	0.000010	0.000020	0.000039	0.000000	**0.000000**	0.000024	0.000050	0.000111	0.000000	**0.000000**
100	Max	0.400856	0.400852	0.400857	0.400857	**0.400857**	0.018194	0.018225	0.018278	0.018192	**0.018192**	0.045415	0.045474	0.045627	0.045382	**0.045382**
	Min	0.400624	0.400783	0.397599	0.400857	**0.400857**	0.018143	0.018149	0.017879	0.018192	**0.018192**	0.045265	0.045284	0.044967	0.045382	**0.045382**
	Median	0.400826	0.400832	0.400598	0.400857	**0.400857**	0.018182	0.018174	0.018278	0.018192	**0.018192**	0.045360	0.045339	0.045627	0.045382	**0.045382**
	Mean	0.400801	0.400826	0.400376	0.400857	**0.400857**	0.018179	0.018180	0.018212	0.018192	**0.018192**	0.045356	0.045356	0.045488	0.045382	**0.045382**
	SD	0.000073	0.000025	0.000984	0.000000	**0.000000**	0.000016	0.000022	0.000124	0.000000	**0.000000**	0.000041	0.000056	0.000216	0.000000	**0.000000**

Table B2
Computational values of Sharpe ratio, Return, and Risk for RBPS model using proposed IWO approach compared with GA, SA, DE, and PSO for NSE data for different RB% values

RB%		Sharpe Ratio					Return					Risk				
12.5		GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO
	Max	0.377837	0.380247	0.379166	0.380330	**0.380342**	0.014381	0.015530	0.014802	0.015274	0.015267	0.038060	0.040875	0.039039	0.040161	0.040200
	Min	0.375761	0.379638	0.378391	0.377779	**0.379947**	0.014151	0.014847	0.014574	0.014400	0.015073	0.037608	0.039098	0.038515	0.038118	0.039651
	Median	0.376455	0.379962	0.378794	0.379628	**0.380123**	0.014219	0.015163	0.014686	0.014970	0.015234	0.037769	0.039891	0.038772	0.039408	0.040074
	Mean	0.376624	0.379959	0.378768	0.379297	**0.380106**	0.014249	0.015127	0.014680	0.014878	0.015264	0.037834	0.039811	0.038756	0.039225	0.040158
	SD	0.000666	0.000240	0.000226	0.000923	**0.000099**	0.000088	0.000142	0.000061	0.000269	0.000122	0.000178	0.000361	0.000140	0.000619	0.000326
17.5	Max	0.407393	0.409807	0.409853	0.409904	**0.409867**	0.015184	0.015841	0.015814	0.016055	0.015607	0.037329	0.038655	0.038598	0.039240	0.038079
	Min	0.405806	0.409012	**0.409840**	0.409152	0.409489	0.014834	0.015376	0.015388	0.015497	0.015579	0.036525	0.037561	0.037579	0.037829	0.038013
	Median	0.406810	0.409546	**0.409850**	0.409855	0.409732	0.014942	0.015529	0.015642	0.015649	0.015596	0.036766	0.037929	0.038183	0.038181	0.038052
	Mean	0.406621	0.409501	**0.409849**	0.409688	0.409721	0.014969	0.015544	0.015635	0.015686	0.015597	0.036812	0.037958	0.038160	0.038289	0.038055
	SD	0.000553	0.000234	**0.000004**	0.000266	0.000109	0.000101	0.000125	0.000119	0.000150	0.000008	0.000234	0.000297	0.000284	0.000384	0.000020
22.5	Max	0.426533	0.428360	0.428429	0.428476	**0.428479**	0.015716	0.016372	0.016448	0.018145	0.016439	0.036890	0.038242	0.038393	0.046489	0.038339
	Min	0.425605	0.427941	0.428252	0.390310	**0.428453**	0.015555	0.016112	0.016354	0.016165	0.016377	0.036539	0.037651	0.038180	0.037929	0.038150
	Median	0.425982	0.428228	0.428390	0.428380	**0.428472**	0.015630	0.016335	0.016427	0.016390	0.016394	0.036699	0.038139	0.038344	0.038254	0.038234
	Mean	0.425962	0.428196	0.428373	0.424369	**0.428472**	0.015638	0.016289	0.016408	0.016568	0.016398	0.036711	0.038042	0.038304	0.039108	0.038239
	SD	0.000258	0.000137	0.000061	0.011988	**0.000008**	0.000047	0.000086	0.000036	0.000562	0.000018	0.000097	0.000195	0.000083	0.002600	0.000057
27.5	Max	0.439991	0.440741	0.440920	0.440941	**0.440941**	0.016527	0.016849	0.017011	0.017004	0.016942	0.037563	0.038228	0.038584	0.038568	0.038422
	Min	0.439432	0.440372	0.440792	0.426894	**0.440930**	0.016322	0.016691	0.016895	0.016063	0.016918	0.037137	0.037901	0.038322	0.037628	0.038368
	Median	0.439786	0.440527	0.440859	0.440880	**0.440938**	0.016463	0.016737	0.016954	0.016973	0.016924	0.037427	0.037996	0.038456	0.038515	0.038382
	Mean	0.439717	0.440529	0.440862	0.439384	**0.440938**	0.016437	0.016743	0.016951	0.016885	0.016926	0.037380	0.038006	0.038449	0.038425	0.038387
	SD	0.000203	0.000125	0.000036	0.004402	**0.000003**	0.000065	0.000048	0.000040	0.000289	0.000008	0.000132	0.000101	0.000088	0.000283	0.000019
35.0	Max	0.452288	0.452347	0.452383	0.452391	**0.452391**	0.017583	0.017710	0.017758	0.017967	0.017718	0.038875	0.039152	0.039257	0.039884	0.039165
	Min	0.451880	0.451853	0.452355	0.450478	**0.452390**	0.017430	0.017442	0.017671	0.017695	0.017706	0.038573	0.038590	0.039064	0.039114	0.039138
	Median	0.452061	0.452123	0.452375	0.452391	**0.452391**	0.017502	0.017603	0.017714	0.017711	0.017712	0.038717	0.038934	0.039159	0.039150	0.039151
	Mean	0.452092	0.452132	0.452373	0.452199	**0.452390**	0.017505	0.017586	0.017715	0.017735	0.017711	0.038721	0.038896	0.039161	0.039220	0.039150
	SD	0.000129	0.000165	0.000010	0.000605	**0.000000**	0.000052	0.000083	0.000023	0.000082	0.000004	0.000105	0.000173	0.000050	0.000234	0.000009
50.0	Max	0.457501	0.457425	0.457496	0.457502	**0.457502**	0.018237	0.018383	0.018289	0.018479	0.018215	0.039863	0.040252	0.039983	0.040480	0.039814
	Min	0.457496	0.456703	0.457397	0.456492	**0.457502**	0.018221	0.018051	0.018186	0.018215	0.018215	0.039827	0.039472	0.039753	0.039814	0.039814
	Median	0.457500	0.457339	0.457479	0.457502	**0.457502**	0.018226	0.018253	0.018237	0.018215	0.018215	0.039837	0.039904	0.039864	0.039814	0.039814
	Mean	0.457500	0.457288	0.457465	0.457287	**0.457502**	0.018227	0.018228	0.018235	0.018274	0.018215	0.039841	0.039861	0.039861	0.039962	0.039814
	SD	0.000001	0.000212	0.000035	0.000406	**0.000000**	0.000005	0.000115	0.000036	0.000101	0.000000	0.000012	0.000261	0.000081	0.000256	0.000000
75.0	Max	0.457522	0.457493	0.457515	0.457515	**0.457522**	0.018238	0.018469	0.018299	0.018237	0.018237	0.039863	0.040404	0.040000	0.039861	0.039861
	Min	0.457522	0.457052	0.457451	0.457451	**0.457522**	0.018237	0.018046	0.018175	0.018237	0.018237	0.039861	0.039477	0.039727	0.039861	0.039861
	Median	0.457522	0.457302	0.45749	0.457490	**0.457522**	0.018237	0.018290	0.018245	0.018237	0.018237	0.039861	0.039993	0.039881	0.039861	0.039861

(*continued on next page*)

Table B2 (*continued*)

RB%		Sharpe Ratio					Return					Risk				
	Mean	0.457522	0.457271	0.45749	0.457490	**0.457522**	0.018237	0.018289	0.018241	0.018237	0.018237	0.039861	0.039995	0.039871	0.039861	0.039861
	SD	0.000000	0.000144	0.000021	0.000021	**0.000000**	0.000000	0.000144	0.000042	0.000000	0.000000	0.000000	0.000319	0.000093	0.000000	0.000000
100	**Max**	0.457522	0.457466	0.457522	0.457501	**0.457522**	0.018240	0.018439	0.018291	0.018287	0.018237	0.039867	0.040332	0.039984	0.039978	0.039861
	Min	0.457522	0.457168	0.457428	0.457412	**0.457522**	0.018237	0.018233	0.018163	0.018237	0.018237	0.039861	0.039856	0.039702	0.039861	0.039861
	Median	0.457522	0.457370	0.457522	0.457486	**0.457522**	0.018237	0.018322	0.018228	0.018237	0.018237	0.039861	0.040057	0.039846	0.039861	0.039861
	Mean	0.457522	0.457359	0.457512	0.457478	**0.457522**	0.018238	0.018327	0.018228	0.018244	0.018237	0.039862	0.040072	0.039845	0.039876	0.039861
	SD	0.000000	0.000088	0.000030	0.000027	**0.000000**	0.000001	0.000082	0.000036	0.000016	0.000000	0.000002	0.000184	0.000080	0.000037	0.000000

Table B3
Computational values of Sharpe ratio, Return, and Risk for RBPS model using proposed IWO approach compared with GA, SA, DE, and PSO for DAX data for different RB% values

r%		Sharpe Ratio					Return					Risk				
12.5		**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**
	Max	0.338728	0.338747	0.339440	0.338795	0.338918	0.004980	0.005182	0.005243	0.005398	0.005254	0.014746	0.015374	0.015497	0.016032	0.015537
	Min	0.337161	0.336972	0.337933	0.330136	0.337679	0.004929	0.005049	0.005065	0.004827	0.005149	0.014595	0.014931	0.014933	0.014623	0.015235
	Median	0.338290	0.337877	0.339237	0.336950	0.338246	0.004965	0.005159	0.005099	0.005210	0.005184	0.014668	0.015254	0.015045	0.015489	0.015320
	Mean	0.338180	0.337864	0.338899	0.335831	0.338354	0.004963	0.005148	0.005131	0.005175	0.005199	0.014675	0.015237	0.015141	0.015409	0.015366
	SD	0.000560	0.000532	0.000583	0.002652	0.000391	0.000014	0.000040	0.000069	0.000148	0.000036	0.000043	0.000131	0.000219	0.000392	0.000108
17.5	**Max**	0.354339	0.354411	0.354922	0.354122	0.354676	0.005617	0.005866	0.005911	0.006083	0.005895	0.015890	0.016605	0.016711	0.017302	0.016666
	Min	0.352989	0.353265	0.345893	0.351582	0.353696	0.005581	0.005759	0.005745	0.005695	0.005737	0.015757	0.016251	0.016209	0.016083	0.016181
	Median	0.354111	0.353833	0.354879	0.352736	0.354401	0.005591	0.005804	0.005766	0.005899	0.005812	0.015805	0.016409	0.016257	0.016713	0.016389
	Mean	0.353879	0.353785	0.353606	0.352826	0.354278	0.005594	0.005811	0.005788	0.005883	0.005811	0.015809	0.016424	0.016368	0.016674	0.016402
	SD	0.000459	0.000350	0.002796	0.000960	0.000383	0.000013	0.000038	0.000059	0.000107	0.000047	0.000042	0.000114	0.000206	0.000338	0.000145
22.5	**Max**	0.360664	0.360613	0.361063	0.361037	0.360944	0.006219	0.006303	0.006657	0.006656	0.006362	0.017256	0.017490	0.018579	0.018578	0.017651
	Min	0.359882	0.360145	0.358317	0.357181	0.360072	0.006035	0.006116	0.006273	0.006269	0.006219	0.016745	0.016972	0.017373	0.017364	0.017248
	Median	0.360479	0.360416	0.360547	0.359717	0.360667	0.006063	0.006205	0.006360	0.006410	0.006279	0.016811	0.017216	0.017660	0.017828	0.017411
	Mean	0.360447	0.360396	0.360189	0.359544	0.360634	0.006092	0.006216	0.006377	0.006438	0.006284	0.016900	0.017247	0.017706	0.017906	0.017424
	SD	0.000236	0.000137	0.000938	0.001126	0.000245	0.000066	0.000060	0.000119	0.000111	0.000046	0.000186	0.000166	0.000369	0.000349	0.000132
27.5	**Max**	0.362956	0.362516	0.363000	0.362815	0.362826	0.006335	0.006569	0.006745	0.006761	0.006505	0.017486	0.018136	0.018700	0.018926	0.017954
	Min	0.362258	0.362069	0.358360	0.357253	0.362038	0.006213	0.006286	0.006367	0.006430	0.006314	0.017132	0.017347	0.017579	0.017724	0.017439
	Median	0.362858	0.362249	0.362296	0.362077	0.362588	0.006275	0.006396	0.006432	0.006525	0.006426	0.017292	0.017665	0.017765	0.018027	0.017734
	Mean	0.362805	0.362260	0.361911	0.361674	0.362499	0.006278	0.006419	0.006464	0.006532	0.006430	0.017303	0.017719	0.017861	0.018063	0.017737
	SD	0.000208	0.000147	0.001395	0.001633	0.000274	0.000036	0.000085	0.000114	0.000100	0.000056	0.000102	0.000236	0.000327	0.000352	0.000148
35.0	**Max**	0.363785	0.363460	0.363785	0.363785	0.363539	0.006508	0.006681	0.006600	0.006565	0.006621	0.017889	0.018409	0.018158	0.018071	0.018222
	Min	0.363773	0.362955	0.361485	0.363319	0.363325	0.006482	0.006455	0.006483	0.006483	0.006484	0.017818	0.017765	0.017822	0.017822	0.017841
	Median	0.363785	0.363150	0.363103	0.363785	0.363464	0.006485	0.006553	0.006541	0.006483	0.006518	0.017826	0.018036	0.018021	0.017822	0.017930
	Mean	0.363783	0.363180	0.363139	0.363726	0.363449	0.006486	0.006563	0.006533	0.006496	0.006539	0.017830	0.018070	0.017990	0.017859	0.017991
	SD	0.000004	0.000193	0.000687	0.000148	0.000082	0.000008	0.000066	0.000042	0.000028	0.000048	0.000021	0.000188	0.000135	0.000083	0.000134
50.0	**Max**	0.363785	0.363429	0.363785	0.363785	0.363613	0.006524	0.006702	0.006708	0.006519	0.006698	0.017938	0.018478	0.019121	0.017924	0.018440
	Min	0.363662	0.362607	0.347813	0.363667	0.363169	0.006484	0.006458	0.006483	0.006483	0.006469	0.017824	0.017771	0.017820	0.017822	0.017801
	Median	0.363785	0.362837	0.363006	0.363785	0.363445	0.006485	0.006570	0.006559	0.006484	0.006543	0.017827	0.018113	0.018048	0.017824	0.018006
	Mean	0.363761	0.362976	0.361390	0.363759	0.363420	0.006491	0.006584	0.006570	0.006489	0.006556	0.017843	0.018140	0.018184	0.017838	0.018040
	SD	0.000041	0.000317	0.004848	0.000040	0.000147	0.000012	0.000075	0.000083	0.000011	0.000060	0.000035	0.000217	0.000409	0.000031	0.000169
75.0	**Max**	0.363785	0.363288	0.363604	0.363785	0.363623	0.006523	0.006712	0.006844	0.006597	0.006591	0.017938	0.018489	0.018921	0.018149	0.018140
	Min	0.363662	0.362928	0.361744	0.363508	0.363371	0.006483	0.006461	0.006527	0.006480	0.006435	0.017821	0.017789	0.017950	0.017813	0.017701
	Median	0.363784	0.363119	0.363030	0.363785	0.363541	0.006485	0.006545	0.006588	0.006483	0.006527	0.017827	0.018021	0.018138	0.017822	0.017958
	Mean	0.363747	0.363115	0.362907	0.363757	0.363512	0.006496	0.006551	0.006619	0.006495	0.006516	0.017858	0.018041	0.018239	0.017855	0.017926
	SD	0.000052	0.000113	0.000636	0.000088	0.000092	0.000017	0.000069	0.000101	0.000036	0.000055	0.000048	0.000191	0.000305	0.000103	0.000154
100	**Max**	0.363785	0.363278	0.363453	0.363785	0.363517	0.006523	0.006677	0.006743	0.006538	0.006601	0.017938	0.018421	0.018623	0.018013	0.018164
	Min	0.362946	0.362468	0.362055	0.362984	0.363059	0.006485	0.006450	0.006540	0.006483	0.006431	0.017826	0.017764	0.018025	0.017821	0.017698
	Median	0.363758	0.363118	0.362832	0.363760	0.363404	0.006489	0.006554	0.006591	0.006487	0.006538	0.017838	0.018046	0.018163	0.017833	0.017987
	Mean	0.363659	0.363070	0.362793	0.363610	0.363376	0.006499	0.006560	0.006612	0.006494	0.006533	0.017872	0.018068	0.018225	0.017861	0.017979
	SD	0.000257	0.000231	0.000432	0.000313	0.000129	0.000018	0.000066	0.000068	0.000019	0.000046	0.000054	0.000190	0.000200	0.000063	0.000129

Table B4
Computational values of Sharpe ratio, Return, and Risk for RBPS model using proposed IWO approach compared with GA, SA, DE, and PSO for FTSE data for different RB% values

r%		Sharpe Ratio					Return					Risk				
12.5		**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**
	Max	0.292153	0.291784	0.292155	0.292163	0.292014	0.005151	0.005156	0.005153	0.005178	0.005164	0.017680	0.017696	0.017649	0.017949	0.017696
	Min	0.287803	0.290869	0.291957	0.285101	0.291631	0.005036	0.005116	0.005135	0.005016	0.005112	0.017498	0.017542	0.017578	0.017560	0.017530
	Median	0.291981	0.291549	0.292152	0.290255	0.291819	0.005139	0.005144	0.005140	0.005145	0.005129	0.017598	0.017652	0.017594	0.017652	0.017572
	Mean	0.291425	0.291449	0.292132	0.289550	0.291833	0.005131	0.005139	0.005140	0.005120	0.005135	0.017605	0.017634	0.017596	0.017684	0.017596
	SD	0.001377	0.000316	0.000062	0.002415	0.000121	0.000034	0.000015	0.000005	0.000058	0.000018	0.000050	0.000057	0.000020	0.000122	0.000060
17.5	**Max**	0.294222	0.293853	0.294227	0.294070	0.294044	0.005309	0.005345	0.005309	0.005426	0.005331	0.018055	0.018203	0.018050	0.018627	0.018131
	Min	0.292017	0.293236	0.294135	0.286702	0.293695	0.005225	0.005262	0.005285	0.005044	0.005294	0.017894	0.017908	0.017963	0.017593	0.018007

(*continued on next page*)

Table B4 (*continued*)

r%		Sharpe Ratio					Return					Risk				
	Median	0.294120	0.293672	0.294227	0.292497	0.293882	0.005285	0.005283	0.005286	0.005327	0.005315	0.017966	0.017991	0.017967	0.018181	0.018091
	Mean	0.293673	0.293672	0.294212	0.291903	0.293902	0.005276	0.005294	0.005289	0.005308	0.005313	0.017966	0.018025	0.017977	0.018182	0.018077
	SD	0.000842	0.000178	0.000031	0.002190	0.000108	0.000026	0.000030	0.000007	0.000102	0.000011	0.000044	0.000104	0.000027	0.000270	0.000038
22.5	**Max**	0.295285	0.295069	0.295301	0.295199	0.295088	0.005419	0.005463	0.005422	0.005600	0.005438	0.018359	0.018533	0.018490	0.019242	0.018430
	Min	0.294398	0.294294	0.293185	0.291040	0.294806	0.005387	0.005362	0.005402	0.005412	0.005398	0.018289	0.018171	0.018292	0.018335	0.018309
	Median	0.295193	0.294746	0.295293	0.293499	0.295004	0.005405	0.005393	0.005405	0.005458	0.005425	0.018328	0.018305	0.018303	0.018605	0.018390
	Mean	0.294988	0.294723	0.295066	0.293417	0.294979	0.005405	0.005408	0.005410	0.005470	0.005422	0.018323	0.018348	0.018333	0.018642	0.018379
	SD	0.000380	0.000245	0.000662	0.001415	0.000091	0.000010	0.000038	0.000009	0.000057	0.000015	0.000023	0.000134	0.000062	0.000269	0.000047
27.5	**Max**	0.295634	0.295383	0.295636	0.295526	0.295486	0.005545	0.005566	0.005672	0.005680	0.005542	0.018783	0.018878	0.019466	0.019504	0.018758
	Min	0.294355	0.294027	0.291403	0.291195	0.295197	0.005510	0.005468	0.005515	0.005521	0.005477	0.018638	0.018534	0.018655	0.018733	0.018544
	Median	0.295501	0.294915	0.295635	0.294172	0.295371	0.005516	0.005525	0.005516	0.005544	0.005508	0.018681	0.018765	0.018659	0.018863	0.018650
	Mean	0.295310	0.294882	0.295149	0.294075	0.295367	0.005521	0.005518	0.005541	0.005560	0.005511	0.018696	0.018711	0.018774	0.018909	0.018658
	SD	0.000414	0.000381	0.001326	0.001308	0.000083	0.000013	0.000031	0.000050	0.000048	0.000023	0.000053	0.000113	0.000253	0.000224	0.000076
35.0	**Max**	0.295635	0.295430	0.295636	0.295636	0.295525	0.005583	0.005618	0.005716	0.005536	0.005588	0.018963	0.019060	0.019508	0.018778	0.018922
	Min	0.294399	0.294250	0.292986	0.294749	0.295101	0.005518	0.005510	0.005508	0.005516	0.005494	0.018666	0.018691	0.018657	0.018657	0.018602
	Median	0.295283	0.294723	0.295575	0.295635	0.295333	0.005535	0.005564	0.005535	0.005518	0.005563	0.018731	0.018890	0.018728	0.018665	0.018848
	Mean	0.295169	0.294712	0.295216	0.295529	0.295306	0.005539	0.005563	0.005549	0.005522	0.005555	0.018765	0.018875	0.018797	0.018685	0.018812
	SD	0.000458	0.000356	0.000827	0.000275	0.000156	0.000022	0.000034	0.000061	0.000008	0.000028	0.000096	0.000122	0.000258	0.000041	0.000101
50.0	**Max**	0.295586	0.295512	0.295636	0.295636	0.295512	0.005621	0.005632	0.005611	0.005538	0.005563	0.019079	0.019081	0.019063	0.018739	0.018830
	Min	0.294056	0.294800	0.294315	0.295526	0.295133	0.005518	0.005479	0.005516	0.005516	0.005473	0.018668	0.018555	0.018657	0.018657	0.018527
	Median	0.294767	0.295082	0.295575	0.295635	0.295434	0.005544	0.005582	0.005536	0.005517	0.005521	0.018805	0.018912	0.018729	0.018661	0.018691
	Mean	0.294883	0.295082	0.295346	0.295611	0.295398	0.005557	0.005570	0.005542	0.005522	0.005520	0.018846	0.018876	0.018764	0.018680	0.018685
	SD	0.000533	0.000222	0.000462	0.000041	0.000114	0.000037	0.000054	0.000034	0.000009	0.000026	0.000147	0.000185	0.000143	0.000035	0.000090
75.0	**Max**	0.295580	0.295353	0.295636	0.295636	0.295516	0.005604	0.005657	0.005603	0.005542	0.005593	0.019068	0.019222	0.018987	0.018782	0.018945
	Min	0.291529	0.294295	0.295129	0.295059	0.294738	0.005510	0.005524	0.005516	0.005516	0.005515	0.018692	0.018731	0.018657	0.018657	0.018671
	Median	0.295011	0.294903	0.295610	0.295636	0.295371	0.005540	0.005584	0.005519	0.005516	0.005547	0.018817	0.018953	0.018669	0.018658	0.018782
	Mean	0.294525	0.294792	0.295525	0.295569	0.295276	0.005546	0.005587	0.005534	0.005523	0.005550	0.018830	0.018951	0.018727	0.018684	0.018797
	SD	0.001312	0.000377	0.000189	0.000180	0.000246	0.000026	0.000042	0.000030	0.000011	0.000024	0.000125	0.000161	0.000113	0.000045	0.000088
100	**Max**	0.295526	0.295297	0.295636	0.295636	0.295426	0.005611	0.005700	0.005572	0.005572	0.005608	0.019057	0.019369	0.018876	0.018876	0.019002
	Min	0.293861	0.294189	0.295059	0.294461	0.295129	0.005530	0.005508	0.005516	0.005511	0.005497	0.018712	0.018656	0.018657	0.018657	0.018609
	Median	0.295085	0.295031	0.295575	0.295635	0.295368	0.005559	0.005598	0.005536	0.005518	0.005548	0.018847	0.018987	0.018730	0.018665	0.018787
	Mean	0.295037	0.294886	0.295486	0.295463	0.295326	0.005565	0.005594	0.005538	0.005525	0.005547	0.018861	0.018969	0.018742	0.018701	0.018783
	SD	0.000531	0.000402	0.000193	0.000375	0.000088	0.000028	0.000055	0.000016	0.000018	0.000034	0.000124	0.000207	0.000064	0.000068	0.000118

Table B5

Computational values of Sharpe ratio, Return, and Risk for RBPS model using proposed IWO approach compared with GA, SA, DE, and PSO for HangSeng data for different RB% values

r%		Sharpe Ratio					Return					Risk				
12.5		**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**
	Max	0.182210	0.182552	0.182512	0.182514	0.182550	0.005274	0.005352	0.005337	0.005628	0.005388	0.028962	0.029333	0.029244	0.030994	0.029525
	Min	0.181791	0.182304	0.182475	0.181585	0.182433	0.005211	0.005289	0.005315	0.005312	0.005315	0.028663	0.028980	0.029124	0.029117	0.029120
	Median	0.182138	0.182513	0.182496	0.182423	0.182516	0.005241	0.005329	0.005321	0.005383	0.005356	0.028768	0.029195	0.029159	0.029506	0.029343
	Mean	0.182075	0.182490	0.182495	0.182348	0.182510	0.005241	0.005328	0.005324	0.005396	0.005354	0.028787	0.029194	0.029174	0.029593	0.029334
	SD	0.000157	0.000073	0.000011	0.000272	0.000038	0.000018	0.000018	0.000008	0.000088	0.000024	0.000093	0.000102	0.000043	0.000524	0.000135
17.5	**Max**	0.193901	0.193996	0.194011	0.194001	0.194008	0.005743	0.005818	0.005819	0.005984	0.005833	0.029617	0.029993	0.029996	0.030894	0.030071
	Min	0.193420	0.193858	0.194005	0.193361	0.193984	0.005651	0.005725	0.005804	0.005664	0.005804	0.029182	0.029529	0.029915	0.029291	0.029916
	Median	0.193759	0.193977	0.194008	0.193894	0.193996	0.005683	0.005792	0.005809	0.005825	0.005820	0.029332	0.029856	0.029942	0.030038	0.030000
	Mean	0.193747	0.193962	0.194008	0.193822	0.193997	0.005687	0.005785	0.005810	0.005827	0.005819	0.029353	0.029825	0.029949	0.030064	0.029997
	SD	0.000141	0.000041	0.000001	0.000187	0.000009	0.000032	0.000030	0.000004	0.000099	0.000009	0.000149	0.000150	0.000023	0.000505	0.000049
22.5	**Max**	0.202107	0.202392	0.202393	0.202394	0.202390	0.006236	0.006344	0.006283	0.006404	0.006310	0.030865	0.031359	0.031045	0.031658	0.031180
	Min	0.201548	0.202310	0.202388	0.202209	0.202320	0.006145	0.006231	0.006268	0.006259	0.006252	0.030427	0.030792	0.030972	0.030932	0.030900
	Median	0.201962	0.202375	0.202392	0.202279	0.202379	0.006174	0.006273	0.006279	0.006368	0.006275	0.030575	0.030999	0.031022	0.031483	0.031008
	Mean	0.201933	0.202367	0.202391	0.202292	0.202367	0.006176	0.006278	0.006278	0.006349	0.006280	0.030583	0.031022	0.031018	0.031384	0.031032
	SD	0.000167	0.000027	0.000002	0.000057	0.000027	0.000031	0.000032	0.000005	0.000052	0.000020	0.000143	0.000159	0.000025	0.000265	0.000095
27.5	**Max**	0.206852	0.207144	0.207146	0.207146	0.207145	0.006622	0.006852	0.006859	0.006862	0.006862	0.032013	0.033078	0.033112	0.033127	0.033125
	Min	0.206299	0.207076	0.207146	0.207143	0.207143	0.006461	0.006757	0.006857	0.006850	0.006843	0.031317	0.032631	0.033104	0.033068	0.033036
	Median	0.206636	0.207129	0.207146	0.207145	0.207145	0.006554	0.006812	0.006859	0.006858	0.006859	0.031717	0.032886	0.033111	0.033105	0.033111
	Mean	0.206640	0.207120	0.207146	0.207145	0.207145	0.006553	0.006806	0.006859	0.006857	0.006858	0.031713	0.032859	0.033111	0.033102	0.033105
	SD	0.000160	0.000024	0.000000	0.000001	0.000001	0.000051	0.000034	0.000000	0.000004	0.000005	0.000222	0.000160	0.000002	0.000018	0.000025
35.0	**Max**	0.210207	0.210207	0.210207	0.210207	0.210207	0.007022	0.007029	0.007021	0.007022	0.007022	0.033407	0.033438	0.033401	0.033407	0.033407
	Min	0.210198	0.210201	0.210207	0.210207	0.210207	0.007010	0.007011	0.007021	0.007019	0.007020	0.033350	0.033352	0.033400	0.033392	0.033398
	Median	0.210205	0.210205	0.210207	0.210207	0.210207	0.007015	0.007018	0.007021	0.007021	0.007022	0.033372	0.033386	0.033401	0.033401	0.033403
	Mean	0.210204	0.210205	0.210207	0.210207	0.210207	0.007015	0.007019	0.007021	0.007021	0.007021	0.033374	0.033392	0.033401	0.033401	0.033402
	SD	0.000003	0.000002	0.000000	0.000000	0.000000	0.000004	0.000006	0.000000	0.000001	0.000001	0.000019	0.000028	0.000000	0.000006	0.000003
50.0	**Max**	0.210442	0.210441	0.210442	0.210442	0.210442	0.007106	0.007111	0.007106	0.007106	0.007106	0.033767	0.033792	0.033767	0.033767	0.033768
	Min	0.210442	0.210433	0.210442	0.210442	0.210442	0.007106	0.007096	0.007106	0.007106	0.007106	0.033767	0.033719	0.033767	0.033767	0.033767
	Median	0.210442	0.210439	0.210442	0.210442	0.210442	0.007106	0.007106	0.007106	0.007106	0.007106	0.033767	0.033768	0.033767	0.033767	0.033767
	Mean	0.210442	0.210438	0.210442	0.210442	0.210442	0.007106	0.007105	0.007106	0.007106	0.007106	0.033767	0.033764	0.033767	0.033767	0.033767
	SD	0.000000	0.000003	0.000000	0.000000	0.000000	0.000000	0.000005	0.000000	0.000000	0.000000	0.000000	0.000022	0.000000	0.000000	0.000000
75.0	**Max**	0.210442	0.210441	0.210442	0.210442	0.210442	0.007106	0.007126	0.007106	0.007106	0.007106	0.033767	0.033864	0.033767	0.033767	0.033768
	Min	0.210442	0.210433	0.210442	0.210442	0.210442	0.007106	0.007096	0.007106	0.007106	0.007106	0.033767	0.033720	0.033767	0.033767	0.033767

(*continued on next page*)

Table B5 (*continued*)

	Median	0.210442	0.210439	0.210442	0.210442	0.210442	0.007106	0.007108	0.007106	0.007106	0.007106	0.033767	0.033778	0.033767	0.033767	0.033767
	Mean	0.210442	0.210438	0.210442	0.210442	0.210442	0.007106	0.007109	0.007106	0.007106	0.007106	0.033767	0.033781	0.033767	0.033767	0.033767
	SD	0.000000	0.000003	0.000000	0.000000	0.000000	0.000000	0.000009	0.000000	0.000000	0.000000	0.000000	0.000042	0.000000	0.000000	0.000000
100	**Max**	0.210442	0.210441	0.210442	0.210442	0.210442	0.007106	0.007113	0.007106	0.007106	0.007106	0.033767	0.033799	0.033767	0.033767	0.033768
	Min	0.210442	0.210433	0.210442	0.210442	0.210442	0.007106	0.007089	0.007106	0.007106	0.007106	0.033767	0.033687	0.033767	0.033767	0.033767
	Median	0.210442	0.210439	0.210442	0.210442	0.210442	0.007106	0.007108	0.007106	0.007106	0.007106	0.033767	0.033777	0.033767	0.033767	0.033767
	Mean	0.210442	0.210439	0.210442	0.210442	0.210442	0.007106	0.007105	0.007106	0.007106	0.007106	0.033767	0.033763	0.033767	0.033767	0.033767
	SD	0.000000	0.000002	0.000000	0.000000	0.000000	0.000000	0.000008	0.000000	0.000000	0.000000	0.000000	0.000037	0.000000	0.000000	0.000000

Table B6
Computational values of Sharpe ratio, Return, and Risk for RBPS model using proposed IWO approach compared with GA, SA, DE, and PSO for Nikkei data for different RB% values

r%		**Sharpe Ratio**					**Return**					**Risk**				
		GA	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**
12.5	**Max**	0.132679	0.126640	0.132920	0.129393	0.130365	0.003366	0.003143	0.003380	0.003248	0.003306	0.025367	0.025896	0.025999	0.026331	0.025814
	Min	0.129022	0.118895	0.128848	0.107437	0.124089	0.003184	0.002880	0.003231	0.002740	0.003182	0.024515	0.024227	0.025039	0.024815	0.024912
	Median	0.132387	0.121534	0.131424	0.120963	0.128135	0.003306	0.003084	0.003367	0.003061	0.003217	0.025026	0.024933	0.025424	0.025507	0.025394
	Mean	0.131865	0.122052	0.131074	0.119932	0.127467	0.003301	0.003052	0.003336	0.003057	0.003234	0.025031	0.025007	0.025455	0.025494	0.025375
	SD	0.001295	0.002701	0.001963	0.006200	0.002309	0.000055	0.000094	0.000056	0.000144	0.000042	0.000241	0.000547	0.000263	0.000424	0.000359
17.5	**Max**	0.136091	0.127047	0.136125	0.136102	0.134899	0.003464	0.003274	0.003468	0.003538	0.003438	0.025721	0.026209	0.026150	0.026525	0.026020
	Min	0.130105	0.118088	0.128963	0.123559	0.130698	0.003346	0.003045	0.003372	0.003227	0.003277	0.024931	0.024457	0.025038	0.024924	0.024939
	Median	0.136015	0.125141	0.136104	0.134826	0.132417	0.003397	0.003104	0.003409	0.003427	0.003372	0.024977	0.025318	0.025041	0.025551	0.025435
	Mean	0.135380	0.123779	0.135278	0.132341	0.132485	0.003398	0.003130	0.003417	0.003406	0.003367	0.025102	0.025292	0.025262	0.025748	0.025410
	SD	0.001863	0.003317	0.002233	0.004634	0.001109	0.000028	0.000076	0.000029	0.000087	0.000056	0.000291	0.000612	0.000389	0.000588	0.000332
22.5	**Max**	0.137852	0.130666	0.137882	0.137882	0.135959	0.003461	0.003330	0.003464	0.003538	0.003502	0.025539	0.026812	0.025196	0.026105	0.025825
	Min	0.131410	0.120238	0.136934	0.128470	0.133505	0.003356	0.003101	0.003419	0.003195	0.003348	0.024680	0.024319	0.024795	0.024794	0.024846
	Median	0.137802	0.124876	0.137680	0.136616	0.134886	0.003407	0.003224	0.003434	0.003448	0.003411	0.024737	0.025880	0.024995	0.025195	0.025384
	Mean	0.137045	0.125095	0.137626	0.135111	0.134753	0.003411	0.003215	0.003440	0.003426	0.003414	0.024890	0.025717	0.024995	0.025362	0.025332
	SD	0.002003	0.003322	0.000315	0.003277	0.001021	0.000027	0.000067	0.000023	0.000089	0.000048	0.000289	0.000885	0.000211	0.000467	0.000319
27.5	**Max**	0.138896	0.132484	0.138901	0.138901	0.135772	0.003479	0.003395	0.003531	0.003530	0.003474	0.025138	0.026743	0.025977	0.027085	0.025783
	Min	0.134551	0.122712	0.131836	0.129945	0.131480	0.003382	0.003006	0.003371	0.003402	0.003261	0.024612	0.024482	0.024664	0.024687	0.024805
	Median	0.138888	0.125658	0.138744	0.134275	0.134555	0.003423	0.003266	0.003430	0.003455	0.003403	0.024651	0.025660	0.024839	0.025811	0.025229
	Mean	0.138343	0.126216	0.137325	0.134466	0.134327	0.003427	0.003241	0.003439	0.003458	0.003395	0.024774	0.025677	0.025053	0.025740	0.025274
	SD	0.001359	0.003059	0.002793	0.003981	0.001220	0.000025	0.000122	0.000041	0.000048	0.000059	0.000218	0.000813	0.000490	0.000849	0.000289
35.0	**Max**	0.139361	0.134481	0.139361	0.139338	0.137626	0.003458	0.003441	0.003475	0.003517	0.003479	0.024852	0.027534	0.025297	0.026943	0.025946
	Min	0.139131	0.121525	0.133245	0.127227	0.133261	0.003429	0.003073	0.003371	0.003313	0.003300	0.024605	0.024210	0.024622	0.024610	0.024656
	Median	0.139361	0.128533	0.139222	0.132473	0.135321	0.003430	0.003296	0.003441	0.003451	0.003403	0.024609	0.025039	0.024833	0.025714	0.024937
	Mean	0.139310	0.128924	0.138564	0.133280	0.135371	0.003438	0.003291	0.003439	0.003438	0.003397	0.024675	0.025549	0.024823	0.025807	0.025098
	SD	0.000085	0.004437	0.001887	0.003859	0.001305	0.000013	0.000111	0.000028	0.000067	0.000057	0.000109	0.001169	0.000217	0.000708	0.000398
50.0	**Max**	0.139380	0.135202	0.139380	0.139380	0.138079	0.003453	0.003478	0.003501	0.003458	0.003502	0.024792	0.028034	0.025314	0.026318	0.025942
	Min	0.138157	0.122807	0.133233	0.131394	0.131250	0.003387	0.003200	0.003357	0.003403	0.003316	0.024385	0.024674	0.024599	0.024599	0.024852
	Median	0.139265	0.125953	0.139016	0.139024	0.135330	0.003430	0.003350	0.003453	0.003429	0.003427	0.024611	0.026333	0.024865	0.024702	0.025284
	Mean	0.139104	0.126672	0.138398	0.137776	0.135165	0.003427	0.003361	0.003450	0.003435	0.003414	0.024639	0.026551	0.024930	0.024941	0.025261
	SD	0.000408	0.003707	0.001862	0.002889	0.002084	0.000023	0.000092	0.000043	0.000019	0.000054	0.000124	0.001117	0.000287	0.000560	0.000330
75.0	**Max**	0.139380	0.133256	0.139380	0.139360	0.136827	0.003501	0.003630	0.003453	0.003501	0.003452	0.025286	0.030023	0.025527	0.026186	0.026148
	Min	0.134684	0.119510	0.132988	0.131629	0.129630	0.003378	0.003226	0.003370	0.003370	0.003251	0.024599	0.024808	0.024599	0.024599	0.024614
	Median	0.139265	0.123651	0.139265	0.138553	0.135339	0.003440	0.003440	0.003429	0.003452	0.003411	0.024787	0.027282	0.024792	0.025055	0.025332
	Mean	0.138685	0.125233	0.137541	0.137039	0.134187	0.003445	0.003426	0.003423	0.003446	0.003396	0.024845	0.027426	0.024895	0.025154	0.025314
	SD	0.001454	0.005668	0.002852	0.002929	0.002550	0.000036	0.000149	0.000033	0.000040	0.000059	0.000274	0.001978	0.000318	0.000457	0.000485
100	**Max**	0.139380	0.130138	0.139380	0.139380	0.136579	0.003453	0.003780	0.003481	0.003501	0.003471	0.024792	0.034988	0.026101	0.026101	0.025857
	Min	0.138764	0.108042	0.132413	0.132413	0.130196	0.003401	0.003174	0.003375	0.003356	0.003248	0.024440	0.025497	0.024611	0.024611	0.024784
	Median	0.139323	0.120970	0.138408	0.138282	0.134608	0.003430	0.003435	0.003446	0.003453	0.003405	0.024611	0.027634	0.024887	0.025242	0.025401
	Mean	0.139261	0.119319	0.136966	0.136422	0.133956	0.003431	0.003494	0.003440	0.003440	0.003395	0.024640	0.029467	0.025121	0.025223	0.025344
	SD	0.000191	0.007895	0.002655	0.003022	0.001974	0.000018	0.000203	0.000029	0.000056	0.000065	0.000116	0.003420	0.000496	0.000416	0.000376

Table B7
Computational values of Sharpe ratio, Return, and Risk for RBPS model using proposed IWO approach compared with GA, SA, DE, and PSO for S&P data for different RB% values

***r*%**		**Sharpe Ratio**					**Return**					**Risk**				
		GA	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**	**GA**	**SA**	**DE**	**PSO**	**IWO**
	Max	0.319592	0.318888	0.319543	0.318729	0.319331	0.005227	0.005277	0.005247	0.005311	0.005260	0.016371	0.016581	0.016477	0.016703	0.016486
	Min	0.318726	0.317676	0.318456	0.313896	0.318880	0.005155	0.005133	0.005204	0.005108	0.005148	0.016174	0.016140	0.016292	0.016222	0.016141
12.5	**Median**	0.319540	0.318389	0.319305	0.317680	0.319092	0.005187	0.005202	0.005223	0.005225	0.005208	0.016231	0.016322	0.016365	0.016455	0.016313
	Mean	0.319393	0.318357	0.319200	0.317114	0.319108	0.005193	0.005196	0.005222	0.005221	0.005204	0.016258	0.016321	0.016361	0.016463	0.016309
	SD	0.000291	0.000380	0.000359	0.001637	0.000133	0.000021	0.000041	0.000018	0.000070	0.000034	0.000062	0.000123	0.000064	0.000173	0.000106
17.5	**Max**	0.319683	0.318932	0.319684	0.319684	0.319431	0.005261	0.005340	0.005263	0.005281	0.005279	0.016491	0.016837	0.016495	0.016560	0.016553

(*continued on next page*)

Table B7 (*continued*)

r%		Sharpe Ratio					Return					Risk				
		GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO	GA	SA	DE	PSO	IWO
	Max	0.319592	0.318888	0.319543	0.318729	0.319331	0.005227	0.005277	0.005247	0.005311	0.005260	0.016371	0.016581	0.016477	0.016703	0.016486
	Min	0.318726	0.317676	0.318456	0.313896	0.318880	0.005155	0.005133	0.005204	0.005108	0.005148	0.016174	0.016140	0.016292	0.016222	0.016141
	Min	0.318579	0.317160	0.319041	0.317961	0.318925	0.005222	0.005243	0.005222	0.005222	0.005191	0.016336	0.016445	0.016335	0.016335	0.016260
	Median	0.319553	0.318804	0.319553	0.319440	0.319327	0.005231	0.005301	0.005257	0.005226	0.005231	0.016384	0.016626	0.016451	0.016375	0.016381
	Mean	0.319284	0.318481	0.319492	0.319260	0.319299	0.005236	0.005293	0.005244	0.005238	0.005229	0.016398	0.016620	0.016414	0.016406	0.016375
	SD	0.000462	0.000583	0.000262	0.000568	0.000142	0.000014	0.000040	0.000018	0.000022	0.000031	0.000052	0.000138	0.000068	0.000081	0.000100
22.5	**Max**	0.319684	0.319046	0.319553	0.319683	0.319464	0.005277	0.005337	0.005274	0.005257	0.005276	0.016521	0.016763	0.016507	0.016462	0.016523
	Min	0.319088	0.318342	0.319495	0.318412	0.319030	0.005222	0.005238	0.005238	0.005196	0.005166	0.016335	0.016419	0.016395	0.016296	0.016193
	Median	0.319639	0.318545	0.319502	0.319500	0.319310	0.005235	0.005273	0.005257	0.005238	0.005235	0.016377	0.016554	0.016452	0.016399	0.016394
	Mean	0.319532	0.318576	0.319523	0.319238	0.319283	0.005244	0.005275	0.005259	0.005236	0.005233	0.016412	0.016557	0.016460	0.016401	0.016389
	SD	0.000200	0.000230	0.000029	0.000490	0.000120	0.000021	0.000034	0.000013	0.000020	0.000031	0.000075	0.000111	0.000040	0.000056	0.000093
27.5	**Max**	0.319684	0.319135	0.319684	0.319683	0.319443	0.005248	0.005333	0.005339	0.005317	0.005274	0.016425	0.016732	0.016753	0.016686	0.016525
	Min	0.319164	0.318180	0.318702	0.317272	0.319169	0.005222	0.005189	0.005222	0.005184	0.005190	0.016336	0.016292	0.016336	0.016272	0.016249
	Median	0.319536	0.318733	0.319495	0.319419	0.319299	0.005229	0.005252	0.005236	0.005257	0.005236	0.016382	0.016459	0.016398	0.016458	0.016402
	Mean	0.319466	0.318704	0.319368	0.319030	0.319288	0.005230	0.005255	0.005260	0.005256	0.005234	0.016370	0.016489	0.016471	0.016474	0.016394
	SD	0.000232	0.000303	0.000330	0.000779	0.000090	0.000009	0.000047	0.000039	0.000041	0.000029	0.000035	0.000152	0.000139	0.000144	0.000094
35.0	**Max**	0.319684	0.318989	0.319639	0.319684	0.319422	0.005260	0.005344	0.005317	0.005292	0.005282	0.016465	0.016793	0.016677	0.016620	0.016565
	Min	0.318458	0.317885	0.318368	0.318425	0.318842	0.005185	0.005182	0.005200	0.005189	0.005208	0.016281	0.016302	0.016323	0.016288	0.016317
	Median	0.319639	0.318585	0.319159	0.319298	0.319288	0.005224	0.005257	0.005235	0.005237	0.005234	0.016342	0.016502	0.016377	0.016401	0.016393
	Mean	0.319417	0.318571	0.319081	0.319224	0.319243	0.005228	0.005251	0.005242	0.005239	0.005244	0.016366	0.016483	0.016427	0.016412	0.016425
	SD	0.000431	0.000350	0.000463	0.000461	0.000171	0.000020	0.000049	0.000036	0.000029	0.000026	0.000058	0.000156	0.000112	0.000098	0.000088
50.0	**Max**	0.319684	0.318953	0.319684	0.319553	0.319422	0.005274	0.005291	0.005292	0.005307	0.005276	0.016506	0.016608	0.016574	0.016660	0.016533
	Min	0.319495	0.317681	0.318694	0.318261	0.318949	0.005221	0.005153	0.005222	0.005229	0.005195	0.016332	0.016157	0.016335	0.016382	0.016279
	Median	0.319639	0.318374	0.319500	0.319299	0.319298	0.005235	0.005253	0.005238	0.005257	0.005236	0.016377	0.016509	0.016421	0.016452	0.016397
	Mean	0.319612	0.318344	0.319423	0.319108	0.319235	0.005242	0.005245	0.005249	0.005262	0.005228	0.016400	0.016475	0.016434	0.016491	0.016376
	SD	0.000078	0.000386	0.000300	0.000453	0.000150	0.000021	0.000043	0.000024	0.000027	0.000025	0.000070	0.000143	0.000080	0.000103	0.000080
75.0	**Max**	0.319683	0.319136	0.319684	0.319683	0.319382	0.005257	0.005313	0.005331	0.005344	0.005295	0.016454	0.016696	0.016725	0.016895	0.016589
	Min	0.318779	0.318208	0.318711	0.316337	0.318976	0.005222	0.005173	0.005222	0.005219	0.005186	0.016334	0.016246	0.016335	0.016325	0.016257
	Median	0.319548	0.318559	0.319538	0.319499	0.319205	0.005242	0.005238	0.005235	0.005243	0.005231	0.016410	0.016444	0.016383	0.016414	0.016384
	Mean	0.319442	0.318639	0.319471	0.319128	0.319197	0.005239	0.005246	0.005253	0.005250	0.005231	0.016400	0.016463	0.016443	0.016452	0.016389
	SD	0.000297	0.000326	0.000300	0.001062	0.000137	0.000014	0.000041	0.000035	0.000037	0.000030	0.000052	0.000134	0.000123	0.000171	0.000095
100	**Max**	0.319684	0.318912	0.319639	0.319639	0.319399	0.005274	0.005301	0.005297	0.005346	0.005253	0.016506	0.016637	0.016627	0.016800	0.016451
	Min	0.319299	0.318115	0.318271	0.318190	0.318868	0.005222	0.005177	0.005235	0.005235	0.005159	0.016336	0.016248	0.016378	0.016377	0.016167
	Median	0.319553	0.318586	0.319146	0.319394	0.319280	0.005256	0.005243	0.005274	0.005269	0.005226	0.016447	0.016470	0.016507	0.016496	0.016369
	Mean	0.319497	0.318556	0.319104	0.319178	0.319231	0.005249	0.005245	0.005268	0.005273	0.005220	0.016429	0.016465	0.016508	0.016520	0.016351
	SD	0.000138	0.000282	0.000486	0.000498	0.000150	0.000017	0.000041	0.000022	0.000037	0.000027	0.000055	0.000131	0.000083	0.000140	0.000085

Data availability

Data generated during this study can be provided only on request to the corresponding author. However, the benchmark datasets of foreign stock markets used in this paper can be accessed at http://people.brunel.ac.uk/mastjjb/jeb/orlib/portinfo.html.

APPENDIX

The appendix of the paper consists of two sections: Appendix A and Appendix B. Appendix A summarizes the mathematical and conceptual background of genetic algorithm (GA), simulated annealing (SA), differential evolution (DE), and particle swarm optimization (PSO) algorithms. Appendix B contains all tables that present the results. These tables present the computational values of the Sharpe ratio, Return, and Risk for the RBPS model using the proposed IWO approach compared with GA, SA, DE, and PSO for BSE, NSE, DAX, FTSE, Hang Seng, S&P, and Nikkei datasets for different RB% values.

References

[1] A. Salo, M. Doumpos, J. Liesiö, C. Zopounidis, Fifty years of portfolio optimization, Eur. J. Oper. Res. 318 (2024) 1–18, https://doi.org/10.1016/j.ejor.2023.12.031.

[2] H.M. Markowitz, Portfolio selection: efficient diversification of investments, Yale university press, 2008.

[3] H. Markowitz, Portafolio Selection, J. Financ. 7 (1952) 77–91. ⟨http://www.performance-measurement.org/Markowitz1952.pdf%0Ahttp://links.jstor.org/sici?sici=0022-1082%28195203%297%3A1%3C77%3APS%3E2.0.CO%3B2-1⟩.

[4] W.F. Sharpe, Portfolio Analysis, J. Financ. Quant. Anal. 2 (1967) 76, https://doi.org/10.2307/2329895.

[5] E.K.W. Leow, B.P. Nguyen, M.C.H. Chua, Robo-advisor using genetic algorithm and BERT sentiments from tweets for hybrid portfolio optimisation, Expert Syst. Appl. 179 (2021) 115060, https://doi.org/10.1016/j.eswa.2021.115060.

[6] J. Samuel Baixauli-Soler, E. Alfaro-Cid, M.O. Fernandez-Blanco, A naïve approach to speed up portfolio optimization problem using a multiobjective genetic algorithm, Investig. Eur. Dir. Y. Econ. La Empres 18 (2012) 126–131, https://doi.org/10.1016/S1135-2523(12)70002-3.

[7] H. Babazadeh, A. Esfahanipour, A novel multi period mean-VaR portfolio optimization model considering practical constraints and transaction cost, J. Comput. Appl. Math. 361 (2019) 313–342, https://doi.org/10.1016/j.cam.2018.10.039.

[8] G.A. Vijayalakshmi Pai, T. Michel, Metaheuristic optimization of Risk Budgeted global asset allocation portfolios, in: World Congr. Inf. Commun. Technol, 2011, IEEE, 2011, pp. 154–159, https://doi.org/10.1109/WICT.2011.6141235.

[9] M. Shahid, Z. Ashraf, M. Shamim, M.S. Ansari, Solving constrained portfolio optimization model using stochastic fractal search approach, Int. J. Intell. Comput. Cyber 16 (2023) 223–249, https://doi.org/10.1108/IJICC-03-2022-0086.

[10] G. di Tollo, F. Lardeux, R. Pesenti, M. Petris, A SAT encoding for the portfolio selection problem, Soft Comput. (2023), https://doi.org/10.1007/s00500-023-09484-z.

[11] C.B. Kalayci, O. Ertenlice, M.A. Akbay, A comprehensive review of deterministic models and applications for mean-variance portfolio optimization, Expert Syst. Appl. 125 (2019) 345–368, https://doi.org/10.1016/j.eswa.2019.02.011.

[12] O. Ertenlice, C.B. Kalayci, A survey of swarm intelligence for portfolio optimization: Algorithms and applications, Swarm Evol. Comput. 39 (2018) 36–52, https://doi.org/10.1016/j.swevo.2018.01.009.

[13] K. Erwin, A. Engelbrecht, Meta-heuristics for portfolio optimization, Soft Comput. 27 (2023) 19045–19073, https://doi.org/10.1007/s00500-023-08177-x.

[14] P. Xidonas, R. Steuer, C. Hassapis, Robust portfolio optimization: a categorized bibliographic review, Ann. Oper. Res. 292 (2020) 533–552, https://doi.org/10.1007/s10479-020-03630-8.

[15] B. Bruder, T. Roncalli, Managing Risk Exposures Using the Risk Budgeting Approach, SSRN Electron. J. (2012) 1–33, https://doi.org/10.2139/ssrn.2009778.

[16] M. Haugh, G. Iyengar, I. Song, A generalized risk budgeting approach to portfolio construction, J. Comput. Financ 21 (2017) 29–60, https://doi.org/10.21314/JCF.2017.329.

[17] R. Ji, M.A. Lejeune, Risk-budgeting multi-portfolio optimization with portfolio and marginal risk constraints, Ann. Oper. Res. 262 (2018) 547–578, https://doi.org/10.1007/s10479-015-2044-9.

[18] H.P. Ramos, M.B. Righi, P.C. Guedes, F.M. Müller, A comparison of risk measures for portfolio optimization with cardinality constraints, Expert Syst. Appl. 228 (2023) 120412, https://doi.org/10.1016/j.eswa.2023.120412.
[19] A.R. Mehrabian, C. Lucas, A novel numerical optimization algorithm inspired from weed colonization, Ecol. Inf. 1 (2006) 355–366, https://doi.org/10.1016/j.ecoinf.2006.07.003.
[20] Y.R. Naidu, A solution to common fixed point problems using a hybrid method of invasive weed optimization and jaya algorithm, Soft Comput. 28 (2024) 13389–13406, https://doi.org/10.1007/s00500-024-10315-y.
[21] A. Ibrahim, F. Anayi, M. Packianather, O.A. Alomari, New Hybrid Invasive Weed Optimization and Machine Learning Approach for Fault Detection, Energies 15 (2022) 1488, https://doi.org/10.3390/en15041488.
[22] A.K. Mishra, I.K. Gupta, T.D. Diwan, S. Srivastava, Cervical precancerous lesion classification using quantum invasive weed optimization with deep learning on biomedical pap smear images, Expert Syst. 41 (2024), https://doi.org/10.1111/exsy.13308.
[23] A. Hajnoori, M. Amiri, A. Alimi, Forecasting stock price using grey-fuzzy technique and portfolio optimization by invasive weed optimization algorithm, Decis. Sci. Lett. (2013) 175–184, https://doi.org/10.5267/j.dsl.2013.04.004.
[24] A. Rezaei Pouya, M. Solimanpur, M. Jahangoshai Rezaee, Solving multi-objective portfolio optimization problem using invasive weed optimization, Swarm Evol. Comput. 28 (2016) 42–57, https://doi.org/10.1016/j.swevo.2016.01.001.
[25] A. Goli, H. Khademi Zare, R. Tavakkoli-Moghaddam, A. Sadeghieh, Application of robust optimization for a product portfolio problem using an invasive weed optimization algorithm, Numer. Algebr Control Optim. 9 (2019) 187–209, https://doi.org/10.3934/naco.2019014.
[26] P. Swain, A.K. Ojha, Portfolio Optimization Using Particle Swarm Optimization and Invasive Weed Optimization, in: 2020: pp. 307–314. https://doi.org/10.1007/978-981-15-3215-3_30.
[27] M. Shahid, M.S. Ansari, M. Shamim, Z. Ashraf, A Risk-Budgeted Portfolio Selection Strategy Using Invasive Weed Optimization, in: 2022: pp. 363–371. https://doi.org/10.1007/978-981-16-3802-2_30.
[28] M. Shahid, Z. Ashraf, M. Shamim, M.S. Ansari, F. Ahmad, A Risk-Budgeted Portfolio Selection Strategy Using Novel Metaheuristic Optimization Approach, in: 2022: pp. 193–204. https://doi.org/10.1007/978-981-16-9416-5_14.
[29] T.-J. Chang, N. Meade, J.E. Beasley, Y.M. Sharaiha, Heuristics for cardinality constrained portfolio optimisation, Comput. Oper. Res. 27 (2000) 1271–1302, https://doi.org/10.1016/S0305-0548(99)00074-X.
[30] Y. Crama, M. Schyns, Simulated annealing for complex portfolio selection problems, Eur. J. Oper. Res. 150 (2003) 546–571, https://doi.org/10.1016/S0377-2217(02)00784-1.
[31] K.P. Anagnostopoulos, G. Mamanis, The mean–variance cardinality constrained portfolio optimization problem: An experimental evaluation of five multiobjective evolutionary algorithms, Expert Syst. Appl. (2011), https://doi.org/10.1016/j.eswa.2011.04.233.
[32] L. Yu, S. Wang, F. Wen, K.K. Lai, Genetic algorithm-based multi-criteria project portfolio selection, Ann. Oper. Res. 197 (2012) 71–86, https://doi.org/10.1007/s10479-010-0819-6.
[33] D. Ardia, K. Boudt, P. Carl, K.M. Mullen, B.G. Peterson, Differential Evolution with DEoptim An Application to Non-Convex Portfolio Optimiza-tion, R. J. 3 (2011) 27–34. ⟨https://ssrn.com/abstract=1584905⟩.
[34] D. Cheong, Y.M. Kim, H.W. Byun, K.J. Oh, T.Y. Kim, Using genetic algorithm to support clustering-based portfolio optimization by investor information, Appl. Soft Comput. 61 (2017) 593–602, https://doi.org/10.1016/j.asoc.2017.08.042.
[35] S.S. Meghwani, M. Thakur, Multi-objective heuristic algorithms for practical portfolio optimization and rebalancing with transaction cost, Appl. Soft Comput. 67 (2018) 865–894, https://doi.org/10.1016/j.asoc.2017.09.025.
[36] R. Pal, T.D. Chaudhuri, S. Mukhopadhyay, Portfolio formation and optimization with continuous realignment: A suggested method for choosing the best portfolio of stocks using variable length NSGA-II, Expert Syst. Appl. 186 (2021) 115732, https://doi.org/10.1016/j.eswa.2021.115732.
[37] B. Li, X. Li, K.L. Teo, P. Zheng, A new uncertain random portfolio optimization model for complex systems with downside risks and diversification, Chaos Solitons Fractals 160 (2022) 112213, https://doi.org/10.1016/j.chaos.2022.112213.
[38] Y. Chen, L. Ye, R. Li, X. Zhao, A Multi-Period Constrained Multi-Objective Evolutionary Algorithm with Orthogonal Learning for Solving the Complex Carbon Neutral Stock Portfolio Optimization Model, J. Syst. Sci. Complex 36 (2023) 686–715, https://doi.org/10.1007/s11424-023-2406-3.
[39] T. Cura, Particle swarm optimization approach to portfolio optimization, Nonlinear Anal. Real. World Appl. 10 (2009) 2396–2406, https://doi.org/10.1016/j.nonrwa.2008.04.023.
[40] M. Corazza, G. Fasano, R. Gusso, Particle Swarm Optimization with Non-Smooth Penalty Reformulation for a Complex Portfolio Selection Problem, SSRN Electron. J. (2011), https://doi.org/10.2139/ssrn.1904873.
[41] H. Zhu, Y. Wang, K. Wang, Y. Chen, Particle Swarm Optimization (PSO) for the constrained portfolio optimization problem, Expert Syst. Appl. 38 (2011) 10161–10169, https://doi.org/10.1016/j.eswa.2011.02.075.
[42] H.R. Golmakani, M. Fazel, Constrained Portfolio Selection using Particle Swarm Optimization, Expert Syst. Appl. 38 (2011) 8327–8335, https://doi.org/10.1016/j.eswa.2011.01.020.
[43] G.-F. Deng, W.-T. Lin, C.-C. Lo, Markowitz-based portfolio selection with cardinality constraints using improved particle swarm optimization, Expert Syst. Appl. 39 (2012) 4558–4566, https://doi.org/10.1016/j.eswa.2011.09.129.
[44] Z. Wang, S. Liu, X. Kong, An Improved Artificial Bee Colony Algorithm for Portfolio Optimization Problem, Int. J. Adv. Comput. Technol. 3 (2011) 67–74, https://doi.org/10.4156/ijact.vol3.issue10.9.
[45] A.H.L. Chen, Y.-C. Liang, C.-C. Liu, An artificial bee colony algorithm for the cardinality-constrained portfolio optimization problems, in: 2012 IEEE Congr. Evol. Comput, IEEE, 2012, pp. 1–8, https://doi.org/10.1109/CEC.2012.6252920.
[46] K. Lwin, R. Qu, J. Zheng, Multi-objective Scatter Search with External Archive for Portfolio Optimization. in: Proc. 5th Int. Jt. Conf. Comput. Intell, SCITEPRESS - Science and and Technology Publications, 2013, pp. 111–119, https://doi.org/10.5220/0004552501110119.
[47] M. Tuba, N. Bacanin, Upgraded Firefly Algorithm for Portfolio Optimization Problem, in: 2014 UKSim-AMSS 16th Int. Conf. Comput. Model. Simul, IEEE, 2014, pp. 113–118, https://doi.org/10.1109/UKSim.2014.25.
[48] D. Suthiwong, M. Sodanil, Cardinality-constrained Portfolio optimization using an improved quick Artificial Bee Colony Algorithm, in: 2016 Int. Comput. Sci. Eng. Conf, IEEE, 2016, pp. 1–4, https://doi.org/10.1109/ICSEC.2016.7859943.
[49] G. di Tollo, Hybrid. Metaheuristic Portf. Sel. Comp. Exact. solver Search Space Anal. (2015) 579–588, https://doi.org/10.15439/2015F7.
[50] M. Mansourinia, A. Momeni, Cardinality Constrained Portfolio Optimization Using Bee Colony Algorithm (Case Study: Tehran Stock Exchange), (2018). https://doi.org/10.20944/preprints201812.0103.v1.
[51] M. Corazza, G. di Tollo, G. Fasano, R. Pesenti, A novel hybrid PSO-based metaheuristic for costly portfolio selection problems, Ann. Oper. Res. 304 (2021) 109–137, https://doi.org/10.1007/s10479-021-04075-3.
[52] A.P. Sahala, G.F. Hertono, B.D. Handari, Implement. Improv. Quick Artif. Bee Colony Algorithm Portf. Optim. Probl. constraints (2020) 030008, https://doi.org/10.1063/5.0008145.
[53] M. Rezaei, H. Nezamabadi-Pour, A taxonomy of literature reviews and experimental study of deepreinforcement learning in portfolio management, Artif. Intell. Rev. 58 (2025) 94, https://doi.org/10.1007/s10462-024-11066-w.
[54] P. Fu, H. Yang, W. Qian, El.I. Mohamed, W.A.J. Almohri, H.M. Alshanbari, Financial engineering and the digital economy: The implementations of machine learning algorithms, Alex. Eng. J. 125 (2025) 311–319, https://doi.org/10.1016/j.aej.2025.03.122.
[55] X. Liu, S. Salem, L. Bian, J.T. Seong, H.M. Alshanbari, Application of machine learning algorithms in the domain of financial engineering, Alex. Eng. J. 95 (2024) 94–100, https://doi.org/10.1016/j.aej.2024.03.058.
[56] Z. Liang, H. Chen, J. Zhu, K. Jiang, Y. Li, Advers. Deep Reinf. Learn. Portf. Manag. (2018), https://doi.org/10.48550/arXiv.1808.09940.
[57] H.-G. Zimmermann, R. Neuneier, R. Grothmann, Active Portfolio-Management based on Error Correction Neural Networks, in: T. Dietterich, S. Becker, Z. Ghahramani (Eds.), Adv. Neural Inf. Process. Syst, MIT Press, 2001, in: ⟨https://proceedings.neurips.cc/paper_files/paper/2001/file/4d8556695c262ab91ff51a943fdd6058-Paper.pdf⟩.
[58] F. Black, R. Litterman, Global Portfolio Optimization, Financ. Anal. J. 48 (1992) 28–43, https://doi.org/10.2469/faj.v48.n5.28.
[59] H. Ko, J. Lee, Portfolio Management Transformed: An Enhanced Black–Litterman Approach Integrating Asset Pricing Theory and Machine Learning, Comput. Econ. (2025), https://doi.org/10.1007/s10614-024-10760-9.
[60] G.-Y. Ban, N. El Karoui, A.E.B. Lim, Machine Learning and Portfolio Optimization, Manag. Sci. 64 (2018) 1136–1154, https://doi.org/10.1287/mnsc.2016.2644.
[61] W. Wang, W. Li, N. Zhang, K. Liu, Portfolio formation with preselection using deep learning from long-term financial data, Expert Syst. Appl. 143 (2020) 113042, https://doi.org/10.1016/j.eswa.2019.113042.
[62] Y. Ma, R. Han, W. Wang, Portfolio optimization with return prediction using deep learning and machine learning, Expert Syst. Appl. 165 (2021) 113973, https://doi.org/10.1016/j.eswa.2020.113973.
[63] A. Guarino, L. Grilli, D. Santoro, F. Messina, R. Zaccagnino, To learn or not to learn? Evaluating autonomous, adaptive, automated traders in cryptocurrencies financial bubbles, Neural Comput. Appl. 34 (2022) 20715–20756, https://doi.org/10.1007/s00521-022-07543-4.
[64] K. Park, H.-G. Jung, T.-S. Eom, S.-W. Lee, Uncertainty-Aware Portfolio Management With Risk-Sensitive Multiagent Network, IEEE Trans. Neural Netw. Learn. Syst. 35 (2024) 362–375, https://doi.org/10.1109/TNNLS.2022.3174642.
[65] M. Lv, J. Wang, S. Wang, J. Gao, H. Guo, Developing a hybrid system for stock selection and portfolio optimization with many-objective optimization based on deep learning and improved NSGA-III, Inf. Sci. (Ny.). 670 (2024) 120549, https://doi.org/10.1016/j.ins.2024.120549.
[66] A. Guarino, D. Santoro, L. Grilli, R. Zaccagnino, M. Balbi, EvoFolio: a portfolio optimization method based on multi-objective evolutionary algorithms, Neural Comput. Appl. 36 (2024) 7221–7243, https://doi.org/10.1007/s00521-024-09456-w.
[67] L. Yan, W. Zhou, O.H. Odhah, A.M. Widyan, H.A.E.-W. Khalifa, H. Alqahtani, On modeling and forecasting the stock risk using a new statistical distribution and time series models, Alex. Eng. J. 122 (2025) 65–76, https://doi.org/10.1016/j.aej.2025.02.071.
[68] Z. Ali, Y. Ansari, M. Bukhari, M. Maqsood, S. Park, S. Rho, CMGM: A novel cross-market assets and multi-market modeling graph neural networks for financial market forecasting leveraging market states dependencies, Alex. Eng. J. 128 (2025) 1101–1124, https://doi.org/10.1016/j.aej.2025.08.024.
[69] G.A.V. Pai, T. Michel, Evolutionary optimization of Risk Budgeted long-short portfolios, in: 2011 IEEE Symp. Comput. Intell. Financ. Eng. Econ, IEEE, 2011, pp. 1–8, https://doi.org/10.1109/CIFER.2011.5953554.
[70] G.A.V. Pai, Multi-objective Differential Evolution Based Optimization of Risk Budgeted Global Asset Allocation Portfolios, in: 2014 2nd Int. Symp. Comput. Bus. Intell, IEEE, 2014, pp. 17–20, https://doi.org/10.1109/ISCBI.2014.11.
[71] G.A.V. Pai, Active Portfolio Rebalancing using Multi-objective Metaheuristics, in: 2018 IEEE Symp. Ser. Comput. Intell, IEEE, 2018, pp. 1845–1852, https://doi.org/10.1109/SSCI.2018.8628875.

[72] M. Shahid, M.S. Ansari, M. Shamim, Z. Ashraf, A Stochastic Fractal Search Based Approach to Solve Portfolio Selection Problem, in: 2022: pp. 453–461. https://doi.org/10.1007/978-981-16-6407-6_41.
[73] Z. Ashraf, M.S. Shamim, R. Noman, M. Shahid, F. Ahmad, L. Sami, Constrained Portfolio Optimization Using Hybrid Nature-Inspired Metaheuristic Algorithm in Stock Market. Nature-Inspired Optimization Algorithms for Cyber-Physical Systems, IGI Global Scientific Publishing, 2025, pp. 53–74.
[74] N.S. Kumar Gandikota, M.H. Hasan, J. Jaafar, An adaptive metaheuristic approach for risk-budgeted portfolio optimization, IAES Int. J. Artif. Intell. 12 (2023) 305, https://doi.org/10.11591/ijai.v12.i1.pp305-314.
[75] S. Jaimungal, S.M. Pesenti, Y. Saporito, R. Targino, Risk Budgeting Allocation for Dynamic Risk Measures, SSRN Electron. J. (2023), https://doi.org/10.2139/ssrn.4452742.
[76] A.R. Cetingoz, J.D. Fermanian, O. Guéant, Risk Budgeting portfolios: Existence and computation, Math. Financ 34 (2024) 896–924, https://doi.org/10.1111/mafi.12419.
[77] C.B. Kalayci, O. Ertenlice, H. Akyer, H. Aygoren, An artificial bee colony algorithm with feasibility enforcement and infeasibility toleration procedures for cardinality constrained portfolio optimization, Expert Syst. Appl. 85 (2017) 61–75, https://doi.org/10.1016/j.eswa.2017.05.018.
[78] U. Chauhan, H. Chhabra, P. Jain, A. Dev, N. Chauhan, B. Kumar, Chaos inspired invasive weed optimization algorithm for parameter estimation of solar PV models, IFAC J. Syst. Control 27 (2024) 100239, https://doi.org/10.1016/j.ifacsc.2023.100239.
[79] H. Sang, Z. Li, M.F. Tasgetiren, An Effective Optimization Method for Integrated Scheduling of Multiple Automated Guided Vehicle Problems, Tsinghua Sci. Technol. 29 (2024) 1355–1367, https://doi.org/10.26599/TST.2023.9010087.
[80] D.H. Wolpert, W.G. Macready, No free lunch theorems for optimization, IEEE Trans. Evol. Comput. 1 (1997) 67–82, https://doi.org/10.1109/4235.585893.